

Weird Tales

The Unique Magazine

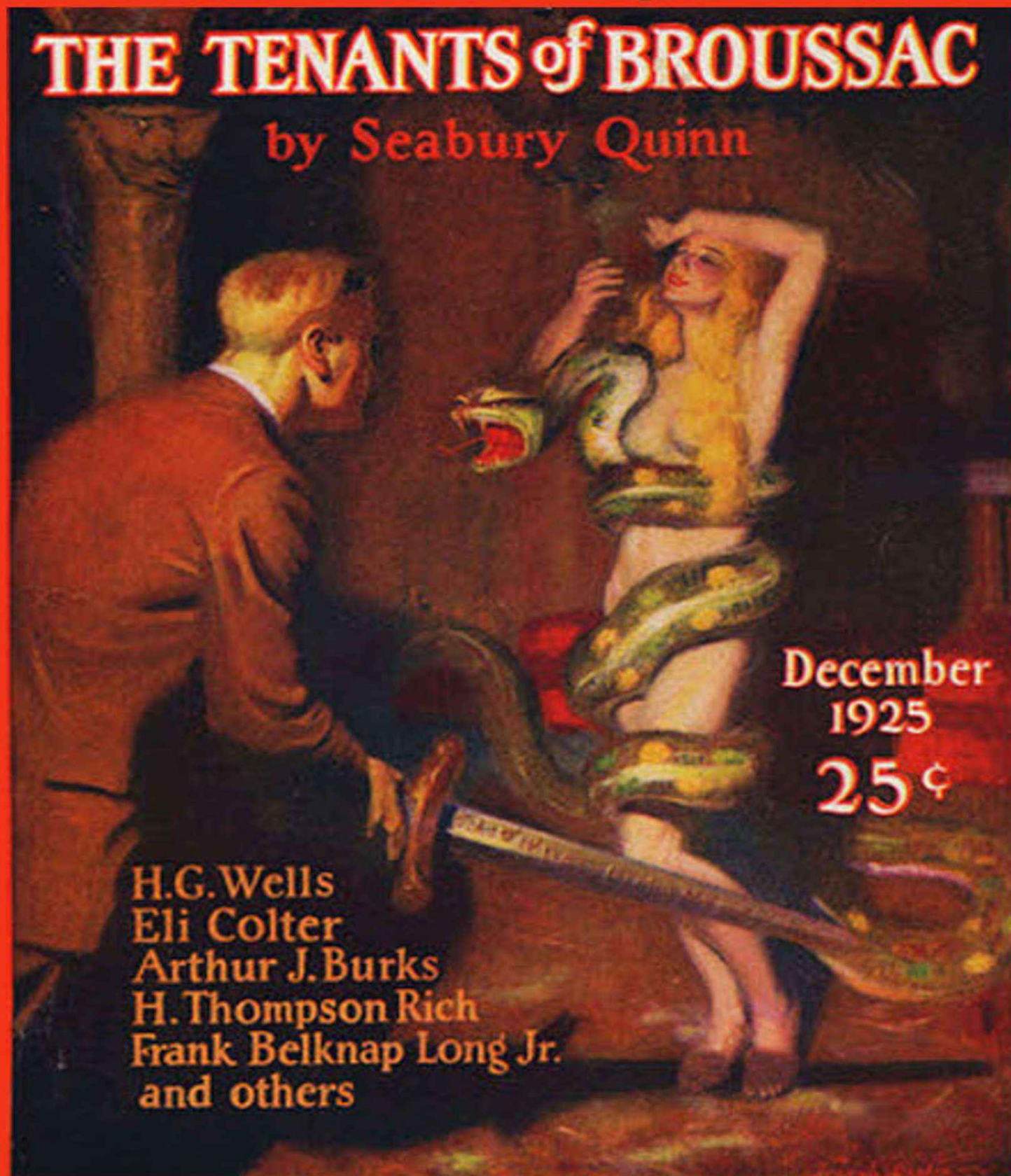
THE TENANTS of BROUSSAC

by Seabury Quinn

December
1925

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H.G. Wells
Eli Colter
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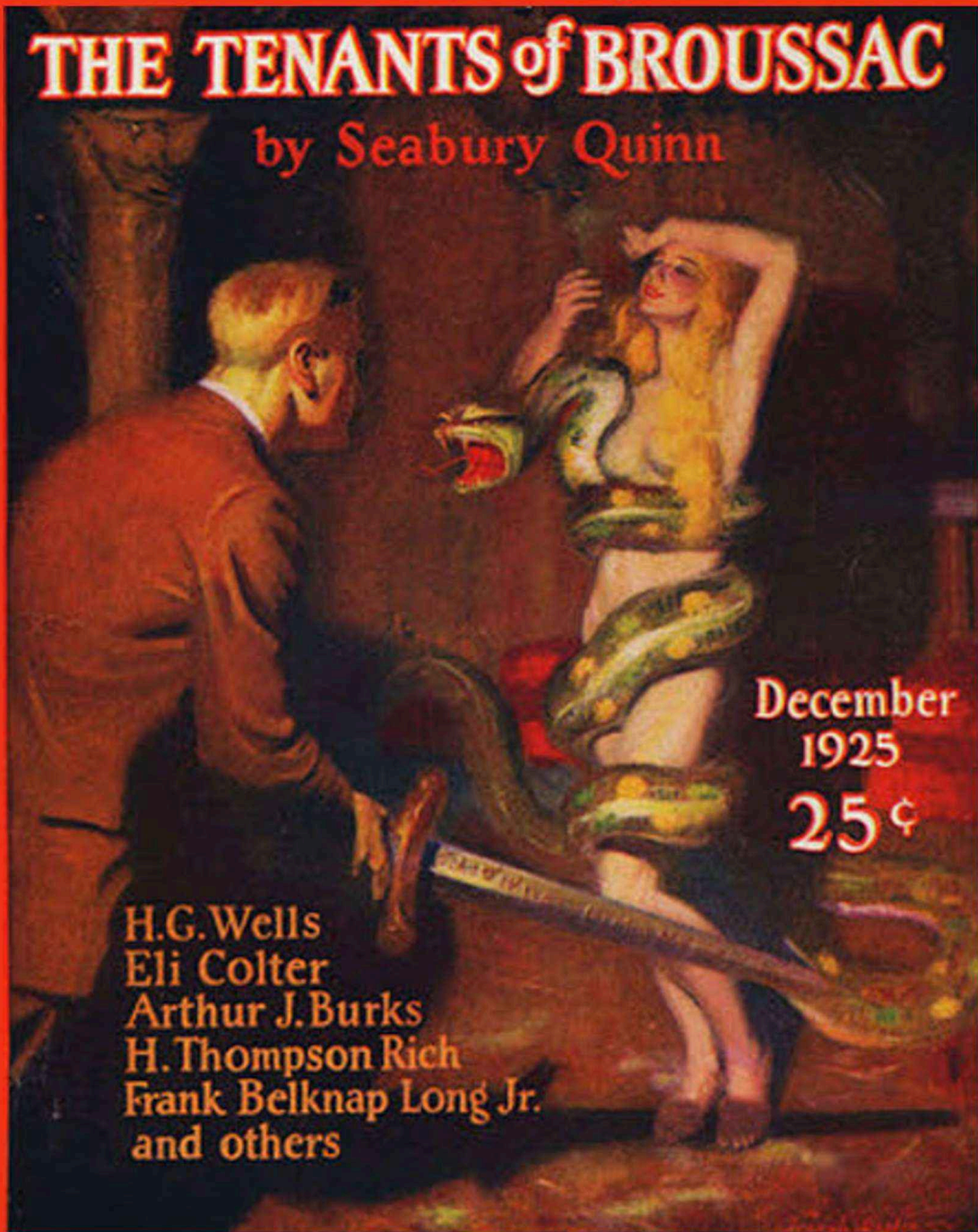
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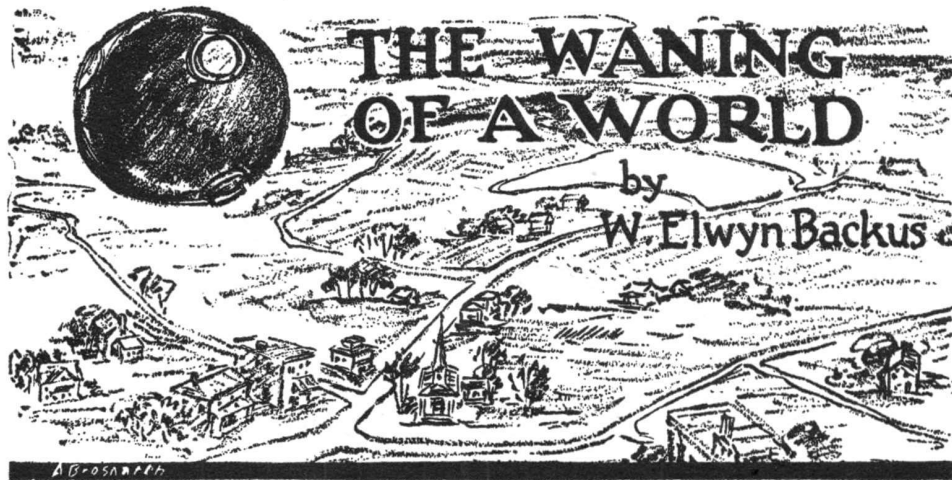
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THE WANING OF A WORLD

by W. Elwyn Backus
Author of "The Hall Bedroom"

It all started over Professor Palmer's book, *Man and the Universe*.

Out of this grew the Palmer-Margard controversy which attracted such wide-spread interest. Profusely illustrated magazine articles abounded on the subject, while Sunday supplements, with imagination rampant, were in their glory. The upshot of this literary duel was the publication of a volume by Professor L. R. Margard, F. R. S., etc., in critical review of his contemporary's deductions.

Public opinion was divided into two camps, each with its chosen champion. The explanation of certain geographical features on a planet some thirty-five million miles away absorbed more of the fickle public's attention for the moment than the outrageous price of a pound of sugar or a dozen of eggs.

In spite of the tax upon credulity which Professor Palmer's theories demanded, they inspired belief among the majority. Perhaps this is because most of us are gifted with an over-supply of imagination; and the Palmer theories appealed strongly to the imagination.

But the majority is not always right; rather the contrary, all of which Professor Margard promptly pointed out. "A challenge to the thinking world," he branded the Palmer theories. To which the eminent Professor Bernard Palmer, A.B., LL. D., retaliated that even Columbus was ridiculed. No doubt, he stated, an astronomer on Mars would have equal difficulty in convincing a Martian public of the possible existence of inhabitants on our earth.

Man and the Universe was written by Professor Palmer after nine years of intensive personal study of the planet Mars. Even his opponents accorded him admiration for his unremitting labors, his perseverance and successful observation.

All of these observations were made from the lonely Palmer observatory constructed near F——, California, 8,000 feet above sea level. Equipped with a giant equatorial telescope having a 48-inch object glass, and situated

ideally as to atmospheric conditions, Professor Palmer was excellently prepared to observe our much-discussed neighbor.

One result of his observations was the careful recording and mapping of curious straight lines visible on the planet. Running from the polar caps down to and across the equator, crossing and recrossing, these lines formed a veritable network over the planet's surface. Here and there round spots appeared at junctures of the lines. Some of the lines were discovered to be double, although these were few, the great majority of them appearing singly.

By continually observing the planet during ensuing seasons, a marked decrease in the size of the polar caps during the Martian summers was noted, with a corresponding darkening of the "canals", as Professor Palmer designated the lines. The spots, or terminals, he called "oases." His deductions were, that owing to the admitted scarcity of water on the arid planet, the Martians transported water from the vast, melting polar snows by means of canals. It was this question of canals, and the much mooted question of sufficient heat to sustain life on the small planet, which caused contention between the two renowned experts.

Despite many years of concentration on technical things, Professor Palmer was a surprisingly human and ordinary-appearing man. The fact that he was considered one of the world's foremost authorities on astronomy, and was a lecturer of world-wide renown, did not detract from his naturally benign disposition. Yet there was something compelling about his personality. Students before whom he delivered lectures accorded him marked attention, and went away with a graphic picture in their minds of the things he wished to convey to them.

"Henry," he remarked quizzically to his young but capable secretary one morning, having just read a particularly seathing criticism of his pet theories, "these gentlemen at least take my hobby seriously. That in itself convinces me that my deductions are worthy of consideration."

"They take advantage of the fact that you have no means of definitely proving your deductions," defended Henry, loyally. "As you have remarked, even Columbus was ridiculed; but he, at least, had means of obtaining concrete proof to silence the scoffers."

"Well put, my friend. A few more earnest advocates of my theories like yourself, and I should require no proof."

"That reminds me," resumed the grateful Henry G. Simms, "there was a young chap in here yesterday who claimed to be an ardent champion of your views. He was extremely anxious to see you. Said he had a matter of great importance to take up with you."

"He'll return?"

"Grant's army couldn't keep him away. He will be here at 10 o'clock."

"So? What sort of a chap is he?" curiously.

"A well set-up fellow of about twenty-four. About six feet tall; light hair, pleasant features, refined manners. Impetuous sort of fellow."

That was how Robert Sprague happened to meet Professor Palmer a half hour later.

A pleasant thrill possessed him as he shook hands with the professor. The kindly, though keen, gray eyes met his encouragingly. This was contrary to

Robert's expectations, for he had assumed that he would be fortunate if he succeeded in seeing so busy and prominent a man for a few minutes. He was prepared even for a curt dismissal. What he did not know was that his evident earnestness and enthusiasm had obtained for him an interview through the redoubtable Henry where others would have failed.

Without realizing how he had commenced, he found himself conversing easily with this learned man as if such interviews were everyday occurrences with him.

The professor was impressed with equal favorableness by his caller. The frank, winning countenance and earnest manner created a profound impression upon him in spite of an extraordinary story.

"Let me get this right," said the professor, finally. "You say that the machine is virtually perfected—that you have succeeded in accomplishing the aim for which your father unsuccessfully spent his life?"

"Not unsuccessfully," defended Robert, quickly; "without what he had accomplished I could never have constructed a machine of its kind."

"But it can actually be controlled as you suggest?"

"It can."

"Pardon my insistence, Mr. Sprague. The idea is so—ah—extraordinary."

"I realize that, professor. I should be happy to have you see for yourself."

Professor Palmer pondered. The young man's story had impressed him, notwithstanding its unusualness. At any rate, he concluded, he would investigate. He could risk no more than disappointment. If there was anything in it, the possibilities for research and discovery were boundless. He found his own enthusiasm rivaling that of his caller as he momentarily allowed it free rein. Why—he might yet prove his own weird theories to the world!

The next moment he smiled at his own indulgence. First he would humor this young man by investigating his wild claims: time enough for dreams afterward.

"Well, Mr. Sprague," he said, "this is an age of strange accomplishments. I'm going to look at that machine of yours. How will tomorrow evening do?"

A feeling of relief and exultation swept over Robert as the professor spoke. At last his absurd-sounding claims for the life-work of his father had

been taken seriously, and recognition of his labors was within reach.

He stammered his thanks, shook hands with the amused professor, and departed.

“That boy believes in the machine; and he is no fool, either,” remarked Professor Palmer after Robert had left.

“The world is full of them,” observed Henry sagely.

Henry even openly questioned his employer’s theories at times. Not that the latter minded, for the ensuing arguments furnished interesting debates, and fresh ideas sometimes; and in the end he usually succeeded in silencing his intrepid secretary—if only temporarily.

But today, Henry’s caustic comment irritated him. He wanted to believe in the weird claims of his caller regarding a strange, gravity-defying machine, in spite of his saner judgment to the contrary. He subconsciously resented any expression of his own disbelief.

Professor Palmer slept poorly that night, though he rarely failed to sleep soundly. Try as he would, he could not dismiss from his mind the hope which struggled so persistently with his natural skepticism.

But he was not alone in his sleeplessness. Robert slept not a minute that night. Over and over he reflected on just how he would best explain the intricacies of the *Sphere* in order that he could convince Professor Palmer of its practicability. The fact that he lacked the necessary funds to complete the apparatus gave him considerable concern. For much, therefore, depended upon his ability to convince the professor of the feasibility of mere theories.

It was with considerable relief and expectancy on both sides that Robert and Professor Palmer shook hands in the big, high-ceilinged parlor of the old Sprague manor.

Their footsteps echoed eerily through the house as they tramped back through the long dark hallway to a big barnlike addition which had long served as a workshop. Here Robert's father had spent countless weary hours, to the despair of his good wife, who had already followed him to his reward.

As they entered the doorway the professor became aware of an immense gray-black sphere in the dusk of the far end of the shop. The top of this sphere reached within a few inches of the lofty ceiling. It was probably twenty-five feet in diameter, and rested upon a short scaffold. What appeared to be curious round windows in its side, like portholes in a ship's hull, gave it the appearance of a gigantic diver's helmet.

Robert approached the *Sphere*. Without hesitancy he selected and pressed upon what appeared to be an ordinary rivet-head like hundreds of others over the *Sphere's* shell. A round hatch, large enough to admit a man, swung open, disclosing a black and uninviting interior. Flustered, he courteously invited Professor Palmer to enter first.

For an instant the professor hesitated. The weirdness of the whole affair suddenly struck him forcibly. This young man's queer claims, the big manor with its eery echoes and atmosphere of dismal loneliness—all seemed to cry out to him to beware. The dull gray shape looming above them in the gathering twilight looked disquietingly like some freak prison, such as a madman might invent.

Robert, sensing the professor's misgivings, apologized for not having considered his difficulty in negotiating the unfamiliar interior in the darkness, and relieved him by entering first. A sharp click, and a comfortable glow of light suffused the interior. They passed up a brief, winding stairway into a long chamber.

“This is the gyrostatic control which neutralizes the force of gravity,” Robert began, calmly, as if this assertion were the simplest thing in the world. He indicated a complicated mass of glittering machinery in the center of the compartment in which they stood.

He reached for a small lever, and pulled it toward him. Simultaneously there was a soft whirring sound. For a moment the floor tilted slightly, then steadied again.

“And the power for this?” queried the professor.

“Furnished by storage batteries,” Robert explained. “The batteries are recharged by petrol-driven dynamos.”

“But your supply of petrol? Where have you sufficient space for a supply that will last any considerable length of time?”

“All round us.”

The professor swept their surroundings with his sharp eyes. No receptacle was visible. Two full-size doors and several small ones appeared in the partitions; but nothing suggested a receptacle for a large supply of fuel. Then quite suddenly it dawned upon him that there was a vast amount of space unaccounted for between the partitions, floor and ceiling, and the *Sphere*’s outside shell. His respect for Robert’s claims was growing. So far, at least, the young inventor seemed quite confident.

“What is this?” asked the professor, indicating what resembled the breech of a dreadnaught’s gun protruding from the floor. Electric wires, dials, and other curious devices were connected to it.

“That’s the Norrensen Tube, so named by my father after its inventor, an old friend of his, now deceased. It is capable of terrible destruction. It will produce a bolt of lightning rivaling the elements, which will strike up to twelve miles away—and it can be aimed with startling accuracy. I remember seeing a giant oak blasted into pulp with it in a test across a valley four miles wide, when I was a boy.”

“But, how is it that the world has never heard of this remarkable invention?”

“Norrensen was an eccentric character whom the world had wronged grievously. He insisted on conducting the tests with greatest secrecy. Overtaken suddenly by a fatal illness, he exacted a promise from my father to retain the secret of this weapon till his death.”

“What a terrible weapon that would make in the hands of a man bent on destruction!” mused the professor.

The compartment they were in looked to be about twelve by twenty-five feet, and some ten feet in height. There were three round ports at either of its rounded ends; these, being located below the bulge of the *Sphere*’s greatest girth, enabled one to obtain a good view downward as well as outward. The straight sidewalls and ceiling were windowless, but a vertical well extending from the floor, beside the controls, to the outer shell, with heavy, circular glass panes at either end, enabled the operator to see out directly below. The compartment was flooded with soft, mellow light from a dozen frosted incandescent lamps.

“Deducting for this compartment, two small storerooms, the cupboards, and the water and oxygen tanks,” Robert was saying, “the net capacity of the petrol reservoirs is more than 40,000 gallons. That and the full storage power of the batteries is sufficient to operate the high-speed, but delicately balanced gyrostats, more than fifty days and nights continuously.”

“You say that gravity is completely neutralized?”

“Almost entirely so, even with all reservoirs filled to capacity. The stability of the gyrostatic device is so powerful that weight becomes a negligible factor. If you will follow me I can prove this to you.”

The professor quivered with suppressed excitement as he followed Robert down the flight of steps leading to the outer manhole through which they had entered. At last he was about to know beyond doubt whether the remarkable claims made by his guide had any foundation. If they had, a new era would be unfolded. Again his common sense reacted against hope, blasting his short-lived credulity. That either this boy or his father should have mastered the problem of the fifth dimension after experts of centuries had failed, seemed unbelievable. And yet—

In the deepening twilight the *Sphere* seemed to loom above them larger than ever. Its lighted portholes, contrasting strangely with its shadowy bulk, gave it a weird, fantastic, almost unearthly aspect.

“The *Sphere* is now in almost perfect equilibrium from every direction,” Robert explained, pride creeping into his voice unconsciously. He indicated two iron rungs near the bottom of the *Sphere*. “If you will take hold here, you will be able to move it in any direction without effort. Softly though—keep a firm hold upon it.”

Doubting still, Professor Palmer grasped the rungs, fully expecting to find the vast bulk an immovable weight.

To his intense surprize it rose from the floor as if it were an air-filled balloon! He had exerted himself not the slightest bit. The *Sphere* had simply risen at his first slight lift, and had continued to rise until a slight tug upon his arm stopped it. He extended his right arm, still gripping one rung. The *Sphere* followed easily, its only resistance apparently that of the atmosphere surrounding it.

“Now release it,” suggested Robert.

The astonished professor did so, half expecting to see it crash to the floor.

But nothing of the sort occurred. For several seconds the giant ball continued to rise very slowly, like a sluggish soap-bubble. Doubtless he had unwittingly allowed his hand to waver slightly when releasing it.

Then very, very slowly the *Sphere* began to descend, finally settling softly and with scarcely a sound. Though it had been but a few feet above the floor, it required fully a minute to come to rest. One noticeable feature was its vertical stability. It neither rolled in its descent nor wobbled in settling, but simply came down with a paradoxical combination of majestic ponderance and zephyrlike softness.

“Remarkable!” ejaculated the professor, feeling the inadequacy of the word when applied to this marvelous achievement.

“The rigid stability,” Robert explained, “is automatically controlled by a delicate device attached to the central upright gyroscope.”

“I was under the impression that the entire apparatus was unaffected by gravity.”

“This device is the exception. The *Sphere*’s weight is neutralized to an absolute minimum by the gyroscopic control, but it was necessary to maintain one point of gravitational contact in order to establish some permanent upright stability; otherwise, the *Sphere* would revolve at random when in midair.”

“Manifestly.”

“This device also makes it possible to maintain the observation ports at the ends of the main compartment in any desired direction horizontally. It may surprize you to hear that this device was the last part perfected. My father’s final prostration was largely due to its intricacies. He passed away

just as he was about to achieve its perfection.” A slight quaver in Robert’s voice betrayed his grief and his deep regard for his departed parent.

Professor Palmer’s eyes kindled sympathetically.

“On the contrary,” he replied, “I can readily understand the difficulties encountered.”

A silence ensued during which each was busy with his own thoughts. Robert was thinking of the most important feature of all—the propulsion of the *Sphere*, and its control. This principle had been worked out on a small scale, but owing to its prohibitive cost on a larger scale he had been unable to perfect its application to the *Sphere*. Professor Palmer, with his personal resources and backing, could finance it, but even then, Robert estimated, it would tax his total resources heavily. Robert held no illusions on this point, and he was wondering how best to present his plea for financial aid.

Professor Palmer was trying to visualize the possibilities in the *Sphere*. In it he saw a possibility of proving his own theories regarding the planet Mars, and this brought him round to the very feature on which Robert’s thoughts were concentrated at that moment.

“Hm-m,” mused the professor. “And you claim to have worked out a scheme of magnetic propulsion requiring a minimum of internal energy?”

Robert drew a full breath and prepared to retrench.

“Only on a small scale, professor. I have a miniature model over here, illustrating the practicability of the idea.”

He switched on the light over a work-bench, revealing a curious contrivance about five inches high. A dull black rod, terminating in a tiny blunt bell-like device, hung suspended from a universal joint. The whole was supported by a small frame bolted to the table. Examination of the bell-like bulb showed that its larger end was flat, and composed of a dull, whitish metallic substance similar in appearance to aluminum. Its outer surface was a brilliant silver. This bulb seemed of extraordinary weight for so small an object, swinging heavily back to its former position when released, where it came to rest quickly over the center of the disk almost as if bound in that position with a strong, invisible elastic band.

“This pendulum,” Robert explained, in response to the compelling and unconcealed curiosity in the professor’s eyes, “contains a rare, and hitherto unknown, element which my father named ‘mythonite.’ A good part of his life was devoted to the accumulation of this small quantity for experimental

purposes. It was obtained bit by bit through a difficult and costly process from vast amounts of river-gravel, in conjunction with platinum, to which, strangely enough, it has a strong antipathy. This condition is responsible for the most curious discovery of all. The effect of gravity upon mythonite is almost entirely annulled through platinum!”

“Remarkable,” said the professor; but there was a trace of incredulity in his voice which was not lost on Robert’s alert ears.

“This casing,” resumed Robert, tapping on the polished side casing of the pendulum, “is a very thin layer of platinum. With the pendulum inverted, the earth’s attraction is intercepted by the casing. At the same time the attraction of any other heavenly body within the radius of the uncovered surface of the mythonite is unchanged. Further, I have discovered that the free attraction of mythonite is greatly intensified by electricity, without any corresponding increase in its gravity through the film of platinum.”

He clicked on a small switch attached to the base of the frame. An odd phosphorescence suffused the disk-like surface of the pendulum.

“Now, professor, will you raise the pendulum to a vertical position? Take hold of the insulated rod, here.”

Professor Palmer raised the pendulum slowly. Its original weight, extraordinary as it had seemed before, was now several times greater, to his astonishment. It now seemed almost as if it were riveted into position.

But gradually, as the glowing disk was pointed upward, its weight decreased. At an angle of ninety degrees its weight had virtually ceased to exist. As it neared an upright position it felt as light as a feather. In an upright position it seemed poised between the professor’s fingers as if about to take flight.

He released it softly. It wavered unsteadily for a moment like a flower balancing in a light breeze, then steadied. The professor’s fingers, clumsy from pent-up excitement, collided with it. With a sudden swoop, it dropped heavily into its former pendent position, coming to rest abruptly.

Professor Palmer drew a sharp breath excitedly.

“Young man,” he said, extending his hand, “you have convinced me, even as I hope to convince a lot of other doubting Thomases and scoffers some day. Apparently you have evolved the greatest discovery of all time; I congratulate you.”

There was no doubting his distinguished visitor's sincerity. Robert's voice was husky as he stammered his appreciation.

"Now, let's get down to brass tacks," continued the professor. "This device installed on a large enough scale in the *Sphere* would make it possible to propel it anywhere in space. The possibilities for research would be virtually boundless. Have you estimated the probable cost of such an apparatus?"

"Often. Even with the aid of improved equipment and sufficient workmen, it would require considerable time and a great expenditure. Fifty thousand dollars is a low estimate—and seven months' time."

Professor Palmer looked thoughtful. Though he was known to be comfortably fixed, his total resources did not quite meet this sum. Slave to science though he might be, he hesitated to gamble his entire fortune on a visionary venture that might prove to be impracticable. As to the deficiency, he could get that as a loan or a gift from one or more of his many wealthy friends who had every confidence in him. Should the scheme fail, he would be penniless—possibly friendless.

"Do you believe a flight to another planet and back could be made successfully in the *Sphere* so equipped?" he asked.

Robert considered carefully. He did, but the professor's question renewed many doubts. Most of all, he hesitated to involve his would-be benefactor in a disastrous venture.

"I do," he answered truthfully, at last.

"So do I," supplemented the professor, stoutly, as if to help convince himself. "Would you be willing to undertake such a journey?" he asked suddenly, fixing his eyes keenly upon his host.

"Yes, sir!" responded Robert quickly.

His prompt reply and evident sincerity convinced Professor Palmer that he was in earnest. The professor had decided. Nothing risked, nothing gained. As for Robert, nine months of trench warfare in France had steeled him against fear of anything except women and the devil.

"It is settled, then," concluded the professor, unconsciously authoritative. "You will come to my home, and together we will supervise the completion of the *Sphere*."

"But the *Sphere*—," began Robert, surprised by the professor's quick decision.

“We will have it conveyed to my estate, where the light and space will be much better; and where I can look after you better, my boy.” His face softened. Ah, an old bachelor had not all the advantages. What would he not give to have a son like this!

Something about Robert’s hesitation reminded him abruptly of an important consideration.

“I am forgetting,” he apologized. “Your interest must be fully protected. We will draw up a contract whereby full possession of the *Sphere* and all its equipment, now and always, will remain yours. I will undertake to complete it, defraying all expenses, in return for which I ask the use of the *Sphere* in a flight to Mars and back if possible.”

“That is more than fair,” Robert replied, feeling ashamed of a shortlived, though natural, apprehension.

Robert found Professor Palmer's homestead vastly more cheerful than his own gloomy quarters.

The *Sphere* was placed in a large, well-lighted barn, which had been carefully prepared for its new purpose. The barred windows were frosted to defeat the curiosity of possible busybodies, and reliable locks put on the heavy doors.

Removal of the *Sphere* from its original quarters presented difficulties, because no provision had been made for its exit. It had been constructed piece by piece inside the four walls which housed it so long. It was necessary to hew an opening through the wall, to the acute curiosity of the neighbors.

However, their curiosity went for naught, as a large tarpaulin and protecting crating disguised the object of their interest, which was removed at night. Professor Palmer and Robert were agreed in their decision to keep their project to themselves as much as possible until they had succeeded in perfecting the *Sphere*.

The hauling of the *Sphere* proved an extremely simple task. With the gyrostats running quietly at halfspeed, its weight was rendered to almost nothing. Nevertheless a large, heavy truck was provided for any emergencies. They wisely avoided any unnecessary chance of destroying, at the very beginning of their task, the intricate work of many painstaking years.

A small brick building was put up and the necessary machinery installed for the production of mythonite. Here the tedious process was soon directed by Robert. Eleven skilled metallurgists and chemists labored day after day under his supervision, without knowing for what purpose the curious metal they were producing was to be used.

Weeks passed, and vast quantities of waste material were hauled away daily; but the quantity of the precious mythonite accumulated with discouraging slowness. An addition was built adjoining the first plant, and

the corps of experts increased to an even two dozen. By improving methods and increasing deftness, the former production was trebled.

Nevertheless, it soon became apparent that the desired quantity could not possibly be produced at the present rate within the period which Robert had estimated. The first month's operation had resulted in but two small ingots, each an inch square and three inches long. This was before the plant was enlarged, however. This had been eventually increased to seven ingots a month. But even at this rate, it would require almost four years longer to produce a sufficient quantity. Obviously, the project was doomed to failure unless some means of greatly increasing the production could be devised.

It was four months after the completion of the original plant that Robert and Professor Palmer were discussing this matter with a view to deciding finally whether or not to abandon the project. During this time Professor Palmer had come to look upon Robert as a son. His untiring energy, his frank, cheerful personality and intelligence, had made a profound impression upon the professor.

"I am going to see this thing to a finish, Robert," he was saying. "But if we are to take advantage of the next favorable apposition of Mars just eight months from now, we must make some radical improvement in our program. Not till fifteen years later will it again approach so close to the earth. Have you any new plans to suggest?"

"This," replied Robert. "We might have the crushing and the crude processes done elsewhere. By concentrating upon the finer processes alone, we should be able to increase our production of mythonite considerably. But we should have to replace the crushing apparatus with additional equipment for the final processes. We must take advantage of every available bit of space and every man's time."

"Our total expenditures to date are what?"

"Approximately \$33,000. But, the sales of excess platinum have reduced that to about \$32,000."

"Not so bad," mused the professor. "However, it is clear that we can not reach our goal without a vastly greater rate of production."

He knit his brows, pondering silently for a little while.

"Robert," he broke out suddenly, "we've got to take a big gamble! We will not only follow out your suggestion, but we will double the present size of our plant."

Robert gasped. He thought of the professor's dwindling resources, wondering if he were suddenly gone mad.

"Why, that would bring the total cost round \$60,000!" he cried.

"Quite so," replied Professor Palmer, calmly; "but a four or five year program would be far more expensive—to say nothing of its impracticability. It's win all or lose all, Robert."

So the Palmer laboratories were enlarged and arrangements successfully made for the crushing and partial separating with a near-by rock plant. The little force of experts was augmented to thirty, and work began in earnest. The next month resulted in a production of forty-one ingots of mythonite!

The following month a minor improvement discovered in the process increased that month's production to fifty ingots. Even this production was bettered somewhat during the following months. At the end of the sixth month after the enlargement of the plant the total production of mythonite had reached more than three hundred ingots—all that were required! A month remained in which to prepare for the great venture into the unknown.

It was with a feeling of overwhelming elation that Robert and the professor gazed upon the little stack of dull, silver-gray bars in the dusk of an early July twilight. Winter and spring had come and gone while they labored. These three hundred tiny ingots were the result. Not entirely, though; for in addition to a sufficient quantity of platinum reserved for their own requirements, the Palmer laboratories had produced and sold enough platinum to defray all expenses incurred. Little wonder that they felt elated.

Professor Palmer put his arm across Robert's broad shoulders with fatherly tenderness.

"My boy," he said, softly, "whatever the *Sphere* accomplishes, it has at least brought us together. To me, our perfect companionship has come to mean more than anything else. I did not realize what a lonely old man I was before you came."

"Old man!" chided Robert. "Fifty-seven years young."

"It is well for me that you had the *Sphere* to occupy you, or some sweet young vision would have taken you in hand ere now. But forgive an old codger's selfishness, Robert."

"Time enough to think about that, professor," smiled Robert.

“Careful. Don’t let them make a bachelor out of you. An old bachelor is a superfluity for which no one really cares. Even an old maid has her cat.”

“Very well. We’ll each make love to a moon-maiden,” laughed Robert, and Professor Palmer joined him heartily.

The following day the small ingots were melted and forced into the big, flattish, circular, platinum-lined and studded mold. Before the pouring was attempted, the mold was securely fastened down as a precaution against the lifting power of the mythonite when freed from the earth’s gravity by the interruption of the platinum beneath it. As an additional precaution, a disk of platinum was suspended over the mass, thereby neutralizing the attraction of heavenly bodies.

With great care, the platinum-incased mass of mythonite was installed in the *Sphere*. A stout steel rod and universal joint connected it to the gyrostatic center, and the wiring and other details of its proper control were quickly completed. The petrol and oxygen tanks were partly filled, the gyrostats tuned up, and the *Sphere* at last was ready for a trial trip.

Henry Simms, much interested, but skeptical to the last, was shown the interior of the *Sphere* on the afternoon set for the first trial. He crawled through the manhole after Robert and the professor, firmly convinced that he was about to witness a flat failure of the *Sphere* for which the professor claimed so much. To do him justice, though, it should be stated that Henry's expectations were not without keen sympathy for the disappointment to which he felt certain the professor was doomed.

"She looks more like a submarine than a blimp, professor," was his first comment as they reached the main compartment.

Indeed, the interior of the *Sphere*, with its intricate mass of machinery and its bull's-eye windows, its riveted partitions and curved walls, and the incandescent lamps, did suggest a typical underseas craft.

"She goes up, Henry, not down," the professor laughed.

"Deal me out, then," cried Henry. "I am not prepared to go up for keeps yet!"

"Rest easy," said Robert. "It will be much easier to drop back, if in doubt, than to continue upward."

Robert proceeded to explain the *Sphere's* important features for Henry's benefit.

"Here is the gage that registers the pull of the disk," he said, finally, after having explained the rudiments of the *Sphere's* operation. He indicated a dial attached to the rod which harnessed the powerful mythonite disk to the core of the *Sphere*.

He pushed the first of a row of switch buttons on the controller. Poor Henry's heart fluttered as a faint scraping sound heralded the mere opening of one of the three cameralike platinum shutters over the mythonite disk's highly magnetic surface. He was already regretting his consent to accompany them on a trial flight. The handle on the dial of the lifting gage suddenly raced from zero and steadied at 605 pounds. The *Sphere* remained at rest.

All three men were now keyed to the highest pitch of excitement. This was the first time the completed apparatus had been tested, and upon its results depended entirely the success of the *Sphere* and its remarkable project planned by the professor.

The registered tension on the strong steel arm removed all doubt from the minds of Robert and Professor Palmer regarding the success of mythonite as a practical power of propulsion. A feeling of wild exultation gripped them both.

“Danger from shock of sudden great pull is avoided by gradual uncovering of the disk’s surface,” resumed Robert as he pushed the next button, sending the hand on the dial up to 1,420. The third button swung it to 3,475, accompanied by a slight tremor perceptible in the floor of the *Sphere*. Their startled glances through the nearest porthole satisfied them, however, that the *Sphere* still rested on *terra firma*.

Robert pushed all three of the corresponding row of buttons directly over the first three, and the hand again registered zero.

“I don’t want to lift the roof off your barn, professor,” exclaimed Robert. “I’ll start the gyrostats now to neutralize the *Sphere*’s weight, and we will get out and push it outside the stable.”

A few minutes later the now thoroughly convinced Henry watched his companions disappear within the *Sphere*’s shell while he debated with himself as to whether he should follow them. A moment later Professor Palmer appeared at a porthole and beckoned him; but Henry shook his head vehemently.

The professor unlatched the window and swung it open.

“Hurry in, Henry,” he called. “Voyage is about to commence.”

“Not I, professor! This suits me real well, right out here.”

“Come on, Henry,” the professor urged. “You aren’t afraid?”

“Not afraid—just a little bit careful. I’m just beginning to find out how nice and solid this ground feels. I’ll watch you do it.”

And no amount of urging would change his mind. He politely but firmly maintained that he felt much healthier outside.

“Stubborn chap, that,” the professor commented to Robert. “Can’t say that I blame him, though.”

“Simply a difference in the values we set on our own carcasses,” suggested Robert. “Henry just takes his more seriously than we.”

They laughed. Both, somehow, felt relieved afterward. Henry had furnished a welcome diversion. The former nervous tension was broken.

“Well, so long, old man,” Robert called out the window, as he prepared to close it.

“Give my regards to Saint Peter,” shouted Henry.

“Cheerful cuss,” contributed the professor, as the heavy glass slammed shut.

Robert stopped the gyrostats.

A deep silence reigned within the heavy walls as he examined carefully the delicate machinery upon which so much depended. Then he pulled the lever, setting them in motion again. Their steady purr was a relief from the oppressive silence.

Professor Palmer’s keen eyes followed him as he moved about. Robert’s excitement of the previous minutes was forgotten as he expertly, almost lovingly, ran his eyes over every detail of the perfect, whirring machinery, most of which his father had produced. His throat contracted strangely as his thoughts dwelt for a moment on his beloved parent. His mother he could scarcely remember, for she had died when he was but a baby of three years. But his father had been his constant companion—his pal. What would he not have given to have him standing by him at this moment, on the eve of his triumph, of the realization of his dreams!

Being a shrewd judge of human nature, the professor rightly guessed his thoughts at that moment. A suspicious moisture in Robert’s eyes confirmed his guess.

Robert’s next move was to adjust the direction of the disk’s covered face toward the zenith. The gyrostats were revolving smoothly. With bated breath, he again pushed the button which partly bared the disk.

The *Sphere* gave a slight lurch. This was followed by a sensation like that felt in an elevator rising suddenly. A faint shout from below. With one impulse Robert’s and the professor’s glances swept eagerly through the ports.

There they saw just what they had expected to see; but the actuality affected them curiously. Oddly enough, they had subconsciously expected till the last moment that the *Sphere* would fail.

The landscape seemed to be dropping from under them. Even the horizon was receding alarmingly.

Robert's hand shot out to the control board, closing the disk's surface. A slight tremor evidenced the abrupt cessation of the disk's pull.

"Six thousand feet," read Professor Palmer from the altimeter.

Robert joined him. A few minutes later it registered seven thousand. They were still rising, but not nearly so rapidly as before. The closing of the disk had checked their speed at once.

"A little more and I'd have boosted her right off the earth," said Robert, breathlessly. "I'll have to use the disk more sparingly on ordinary sight-seeing excursions hereafter."

"You had it opened only to first power, too, hadn't you?"

"Yes; and without the 'juice' turned on. Jove! We didn't realize how much reserve power of propulsion we had. It's well that I experimented first with the minimum. And the current almost quadruples the magnetism of mythonite! Phew!"

Robert paused and read the altimeter again. Eight thousand. He gripped the gyrostatic control, and carefully moved it to half speed.

The *Sphere* seemed to pause a moment, then they could detect its beginning to settle earthward as the neutralization of gravity was modified. Six thousand; five thousand; they were dropping steadily at a rate of nearly a thousand feet a minute.

Robert shoved the lever back to full speed and the *Sphere's* downward momentum was quickly checked. With the disk safely throttled, the *Sphere* became as a rubber balloon. They merely drifted in midair.

Together they peered through the observation well in the floor. Through this they could plainly see the landscape, some three thousand feet below, sliding by sluggishly as they drifted with the light air current. From the side ports they could discern the big Palmer homestead and the laboratories about a mile and a half to the west of them. It was an ideal day for observation. The sky was cloudless, and the air of crystal clearness.

"Well, professor, shall we run back to our stall, or take a little sight-seeing jaunt?" queried Robert.

"Let's see some of the country, by all means," decided the professor, his face aglow with boyish excitement and anticipation.

"All right; here goes," Robert sang out as he deflected the disk to a horizontal position, pointing due north.

The next instant he switched open the first shutter from the disk's surface. There was a jerk, and the landscape suddenly began slipping away to the south with accelerating speed. Another click, and their speed was further increased. Once more the switch clicked, releasing the last shutter from over the disk. The *Sphere* seemed literally to leap ahead. A muffled roar without indicated the great speed at which they were rushing through the air.

Town after town flashed by beneath them with astonishing rapidity. The fact that they were flying at a comparatively low altitude made their speed seem terrific. Robert wisely decided to seek a safer height. He elevated the disk several degrees and the *Sphere* promptly soared higher. At eight thousand feet he checked its upward trend.

Far away to the east they could see a solitary big biplane bound in the same direction as they—probably a fast mail express; but it was quickly left behind, and lost from view in the afternoon haze.

For twenty minutes they roared northward. Then, to their surprise, a vast body of water appeared against the horizon ahead.

“Lake Erie!” gasped Robert, after a moment's reflection. “Two hundred miles in less than half an hour. Why—that's about five hundred miles an hour! And without the aid of electric magnetization of the disk!”

“Marvelous!” exclaimed the professor, enthusiastically.

Already they were soaring over the expanse of water. On the horizon the distant Canadian shore was rapidly taking shape. Beneath them several long, slim lake craft could be discerned, crawling at what appeared, from so great a height, to be a snail's pace. No doubt the *Sphere* would have presented a much more curious sight to those below had its luminous gray shell been more than a faint speck against the brilliant, cloudless sky.

It was at this juncture that Robert's alert ears detected a subtle change in the hitherto soft whirl of the gyrostats.

“What is it, Robert?” whispered Professor Palmer, as he observed Robert's suddenly tense attitude.

“Wait!” anxiously.

Outside, the muffled roar sounded in strange contrast to the still air within. The bright sunshine streamed across the gray door in mock cheerfulness. A single captive fly buzzed drowsily against a windowpane.

These commonplace details registered on Robert's mind indelibly in those fleeting seconds as he listened with palpitating heart for he knew not what.

Taking his cue from Robert, Professor Palmer was listening with equal intensity to the drone of the machinery upon which their lives depended. Even he could now detect the change. The drone was gradually, unmistakably, decreasing in volume. The gyrostats were stopping!

Unconsciously they gripped each other's hands an instant as they realized the seriousness of their plight. Should the gyrostats stop, the *Sphere* would plunge to its doom!

Frantically Robert tortured his mind for a possible solution, or a reason for the unexpected interruption. The altimeter already indicated that they were falling at a steadily increasing speed. The formerly tiny ships below were no longer tiny. The water seemed to be rushing toward them at a terrific rate. Robert remembered afterward a sudden inane conjecture as to how big a splash they would make.

It was at this moment his numbed senses returned to him. Cursing himself silently for a rattle-brained idiot, he spun the wheel madly, thus adjusting the vertical position of the disk. To his tortured mind it seemed an eternity before it finally pointed toward the zenith.

Their downward rush was noticeably checked, but the lift of the disk was not equal to the weight of the *Sphere*. They continued to fall at a dangerous rate. The altimeter registered but two thousand feet!

Fully recovered now from his former temporary inertia, Robert jammed over the switch which connected the disk to the powerful storage batteries. This was the reserve that he had not ventured to utilize before. Thus the lift of the *Sphere* should have been increased more than four-fold, and its descent checked at once.

As the switch swung over, the gyrostats stopped completely. In a flash the explanation of it all occurred to Robert. The batteries were exhausted!

The world was rudely shaken from its customary lethargy.

Having lapsed into a monotonous, smooth-running order of events, the public had long since resigned itself to such. Not since the Great War had newspapers had such an opportunity. Even the steady development of trans-Atlantic and trans-continental air traffic had become commonplace.

Of the myriad readers, perhaps none was so keenly interested in the article which appeared on the front page of every paper in the United States on the morning of the eighteenth as Henry Simms.

Since the *Sphere* had disappeared from his astonished gaze the day before, he had anxiously awaited its return. As hour after hour passed, his fears for its little crew of two grew proportionately. He had little faith in the curious invention to which the professor and his companion had entrusted their lives.

So it was with little spirit that Henry sat down to his breakfast that morning at the Palmer homestead, where he lived. He picked up the morning paper listlessly, hoping it might contain some report of the *Sphere*. He feared that if it did contain such news, it would be fatal news. Henry was a pessimist.

The big heading escaped his notice at first because he was looking for some smaller notice regarding the *Sphere* and its failure to return. Then suddenly it caught his eye. Breathlessly he devoured it.

SENSATION CAUSED BY FLYING SPHERE

Curious Metal Blimp Seen Floating Above Lake Vessels' Mast Tops

STRANGE NEW AIR TRIUMPH

Heavier than air machine with no visible means of ascension or propulsion
possesses marvelous speed

(Special Dispatch to the Morning Chronicle)

ERIE, Pa., July 18.—The freighter, “Mary Ann”, arriving here tonight, reported a remarkable incident.

About 4 o’clock this afternoon a member of the crew descried a small speck over the southern horizon. This speck grew in size rapidly until it became apparent that it was not only approaching the “Mary Ann”, but falling with great velocity from its former immense height. It looked to be a large grayish globe.

During the ensuing moments, it seemed as if a huge cannon ball were launched directly at the vessel. Her destruction seemed certain. Consternation seized the crew and officers, who, by this time, were all aware of the pending disaster.

At a critical moment, however, the big ball was seen to slacken in its downward rush, until finally it hung suspended in the air directly above the mast tops, drifting slowly astern.

At this close range several round windows could be seen in the heavily riveted walls of the sphere. A glimpse of the operator was caught as he busily maneuvered divers levers.

Although evidently of considerable weight, and without visible means of support or propulsion, the sphere seemed to float in midair as lightly as a balloon. It appeared to be nearly thirty feet in diameter.

Suddenly a deep humming was heard. A moment later the sphere rose with gathering speed until it appeared to have reached a height of about half a mile. Then it shot abruptly off toward the south at great speed, disappearing rapidly over the horizon.

While this was of great interest to Henry, it but served to increase his uneasiness. He could think of no good reason for the failure of the *Sphere* to return from its trial trip but a fatal reoccurrence of the mechanical trouble suggested in the freighter’s report.

It was at this point in Henry’s gloomy reflections that a hearty laugh outside startled him. The professor!

A moment later Robert and Professor Palmer entered. Both were in fine spirits.

“Should have been along, Henry,” boomed the professor. “Missed the time of your young life.”

“Been reading about it,” Henry replied, tapping the paper. “Were you really trying to drop into the lake, or couldn’t you help it?”

“Fast work, Robert,” laughed the professor, as together they read over the article; “private trial trip in the afternoon—front page headlines next morning! Not so bad, eh?”

“Just missed the freighter,” gasped Robert. “We didn’t have a chance to see her until we had checked our drop and drifted off astern. Phew!”

“Never mind,” soothed the professor. “Can’t be helped now. Anyway, they will probably conclude that we were merely playing with them.”

His mood would not be denied. He seemed more like a boy at that moment than a dignified professor of fifty-seven.

“You folks seem to have had a dull trip,” remarked Henry, ironically. “Where were you last night?”

“Must we tell you? Had you accepted our invitation, you’d know,” retorted the professor. “Man, don’t ask us so many questions. We’re as hungry as wolves.”

They sat down before the appetizing, crisply fried bacon, and eggs that Jarvis, the peerless, smiling butler had brought in.

“It was this way, Henry,” resumed the professor, after he had partly satisfied the inner man: “Robert and I didn’t expect to be gone long, and unfortunately failed to take any provisions along. Had it not been for a cake of chocolate in Robert’s pocket, which we shared, we should have had nothing to eat since we left.”

“But you haven’t told me where you were last night,” persisted Henry.

“Tell him, Robert.”

“Well, after we ran out of power because the storage batteries had not been fully charged, and narrowly missed sinking that freighter, we had just enough current left to suspend the *Sphere* in midair. Then we started the engines driving the dynamos, and soon had sufficient power to start back. But boy! It was a close shave.” Robert paused reminiscently.

“We started back, but changed our minds and decided to see some more of the country first. You see, at five hundred or more miles an hour, it is quite a temptation to look around a bit.”

Henry’s countenance registered a curious combination of astonishment and disbelief.

“Fact,” put in Professor Palmer. “Could have done much better than that, but didn’t want to heat up the *Sphere* uncomfortably by excessive air friction.”

Henry looked very much as if he thought he might be the victim of a little spoofing. Such wild claims, uttered so coolly, confused him and aroused his natural skepticism.

Robert resumed his narrative, with a touch of pardonable pride. Behind him Jarvis stood spellbound, mouth half open, drinking in every word.

“So we flew over to New York, Boston and Baltimore, and looked them over. Great sport. We became so interested that twilight was upon us before we had given it a thought.

“It was pretty dark by the time we got back this way. We forgot, too, that the sun is visible considerably longer from a great height than it is from the earth’s surface.

“The result was that we could not find our way back here in the dark, without lights to guide us. So after a fruitless attempt, we gave up and landed in a large field. There we stayed until dawn, when, upon ascending again, we discovered that we were only a couple of miles from here.”

“Moral: Carry a searchlight, and ye shall find,” contributed the professor.

“And some sandwiches,” added Robert, returning to his interrupted attack upon the bacon and eggs.

The following weeks were crowded ones for the Palmer household. The account of the *Sphere* and the activities at the laboratories were quickly connected by the sharp newspaper world, and acknowledged by Professor Palmer.

A deluge of newspaper reporters followed. The first were a diversion; the rest quickly became a nuisance. Once more did journalistic imagination run wild. Though both Robert and the professor refused to commit themselves on the subject, the Palmer-Margard feud was revived, colored with a wealth of imaginary data concerning prospective trips to Mars in the *Sphere*.

The *Sphere* was photographed and sketched countless times, as were Professor Palmer and Robert. Even Henry came in for a share of publicity.

But the professor had long since determined to attempt the trip to Mars in the *Sphere*. With this in mind he set about mastering the intricacies of its apparatus.

The prospect of venturing into the unknown regions beyond the Earth's attraction is not one that appeals to the faint-hearted. Even Professor Palmer frequently had moments of indecision when he all but decided to drop the project. It would be so easy, reasoned his weaker self, to drop the matter entirely. The *Sphere's* scope on the Earth was sufficient to make them both a vast fortune, and to bring them great fame.

Nevertheless, he remained steadfast in his decision in spite of the advice and warnings of his friends, which were anything but reassuring. He was willing to be a martyr for the possible enlightenment of the world.

It was Robert, though, who strengthened the professor's determination, for he insisted upon accompanying him on the unusual journey.

"I am but an old man, Robert," Professor Palmer argued, "while you are a young man in your prime, with a long, promising career before you. The chances of the *Sphere's* reaching Mars safely and returning, in spite of its remarkable powers, are extremely uncertain. Who knows what strange phenomena it may encounter in the depths of space? Suppose its apparatus should fail midway. Think of the fate that may await us. Even if we reached

Mars, and found it inhabited with intelligent beings, how do we know we should be permitted to return? Take my advice, my boy, and remain here. You may lose the *Sphere*, but you know its principle, and have proved its practicability. You can command the services of the world's best mechanical skill in the rapid construction of another *Sphere*, and still others. In addition, I shall leave you my entire estate and possessions."

Robert was deeply moved by Professor Palmer's concern over him and by his generosity.

"You have been very good to me," he said. "I appreciate it deeply. But I am going with you. We will share the dangers together, and together we will also share the glory of achievement. I believe we are going to succeed."

And so, with these two declarations was sealed the pact of partnership which was to carry them together on the perilous journey.

When their final intention of attempting to reach Mars was announced, the journalistic world fairly seethed with excitement. Every magazine issue contained portraits of Robert and Professor Palmer, accompanied by cuts of the *Sphere* and the professor's latest maps and photographs of the red planet. Never had any human undertaking even mildly approached theirs in magnitude. They were hailed as the heroes of the hour.

It was agreed that the secrets of the *Sphere* were to be set down and placed in a safety deposit box with a certain great trust company, to be opened and read only in case Robert and the professor failed to return after two years' time. Thus, the world could not lose the secret of this remarkable invention.

Professor Margard, at this point, proved that his opposition to Professor Palmer's theories was entirely impersonal. In published interviews, he highly commended his worthy contemporary's courage, as well as that of his companion; but he deplored the dangerous project in the face of what he considered conclusive evidence against the possible existence of inhabitants on Mars. "Misdirected courage; misplaced martyrdom," he termed their intentions.

"Misdirected fiddlesticks," snorted Professor Palmer when he read this. "We'll show these people a thing or two."

Two weeks were devoted to final preparations for the remarkable adventure. A powerful, adjustable searchlight had now been installed within a socket in the bottom of the *Sphere* to facilitate night travel and landings in

the future. Petrol tanks were filled to capacity, and a supply of water taken on, some of which would be used in the cooling coils of the engines. A liberal quantity of life-giving oxygen was forced into the high-pressure tanks. Without this to constantly freshen the air within the *Sphere*, they could not live, as, after passing beyond the Earth's envelope of atmosphere into the void of space, they would have no means of replenishing their air supply. A small supply of nitrogen was also added as a precaution against the total loss of the little ball of atmosphere guarded by the walls of the *Sphere*.

While oxygen had to be replenished as their respiration consumed it, the supply of nitrogen would remain virtually the same except for a slight seepage through the sealed walls when the protecting pressure of the Earth's atmosphere was removed. The atmospheric pressure within the *Sphere* would be about fifteen pounds to the square inch, with the absolute vacuum of space hungrily enveloping the exterior. An apparatus for absorbing the carbonic acid gas thrown off by their lungs was also a part of the *Sphere's* equipment.

Robert tinkered about the *Sphere*, constantly inspecting every part with painstaking care. The resilient rubber window strips, insuring against the loss of the precious atmosphere, were looked to with especial care. The heavy glass panes were examined minutely for possible signs of fracture, or flaws. Such a defect would prove disastrous if it should give way under the pressure within when they were in space. They would then be placed in a vacuum in which no living body can exist. So sudden would such a disaster be that they would have no opportunity, nor means, of saving themselves. All windows, however, were equipped with double panes for safety as well as warmth. They were also fitted outside with guards of heavy wire net.

The lubricating reservoirs of the gyrostats were filled carefully; the bearings were cleaned perfectly. Engines were tuned, and, in short, every bit of mechanism was tested and regulated to a point of perfection.

On the first day of August everything was in readiness for the start of the momentous journey.

Provisions, chiefly of the non-perishable and concentrated variety, had been generously stored in the *Sphere's* food chests. There was a sufficient quantity to last them for months.

Although the world at large understood that the *Sphere* would start on its trip about this time, Robert and the professor had decided to withhold

information as to the exact day or hour of their departure. Neither one desired a public demonstration. In spite of the pleas of divers reporters who besieged them, they refused to divulge the time set for their departure.

As the last day of their stay on Earth approached, Robert was torn by conflicting emotions. At one moment the venture stood forth in all its glory of achievement and adventure; the next, with appalling realization of its vastness, its unknown terrors. From time immemorial, man has instinctively dreaded the unknown, and Robert was plainly afraid. But, though the possibility of backing out did naturally occur to him with devilish persistence, he always rejected it promptly, determinedly. He would not countenance the thought of deserting the professor.

It had finally been decided to start on the following day, the second of the month.

Anxious reporters hovered about the place, each eager to make a “scoop” for his own paper. The more enterprising tried to wheedle some information out of Henry or the taciturn Jasper.

“Now, young mon, ye’ll kindly bate it. I’ve no time to bother with the likes of ye,” the good-natured but sorely bothered Jarvis finally told them, one after another, as they approached him.

Henry, equally annoyed, decided upon cunning.

“I’m not certain,” he was repeating, confidentially, for the third time that day, “but I understand that they plan starting on the sly tomorrow night.”

The young reporter with the brilliant red hair listened with apparently keen interest. He thanked the secretary politely, and departed. But a curious smile on his face as he turned away would not have exactly reassured Henry had he seen it. Evidently the redhead retained some ideas of his own. His sharp, intelligent features did not give him the appearance of one easily fooled by subterfuge.

And indeed he was not. Hugh Taggert had a trait of always trying to out-think the other fellow—and he usually succeeded. Probably it was this that had made him the most valuable man on the *Morning Chronicle*’s staff of reporters.

That the secretary had tried to mislead him Taggert felt certain. But as to when the *Sphere* was scheduled to start, he knew no more than before. However, Henry’s statement had a significance which suggested something to his alert mind. The night start did not seem unlikely, but that a man of the

character he keenly judged Henry Simms to be should readily give his employer's secret plans away, did seem unlikely. He determined not only to redouble his vigilance, but to remain on watch that very night instead of waiting for the next night.

Henry's mistake was in mentioning anything about night at all. His idea, of course, was merely to induce the troublesome reporters to lose a whole night's sleep uselessly.

As a matter of fact, it mattered little to Robert and the professor whether their departure was observed or not. It simply amused them to evade the persistence of their besiegers if they could.

Despite their determination, and the intense interest in their great project, it was with many secret misgivings that Robert and Professor Palmer stood without the improvised hangar on that memorable night. They were about to embark on the strangest journey that man had ever attempted.

Henry Simms alone accompanied them to see them off. Till the last he had tried to persuade them to abandon the dangerous project, but without avail.

To Robert, the stars had never seemed quite so brilliant, the night so bewitching. The very air seemed to have a special tang and sweetness which he had never before noticed. The myriad sounds of the night possessed a magic power of enchantment over him. He caught himself wondering inconsequently whether he should ever again hear the soothing voice of the crickets and other denizens of the summer twilight; whether such sounds might be heard on Mars if they reached it.

Quietly they took leave of Henry and filed into the *Sphere*. The trap slammed shut, and Robert and the professor were enveloped in the dead, black silence of the *Sphere's* interior. It was at this point that Robert's resolution reached its ebb. Had Professor Palmer turned to him at that moment and again begged him to remain safely on Earth, he could not have resisted the temptation.

Never had a glow of light seemed so comforting as that which flooded the *Sphere* a moment later. The temptation of the previous minutes fled. In its place Robert felt only an eagerness to be on his way. Nevertheless, when they had mounted to the main compartment, he opened one of the windows and leaned out, thirstily drinking in deep breaths of the keen night air.

Toward the east a silver tinge on the horizon heralded the rising of the moon. The two tall stacks of the laboratories were silhouetted sharply against the brightening sky. Their black outlines were registered indelibly in Robert's memory for years afterward. It all seemed like a grotesque dream. Somewhere the shrill scream of a screech-owl cut into the night, breaking the spell.

Final farewells were passed with Henry below, and the window slammed to into its soft rubberstripped socket. The air-purifying devices were put into operation.

With Henry's aid they had already removed the *Sphere* from its stall. Its machinery had been carefully inspected that afternoon. With a final glance over everything, they prepared to start at once. For the first time the full electrified lifting power of the disk was to be used. Storage batteries had been charged to capacity.

"All ready, professor?" called Robert.

"Let her go."

There was a soft jar, and the Earth began dropping away. The altimeter registered three thousand feet when Robert opened the second shutter. Immediately the landscape began receding at a disconcerting rate. With a moment of involuntary hesitation, Robert pushed the third button, entirely baring the disk's surface. An answering roar from without indicated the terrific speed at which they were leaving the Earth's surface.

"Twenty thousand," read Professor Palmer.

Almost as he finished speaking the instrument registered another thousand feet. They were rising at virtually the same rate as they had been traveling parallel to the Earth's surface during their original trial trip.

The moon, nearly full, was now in full view because of their height. It had also risen sufficiently to cast long, grotesque shadows of trees and other objects on the Earth's surface. Roads appeared as narrow, winding ribbons; houses as mere faint blots.

A minute later they had reached a level of 62,000 feet. Doltaire's remarkable and recently established airplane record of 46,800 feet was already eclipsed by more than 15,000 feet! The dusky landscape began to take on a blurred appearance. As yet Robert had not turned the current into the disk, fearing excessive air friction. Time enough for that when they had arisen beyond the belt of atmosphere which enveloped the Earth some 200 miles deep. This figure had been approximated from observations of falling meteors, which become white-hot from air friction as they fall with terrific speed from space into the envelope of atmosphere.

"Ah—pardon me, gentlemen," a quiet voice said suddenly.

Robert and the professor wheeled sharply, thoroughly startled.

To their astonishment, they beheld a man walking toward them!

“W-where did you come from?” stammered Robert, the first to recover his speech.

The newcomer, however, did not seem to share their surprise in the least. Rather he appeared to be very much at ease. His brilliant red hair, the easy and pleasant smile on his intelligent features, stamped him as an ordinary, normal person. But how had he come there?

“My apologies, gentlemen,” spoke the stranger. “I determined to cover this trip for *The Chronicle*, and hid in a storeroom. Hugh Taggart’s my name.”

He advanced and shook hands with them both heartily.

“Thought I might as well get acquainted right away,” he ran on, “since we are going to be companions all the way to Mars. Nifty little ship you’ve got.”

Until now their astonishment had kept Robert and the professor speechless. With the disclosure of the identity of the nervy young reporter, however, the humor of the situation struck them both.

“We hadn’t counted on company,” said the professor, “but now that you’re here, I can’t say that I’m sorry. Kind of livens the trip up, eh, Robert? Not so lonesome. But you’ve got your nerve, young man!”

“You’re certainly welcome, so far as I’m concerned, Taggart,” said Robert, agreeably. “Only you might have to share some scanty rations before we land.”

“Shan’t mind that,” was the reply. “Brought some myself to help out.”

He picked up a good-sized cubical package from where he had set it down a few minutes before.

“Bouillon cubes, malted milk tablets, and chocolate,” he explained, tapping the parcel.

“Fair enough,” said Robert.

“Boys,” interrupted the professor, “take a look at old Mother Earth now.”

With one accord they hurried to the windows to gaze upon the receding Earth, which for a minute they had almost forgotten.

The wavering altimeter indicated a height of more than 125,000 feet—almost twenty miles!

The semi-luminous Earth far below them now presented a dull, nebulous appearance, devoid of landmarks, except that far to the southeast a faint

thread of lighter color wound its way irregularly across the country; this they judged to be the Ohio River. One other distinguishable mark was a small, dimly illuminated patch indicating the city near the laboratories.

“Good old Earth, good-bye,” said Taggert.

His customarily cheerful voice contained a note of awe. Indeed, the sight was sufficient to strike awe into anyone’s heart; but then, Taggert was thinking, too, of a certain dark-haired and brown-eyed lass who would be waiting anxiously for him to return to her.

At the end of a fifteen-minute wait, basing his estimate on their former rate of ascension, when the altimeter was still registering accurately, Robert calculated that they had reached a distance of approximately 150 miles above the Earth. At this distance the atmosphere should be sufficiently thinned to eliminate it as a factor of interference with their course or danger of air friction. He could now safely utilize the disk’s full magnetic power. With the resistance of the atmosphere reduced to nothing, their speed was doubtless already increasing, and with the maximum pull of the disk developed by the current from the storage batteries, their velocity would quickly double and redouble until they were rushing through space at a terrific rate. Thus would they continue exactly like a planet until checked by the attraction of some other body or a readjustment of the disk. Just how great a velocity they might obtain they had been unable to determine accurately, but it was considered not improbable that the *Sphere* might reach Mars within a month.

Under Professor Palmer’s guidance Robert now laid their course for Mars, carefully focusing the disk upon it. The full propulsive force of the disk was about to be used for the first time. All three watched tensely through the windows as Robert prepared to throw on the switch that would charge the mythonite with electricity.

An answering jar was felt as contact was made with the first terminal. By degrees, their velocity was increased until the full energy of the powerful batteries was diverted into the disk.

“Why, the old gourd’s shrinking like a toy balloon!” gasped Taggert, watching the Earth intently.

Indeed, the rapidly changing appearance of the Earth was evidence of the remarkable rate at which they were shooting away from it. Gradually the entire continent took shape before their eyes, presenting an appearance

startlingly like the relief maps one sees in every geography. Here and there, however, fields of clouds hid sections of it.

It was at this point that Robert was possessed with a temporary but almost overwhelming impulse to rush the *Sphere* back to the Earth. He suddenly recalled its many comforts and pleasures; its wonderful scenes, sunsets and countless other beauties. All these things seemed a thousandfold more desirable than the cold, cheerless and mysterious void through which they were rushing. It would be such a simple matter to return now while he knew they could; but later—who knew what would be their fate? A moment later the temptation was gone. The possibilities of the curious planet toward which they were bound filled his imagination. He became anxious only to reach it as quickly as possible.

“Ugh,” he shivered, suddenly realizing that their air in the *Sphere* had become chilled.

“B-r-r,” echoed the professor and Taggert.

“Why, it’s down to freezing,” exclaimed Taggert, as he caught sight of the thermometer near him on the inner wall.

“Forgot all about our stoves,” chuckled the professor, turning on one of the two electric heaters with which the chamber had been equipped.

“The cold is one of our greatest dangers,” the professor told Taggert. “Out here in space the cold is absolute. There is nothing to reflect or retain the heat from the sun’s rays. Even if the gyrostats should stop, the disk is powerful enough to keep the *Sphere* from falling back into the Earth, or on any other planet if we lightened it by throwing out excess weight as we neared the planet where gravitation would be much stronger than it is at this distance. We have enough food to last us for weeks. But we must have warmth. Should our current fail us we should be in danger of freezing to death. Fortunately we have a petrol heater for emergencies.”

“Oy, and me with my overcoat at home!” wailed Taggert, in mock consternation, backing up close to the heater.

By this time the Earth had shrunk greatly. No longer did it constitute the greater part of their view. Suddenly a ribbon of fire appeared along its western rim! Steadily it widened, lighting up the *Sphere* brilliantly. Then the explanation of this phenomenon dawned upon them. The *Sphere* was carrying them beyond the Earth’s shadow into view of the sun, whose

pleasant, warm rays shone cheerfully through the windows, buoying up their spirits considerably.

As the time passed the Earth appeared smaller and smaller. Its farther edge, still obscuring a slice of the sun, produced much the same effect as a partial eclipse of the sun by the moon when seen from the Earth. The physical features of the Earth were no longer visible against the glare of the sun. It simply looked like a black disk, slightly larger than the moon.

About this time their self-invited companion seemed a good deal perplexed over something. He stood shifting his weight from one foot to the other with a look of comical mystification on his ruddy countenance.

“What’s the matter, old man?” asked Robert, much amused at Taggert’s curious antics. “Cootie?”

“Something wrong here,” giving a little hop.

“I’ll admit you show symptoms of it, young man,” remarked the professor, dryly.

“Feel kind of lightish. Maybe I’m going to become an angel when we get a little higher,” went on the redhead, still engrossed in his private calculations.

“Come to think about it,” said Robert, standing up, “I feel somewhat that way myself. It must be contagious.” They both looked toward Professor Palmer keenly, as if expecting him to show similar symptoms.

The professor laughed long and heartily, until the pair became convinced that there was something wrong with him, too.

“Well, boys, it’s this way,” he said at last; “the farther away from the Earth we get, the weaker its attraction for us becomes. Of course you feel lighter—you *are* lighter—and that’s not all. Before we reach Mars, we shall all weigh nothing. We’ll be floating around in here like toy balloons.”

“That’s a fact,” said Robert after a moment’s reflection. “But I hadn’t thought of it until now.”

“Well, you fellows can swim around like goldfish if you want, but I’m going to find an anchor,” declared Taggert, looking round for a likely object of promising bulk and solidity.

“No use,” replied Professor Palmer. “When you float, everything else that is loose floats, too.”

Taggert scratched his red head thoughtfully.

“All right, then,” he said finally, in mock despair, “float it is; we’ll all play tag.”

A little while later the sun appeared unobstructed. The Earth had shrunk so small this time that it could no longer be seen on account of its close proximity to the sun. Well beyond it the moon hung serenely, though considerably reduced in size. Seen from this angle it was now nearer half than full. Oddly enough, in all directions the heavens presented the same appearance as when seen from the Earth at night, though the sun shone brightly upon the *Sphere*.

“But why?” Taggert wanted to know, looking in perplexity toward the blazing sun and then at the stars twinkling in cold, brilliant splendor.

“On the Earth we were enveloped by a layer of bluish atmosphere many miles deep in which minute particles of dust are suspended,” explained Professor Palmer. “When the rays of the sun shine through this, it produces the luminous, azure sky with which we are so familiar. It is this brilliancy in the Earth’s atmosphere during the day that makes the stars invisible. Out here, with no envelope of atmosphere or dust particles, there is nothing to produce a luminousness to outshine the stars.”

“Guess it’s all okeh,” mused Taggert, doubtfully, gazing out into the black sky, which lacked even the softening indigo of our terrestrial nights.

The stowaway proved a welcome recruit. For instead of dividing the twenty-four hours into two watches, they could now have three, of eight hours each.

The trip had settled down into dull monotony. One condition, however, partly relieved the tedium. This was the ever decreasing weight of their bodies. The adventurers found walking a novel sensation. A giddy feeling possessed them, and there was an unsteadiness in their gait which was difficult to control, resulting in a comical semblance of semi-intoxication.

It was the more reckless and experimental reporter who discovered and demonstrated proudly that he could step the entire length of the chamber, with little effort. Robert and the professor quickly and easily duplicated his feat, but he continued blithely to remind them at intervals of his initial discovery. From that time he supplemented his experiments by jumping up and touching the ceiling, and other gymnastics, proclaiming each noisily to the amusement of his new companions. He seemed to be enjoying himself

immensely and to have entirely forgotten his original idea regarding an anchor.

But even this soon ceased to be a diversion and the three of them finally settled down as best they could, to look over the various latest editions of the newspapers which Professor Palmer had brought along. These all contained articles about their venture, and furnished quite interesting sidelights to the daring adventurers themselves.

“Here’s a cheerful fellow who has figured out that it will take us two years to cover the thirty-odd million miles to the Martian deserts,” announced Taggert from a precarious perch on the back of a chair with his feet on the seat. Ever since his initial gymnastics he had evinced a preference for birdlike attitudes.

“That’s nothing; here’s one that makes it five years,” contributed Robert cheerfully. “What does your paper say, professor?”

“The lowest estimate has it seven months. We, who know more of the *Sphere*’s powers than any of them, had figured on about a month; but at the rate we are going now, and faster every second, we ought to reduce our own estimate by half.”

Taggert heaved a sigh of undisguised relief.

“Gosh, professor, that was close. I began to have bright visions of yours truly alongside a harp.”

“No telling what you’ll find yourself alongside when we pull into Mars,” remarked Robert encouragingly.

“I’d rather be by a plate of ‘ham and’ right now than anything else,” answered the scribe. “You gents made me miss my nightly feed.” He felt in his coat pockets and presently fished out a cake of chocolate.

“Why in bedlam didn’t you say so sooner?” admonished Robert, getting up and making his way wobblingly toward a locker. “You might not believe it, but we’ve got a regular restaurant here. I can fill your order right now.”

“Haven’t got a chicken run on board, too?” bantered the reporter.

“Young man, while you and your brethren were busily writing why we would never reach Mars, we were preparing to do it in the right way,” broke in the professor.

“We not only have a substantial supply of fresh eggs put up in silicate of soda for preservation, but cheese, ham, coffee and a number of other good things that you might not have suspected.”

“And you’re going to turn loose a hungry stowaway scribe on all that?” asked Taggert.

“Certainly,” chorused Robert and the professor.

“Do you think we are going to let you starve?” added Robert. “You know we’ve got no undertaker handy.”

“Oh, thanks, thanks!”

A spirit stove was pulled out from a niche in the wall, and presently a generous slice of ham and a couple of eggs were sizzling in the frying pan. An appetizing aroma filled the chamber, causing Taggert to sniff the air hungrily.

“I call this handsome, now,” he commented, gratefully. “I always was a lucky stiff, though, just let me know when I can save your lives or something and I’ll be there.”

This simple repast was supplemented by a round of quickly brewed bouillon.

With their stomachs satisfied a feeling of drowsiness came over them all. Taggert stoutly insisted upon standing first watch, but Robert was adamant in his refusal. He explained that it was important that he maintain watch over the machinery for the first shift until the most likely period for development of mechanical trouble was passed.

Professor Palmer also offered to take the first watch, but owing to Robert’s greater familiarity with the mechanism he allowed himself to be prevailed upon. First, however, he carefully inspected the heavens, correcting the *Sphere*’s course by various constellations, as it had swung a few points away from its objective.

The sun glared in at the windows at the back end of the chamber. The blinds were drawn, darkening the interior to facilitate sleeping. Professor Palmer and Taggert spread the pallets of bedding obtained out of one of the storerooms, and settled themselves to rest.

Through the long hours Robert maintained his lonely vigil.

The machinery continued its musical purr uninterrupted. Once he started the dynamo for a while, causing the temporary opening of a sleepy eye or two. He wished to keep the batteries charged to fullest capacity until they were well on their way, after which their velocity through space could be maintained with a very little expenditure of current.

The prolonged excitement of the past weeks, particularly of the last few days, together with loss of sleep, proved too much for Robert. Several times he caught himself dozing. Lulled by the hum of the machinery, he finally slipped off into oblivion.

Grotesque and confused dreams followed one after the other through his uneasy slumber in seemingly endless fantasy, causing him to mutter incoherently. These finally gave way to a curious vision of a conjured Martian landscape.

Huge cacti and other polypetalous growths formed a dense, forbidding background. As he looked about him it seemed that they had formed a menacing circle round him, which appeared to grow smaller and smaller. Hideous dark growths pushed their thorny leaves up through the loose sand round the edge of the circle, writhing into distorted shapes.

Desperately but fruitlessly his eyes sought some escape from the shrinking circle. The dark wall presented an impregnable barrier. How he had come there he did not know.

Suddenly he was startled by a rustling of the stiff foliage. The agitation of its tops heralded the approach of some being. He momentarily expected to see some dreadful *thing* leap out from the forbidding jungle—just what, he knew not!

Then, to his intense relief and astonishment, a girl of rare, exotic beauty emerged. Her eyes were like the cool depths of a shaded brook, her really golden-hued hair a delight, the perfection of her soft-clad figure goddesslike. Yet she repelled rather than attracted.

Then, indeed, it seemed as if the doors of paradise had opened. Gone was the aloofness of the moment before. She was smiling—at him.

But even as he took a first eager step toward her and she toward him, a mist seemed to come between them. The amazing loveliness of her faded into the drab desert background. He was alone!

Strange to observe, the threatening jungle was no more. Before his bewildered gaze a trackless desert swept from horizon to horizon. Then this, too, faded.

How long he had slept he did not know, but he was awakened suddenly by a blow in the face!

In a flash he was wide awake. His hands groped out in front of him, coming into contact with a smooth, metallic surface. He seemed to be lying on the floor, and immediately formed the conclusion that he had fallen off his chair while sleeping.

As he endeavored to rise to his feet the floor began to recede from him slowly! It was then that he remembered the steadily decreasing attraction of the Earth as the *Sphere* shot farther and farther away into space. He quickly concluded that the “floating” stage had been reached. The *Sphere* would be maintaining its established momentum just like a planet which hurtles through space century after century, impelled by its original momentum by reason of the complete absence of any obstruction in space to hinder it; and of course their bodies followed serenely with the *Sphere* in its interior. They were as a component part of it—little worlds of their own.

As the metal surface continued to recede from him, he suddenly discovered that the engines and dynamo were missing! Yet the steady purr of the gyrostats was plainly, reassuringly audible. Then abruptly the solution of it all dawned upon him. He had fallen on to the *ceiling*—not the floor!

Suddenly a realization of his danger confronted him. He was drifting slowly toward the gyrostats! Should he be caught in their racing mechanism his body would be whipped into shreds!

Desperately he strove to jerk his body over into a convenient position to assist him in grasping one of the four perpendicular rods surrounding the gyrostats. With nothing to brace himself against, his efforts were strangely akin to those of a cat falling through the air, though, for lack of practise, they were not nearly so adept. Luckily they sufficed to turn his body over facing the gyrostats. Fortunately too, one of the uprights was within reach. He clutched this as a drowning man clutches a tossed rope, and the danger was over.

He lowered himself breathlessly to the floor. For the first time he noticed that he was perspiring freely.

“Close shave, that,” he muttered, mopping his face nervously. “Have to rig a guard around this.”

He looked sharply toward where the professor and the reporter had lain. Strangely enough they were still in the same spot. Then he became aware that there was still a slight pull toward the floor. The Earth had not yet entirely released its hold upon the *Sphere* even though it had long since ceased to be visible to them. Evidently he had made some abrupt move in his sleep with sufficient force to send him slowly upward to the ceiling against the dwindling force of terrestrial gravity.

The following days were interesting ones for the adventurers, but inconvenient—though amusing.

With the passage of each day the Earth’s attraction for them became weaker until finally it was completely neutralized by the counter attraction of Mars.

This point was reached by the ninth day out, when, according to Professor Palmer’s reckonings, the *Sphere* had traversed more than half the distance between the two worlds.

Ordinary movement about the compartment became an impossibility. Walking was an accomplishment of the past. In order to move from one end of the chamber to the other it became necessary merely to place a hand against the wall and push. Immediately they were propelled across the room through space as if suspended by a well-oiled trolley conveyor. The chief requisite was a careful sense of direction and control of strength exerted. Otherwise they were apt to find themselves precipitated roughly into one corner, against the ceiling, or headlong into the whirring machinery.

The reckless Taggert was frequently the cause of much merriment, because of his careless or awkward antics. Once he brought their hearts into their mouths by narrowly missing the smashing of a window pane when his shoulder was brought up sharply against the glass. After that even he exercised extreme caution in his movements.

It became necessary to lash each other to some stationary object for protection so that they could sleep safely. When on watch, Robert was obliged to keep hold upon some stable part to maintain any single position for a time.

The managing of fluids was at once ludicrous and exceedingly difficult. The mere task of drinking a cup of coffee called for more skill than the biting of an apple in a tub of water at a Halloween party. One was apt to have more of the beverage applied externally than internally.

A cup of fluid could be kept safely intact only by a centrifugal motion, as by whirling it round in a circle, bottom outward. Otherwise, at the critical moment the contents would drift off in an irregular, pulsating sphere, like a soft little world of its own held together merely by the slight affinity of its molecules.

A scheme of drinking through a tube from a covered bucket by means of a hole drilled through the cover proved fairly successful and became temporarily the vogue.

During the first of these days the *Sphere* evinced a tendency to revolve slowly and at random because the delicate but sensitive stabilizer could no longer find a central point of gravity. Here, then, was a grave danger confronting them; for with the growing inclination of the *Sphere* to revolve at random it became apparent that they would not be able to hold it to any one course. This because, as the *Sphere* revolved, it would be necessary to constantly shift the direction of the disk; and thus it would be almost impossible to continue constantly with accuracy. Consequently they were facing the possibility of drifting about in space through eternity!

Their predicament might have been likened in a way to that of a sailing vessel caught in the doldrums; but in the *Sphere's* case there appeared to be no chance of relief. Curiously enough, this problem had not occurred to Robert and Professor Palmer in their preparations for the journey. In fact, to their minds, there seemed no possible solution.

Even Taggert's hitherto unfailing good spirits deserted him as the three gloomily faced the dreadful prospect of slow death through starvation or suffocation. The fact that they were millions of miles from the Earth in the midst of a great black void did not make their fate any easier to consider.

Professor Palmer now divided his time between frequent corrections in the *Sphere's* wavering course toward Mars, and the writing of the log.

"Who knows," he remarked resignedly to his companions, "some other more successful adventurers may attempt this venture some time. There is just a chance in a billion that they may find the *Sphere* and this document; or the *Sphere* may finally gravitate back to the earth."

Almost constantly he watched and guided the *Sphere* on its intricate course, insisting upon doing the major part of this difficult task, which only his expert knowledge of the heavens made possible to such a degree of accuracy. But even he was beginning to find it almost impossible to keep the *Sphere* on its true course, as it continued to swing more and more widely from its former stability. Loss of sleep and the terrific strain were beginning to tell upon his iron constitution. It was clear to all three that theirs was a losing struggle. The professor's faint hope that they could decrease their distance from Mars sufficiently to establish a substantial stability upon its attraction faded more and more as gradually the little *Sphere* began to swoop in ever-increasing deviations from its course.

Realizing with sinking heart the hopelessness of the situation, Professor Palmer endeavored to conceal the sureness of their fate from Robert and Taggart. But they sensed it intuitively and each bravely sought to steel himself against the realization of the end.

The thought of conserving their energy by stopping the gyrostats occurred to Robert. For with gravitation virtually equalized from all directions, their operation was scarcely of any assistance at this point.

And then came the thought that caused his heart to halt an instant in its beat. Could it be that both the professor and he overlooked the one possible solution? Was it too late?

"Fool—fool!" he expostulated bitterly as he realized the opportunity that had all but slipped away from them by his failure to think of the solution sooner.

"Robert!" cried Professor Palmer, fearing for his mind. "What is it?"

"Can you hold her to the course steadily for a minute?" Robert almost shouted in his excitement, not having heard the professor's question.

Professor Palmer suddenly realized with a flash of hope that Robert had thought of a possible way out of their predicament.

"Yes, yes," he answered eagerly. His waning strength and alertness rallied temporarily under the inspiration of hope. His tired eyes became as keen as ever as he carefully nursed the drunkenly rolling *Sphere* back to its course and managed with supreme skill to steady it there for several seconds consecutively.

"How long, boy?" he cried hoarsely in desperation, gripping his voice as he realized that he could not balance the *Sphere* accurately on its course

more than a few seconds longer. He felt his control slipping. Too bad—too bad. The boy had had the idea, and he had failed—failed. He felt suddenly broken, as a very old man. His gray head nodded wearily. Too bad!

“Professor—professor,” someone was whispering huskily. He recovered from his lapse of semi-consciousness as he felt a hand placed nervously upon his shoulder. He turned to see Robert’s eager young face behind him, smiling! That might mean—but he hesitated even to hope, stifling its ray of comfort almost before it came to him. He waited dully for Robert to go on.

“We are holding our course now,” went on Robert, controlling his voice with an effort. “See?” he pointed to the glittering heavens visible through the observation windows.

Instead of the dizzily swerving canopy of lights with which they were already too familiar, the stars hung stationary.

“How did you do it, Robert?” gasped the professor. As he spoke he was suddenly aware that the gyrostats had stopped!

“You see, it came to me like a flash,” explained Robert, “that it all hinged on velocity. If the disk was suddenly shut off—covered—the *Sphere* would at once cease to be pulled around in various directions. Instead it would then rush ahead only in the direction in which it was last moving when the disk’s power was shut off. Beyond the forces of gravitation and with nothing but void on all sides, we would shoot forward forever until stopped by nearing some planet.”

“Of course, of course,” murmured the professor. “Why didn’t I think of that before? Dunce that I am!”

“That is why I asked you to try to hold the *Sphere* in its course for a little bit—long enough to maintain its momentum toward Mars, when I would stop the wavering interference of the disk. First, I stopped the gyrostats. Then, as I clicked the shutters to cover the disk’s face, the *Sphere* became simply a dead weight already launched with terrific velocity toward our goal. With the influence of the gyrostats removed, the heaviest or bottom side of the *Sphere* became the head of our velocity. Result—the eccentric revolutions of the *Sphere* ceased at once. We have established a temporary stability of our own—velocity.”

“Robert,” said the professor, after a pause, “we owe our lives to your keenness of mind. You thought of what it was my business to have known in the first place. With the *Sphere*’s course automatically maintained now, it

remains but to wait until we are near enough Mars to establish stability based on *its* attraction. Then we can again control the *Sphere* at will. In the meantime we conserve all our power.”

“Just as simple as A-B-C,” broke in Taggert, who had been roused from his sleep by their excited talking, and had been listening interestedly for some time, unobserved.

“But,” interjected Robert, struck by a fresh disquieting thought, “if we are now rushing directly toward Mars, won’t we be left far behind by the time we reach its present location because of its rapid movement along its orbit?”

“Oh, what’s a few million miles or so to us?” said Taggert with extravagant nonchalance.

“Your deductions are partly correct, Robert,” answered Professor Palmer, smiling at their guest’s sally. “There is a curious thing about moving bodies in a void: they will continue in *one* direction indefinitely until attracted or propelled, by some other force. In the case of a propelling force, unless it is in a directly opposite direction, it will simply result in the body going off at a tangent, still maintaining its original rate of velocity in the original direction in conjunction with its new direction. For example: Mars and the Earth moving in virtually parallel directions in their respective orbits at present, it was a comparatively simple matter to lay a straight course for Mars, as the Earth’s orbit velocity was imparted to the *Sphere* when we left it. But since the Earth moves somewhat faster along its orbit than Mars, we would gradually forge *ahead* of Mars if we had laid what at first appeared to be a direct course toward it, and would only have arrived finally by continually correcting our course, and having swung round in a vast curve. Instead, by calculating the difference in the known orbit velocities of the two planets, and accordingly laying a course which at first appeared to be toward a point already passed by Mars, we promptly found ourselves on very nearly a direct course toward the planet.”

Notwithstanding his comparative ignorance of astronomy, Taggert unconsciously echoed Robert’s sigh of relief over this assurance that they were on the correct course. It was clear, even to him, that with no basis of stability they would be in a bad way should they pass Mars at a distance too great to establish gravitational contact with it. With their limited reserve of power and provisions they could not afford to knock around the universe at random.

Thee thirteenth “day” saw them nearing Mars rapidly. The ruddy-hued planet gleamed at them with magnificent brilliance, its cold glitter thrilling them and filling them with vague misgivings.

At this distance it became apparent that Professor Palmer’s feverishly corrected course would carry the *Sphere* at least several thousand miles ahead of the planet. This variation was to be expected, however; the miracle of it was that he had been able to judge direction so closely in those few nerve-racking and never-to-be-forgotten moments in the reeling *Sphere*.

Fearing to disturb the steady course and velocity of the *Sphere*, and content so long as they were still nearing the planet, they made no effort to utilize the disk’s power yet. Professor Palmer estimated that they were approaching Mars at a rate of 100,000 miles an hour. Already they were within some 3,000,000 miles of it, compared with the original distance of roughly 35,000,000 miles.

Brighter and brighter shone the mysterious planet with the passage of each watch. With the rapid reduction of the intervening distance, a faint restoration of gravitation began to be felt. Even in this extremely weak form it proved a very welcome relief to the weary Spherites. No longer did they float about like gas-filled balloons, though walking was still a difficult feat.

The beginning of the fourteenth “day” found them within a few hundred thousand miles of Mars. The *Sphere*’s course had now mechanically corrected itself as anticipated by the professor, and they were “falling” directly toward the planet. All eyes were now kept eagerly trained upon it through the observation well. Its spreading disk almost entirely filled the glass-enclosed tube. The professor watched it with boyish excitement, as feature after feature developed with their swift approach.

Indeed, the planet presented an awe-inspiring and wonderful sight.

Its great, snow-white polar caps, and vast, rose, rich ocher and purplish-bronze expanses between these caps surpassed in grandeur anything their eyes had ever beheld. Crossing and recrossing these expanses or plains was a curious network of straight and some slightly curved, dark-hued lines. All

of these lines appeared to originate at the edges of the polar caps, or to connect indirectly with them by means of juncture with other lines. The northern or smaller cap was surrounded by a border of deep, bluish tint. This cap had shrunk noticeably even during their recent observation. Professor Palmer attributed this phenomenon to the advanced summer in the northern hemisphere, accompanied by a melting of that polar cap and an accumulation of water around it from which the “canals” were fed.

At numerous junctures dark spots occurred. The whole presented much the appearance of a crude map, upon a globe, of some gigantic railroad system, the dots representing the terminals or large cities. Several of the lines were double, running parallel with startling regularity and joining together again at the terminals. The northern extremities of the “canals” were now plainly darker than elsewhere, and strengthened the professor’s well-known theories regarding the purpose of the “canals.”

Another outstanding feature was certain large, bluish-green blotches interrupting the general rose-and-ocher hue of the plains. Some of the lines ended in these blotches. Wherever this occurred there was a caret-shaped junction, the line connecting directly at the point of the caret. At this point the color was deepest; from there it faded, gradually changing in color, the blotch blending into the ocher plain.

All of these features stood out in increasing vividness as the strange planet drew nearer, proving that the Martian atmosphere contained little or no moisture in the form of clouds. With the further expansion of the surface beneath them, new and fainter lines were discovered. The great disk had by this time grown so large that its outer edges could be viewed readily from the sloping ports.

The professor compared painstakingly the actual features of the planet before him with his own maps of it, the result of years of faithful observation and study at the lonely California observatory, that an ever-skeptical world might be further enlightened. The sight of him poring intently over his maps and notes, oblivious for the while of all else, was not without a certain pathos. At last, he stood at the threshold of vindication.

“Gosh,” ejaculated the redhead, breaking a long silence with startling abruptness, “I hope we don’t drop into that snow on the north pole there.”

“We can land at any point we prefer, old man,” Robert assured him.

“That happens to be the south pole, my friends,” said the professor, referring to the large polar cap over the edge of which the *Sphere* then hung. “See, the planet has been turning this way—toward what we shall call the east, which makes this pole the south.”

“Give me the desert,” replied Taggert, indulging in the luxury of a shiver.

“From the looks of things,” said Robert, “we shall have all the desert we want. That ocher shade seems entirely too popular. It makes me thirsty to look at it.”

The gyrostats had been started again for safety as they drew near the planet, and the *Sphere*’s comet-like velocity retarded by cautiously focusing the disk in the opposite direction temporarily.

A deflection of the disk swung the *Sphere* away from the pole and nearer the Martian equator. With the possibility of landing within a few hours, a keen watch was kept for the most promising region in which to land.

“If our belief that Mars is inhabited be correct,” said Robert, “it may be well to avoid the population centers. The Martians may fear that we are seeking to do them harm with some machine of destruction, and destroy us.”

“That’s a good suggestion,” exclaimed Taggert. “It would be just like one of those dudes to take a crack at us with some kind of a howitzer for luck.”

“Let us first fly over the surface at a safe height and examine it carefully, then select a landing site,” suggested Professor Palmer. “If it seems to be inhabited we had best land outside some smaller village where we will have an opportunity to interview a few of the natives without so much danger of being overpowered if they prove antagonistic.”

Sluggishly, the vast map slid westward before their gaze, in panoramic review as the planet rotated in its axis. Thus its entire surface from pole to pole was gradually presented to their view as they continued to descend at a considerably reduced speed.

The fifteenth “day” found them within about 25,000 miles of the planet’s surface. At this elevation Professor Palmer commenced a sharp lookout for the two moons which were known to revolve about Mars at great speed. These moons had been glimpsed on rare occasions by a few of the earth’s astronomers through powerful telescopes, but only when conditions for observation had been ideal. Professor Palmer had seen them more than

once, although they were each approximately but ten miles in diameter, as he explained to Robert and Taggert.

“Ten miles?” repeated Taggert, doubtfully. “Why—dammit, these blamed Martian moons aren’t much more than balls of mud! Say, I’d rather have our moon than half a dozen like those. Ours is more than 2,000 miles in diameter, isn’t it?”

“You must remember that the Earth’s moon is about 240,000 miles away from it, while Mars’ nearest moon is only about 3,700 miles above its surface.”

“How high is the other one?”

“Approximately 13,000 miles; but it is doubtful whether it would be visible to the naked eye from Mars—certainly not plainly. The one nearer Mars, however, should present an interesting spectacle from that planet. It requires eleven hours to cross Mars’ heavens, by reason of the moon’s own swift revolution and the slower rotation of Mars, going through all its phases and half again during that time.”

“It’s odd that it doesn’t fall into the planet from that height,” said Robert.

“Rapid revolution in its orbit round Mars produces sufficient centrifugal force to balance the planet’s attraction,” said Professor Palmer. “The same principle is involved as in the swinging of a pail of water over one’s head without spilling it. Yet, some day one of these moons may be drawn into the planet. Should that happen, the blow may be great enough to change the planet’s own orbit.”

“How about landing on one of these moons?” said Robert.

“Ha! Novel thought, that,” exploded Taggert. “Say, if we’d play catchers on it, we would be apt to run right off.”

“I don’t think catchers would be a very good game to play on these moons,” chuckled Professor Palmer. “You see, a person wouldn’t weigh much more than a feather on one of them, and if he took a real good leap on the under side he might find himself on his way to Mars. Not only that; there is no atmosphere on them, and you could not exist an instant in the vacuum and intense cold prevailing there, even if you were equipped with respiration tanks.”

“That’s enough for me,” Taggert decided with a grimace.

“How cold do you think it is out in space like that?” asked Robert.

“Absolute zero—the nadir of temperature—is said to be 459 degrees below zero Fahrenheit.”

“Enough, enough,” shouted Taggert. “I said I was satisfied without getting out on a moon.”

“There’s Deimos over there now,” cried the professor suddenly, pointing.

“Don’t know the lady,” commented the reporter. Nevertheless he looked interestedly, and a trifle uneasily in the direction indicated.

“Deimos, or ‘Fear,’ is the moon farthest from Mars; Phobus, meaning ‘Panic,’ is the other,” the professor explained.

Off toward the northwest, just beyond the outer rim of the planet, a small star was suspended. It appeared much like any ordinary star in spite of its small bulk, for it was much closer than the vastly larger stars which we are accustomed to see.

“Now we must keep clear of that one, and watch closely for the other,” cautioned Professor Palmer.

At an elevation of about 3,000 miles nothing of the other moon had been seen, and it was concluded that it was then on the opposite side of the planet. But they were soon to find out their mistake—one that nearly proved fatal.

It happened while all three were intent upon the nearing planet, which presented a sight of such interest that even Professor Palmer had temporarily forgotten his own recent warning.

Their sole warning was a sudden cessation of the sun’s rays. Darkness reigned, except for the reflection from the Martian disk below. The abruptness of it made them gasp in unreasoning terror.

Above and around them the blackness of space was unbroken. The sun seemed simply to have gone out completely. Only the vivid sight of Mars reassured them.

“The moon!” yelled Taggert, first to think of it.

Hard on the heels of the realization of what had caused the sudden darkness came to Robert and Professor Palmer the knowledge that the satellite must be dangerously close to them. Otherwise, its small diameter would not have so completely hidden the sun from them. Passing between them and the sun, there was no longer danger of its crashing into them, but it was not entirely improbable that it might attract the *Sphere* toward it, with disastrous results.

Almost the next instant they were plunged again into the joyous rays of the sun. All of this had happened in a few seconds' time, but to their startled minds it had seemed much longer. Looking up furtively they had glimpsed, for just a moment, a slice of blackness slipping from the edge of the sun. The miniature satellite had passed from their vision into the sightless black background of space. But, no; there *was* something there!

The reflection from Mars produced a barely perceptible glow upon the jagged face of the big ball above them. It reminded Robert exactly of the faint, broken reflection of lantern light on the ceiling of a large chamber in a cave which he had once explored. There was something threatening and terrifying about the rapidly dissolving apparition.

"I've seen enough of that baby," sighed Taggert. "Oy, let's go down on Mars and meet some nice, fresh cannibals before we hook up with another one of those things."

"As a matter of fact, my esteemed friend, we are still scurrying toward yon cannibalistic region to the tune of some thousand miles an hour," volunteered Robert, adopting Taggert's vernacular.

The curious lines and dots now stood out vividly on the ocher-tinted background. So obviously of artificial origin did they appear that neither Taggert nor Robert entertained any further doubts as to the planet's being inhabited. Soon they were all three engrossed in a discussion concerning the probable appearance of these remarkable people, who were engaged in a desperate struggle for life against the waning of their world.

"I think they are a race of giants, because no people of ordinary strength could succeed in constructing such a vast system of canals," Taggert suggested.

"I've heard it claimed that the Martians are probably a race of smaller stature than ourselves, in proportion to the size of their planet," said Robert.

"On the other hand, I think it quite probable," said the professor, "that these people are much like ourselves, in size as well as formation. That they possess extraordinary intelligence I feel certain, for they are a far older race than our own, and their wits have doubtless been sharpened by their ceaseless combat with nature."

"Then you believe that their unusual skill, and not physical strength, has enabled them to achieve these wonderful feats!"

“Not entirely. The force of gravity at the surface of Mars is approximately but three-eighths of that on the Earth. The result is that a being of our strength on Mars would be capable of about seven times as much work as on Earth. For example, he could dig a trench on Mars seven times as long as he could one of the same size on the Earth in the same length of time. The weight of the soil, of his arms and body, would be much less than on Earth, and the ratio of his strength over the reduced task would be augmented greatly.

“Then again, haunted, century after century, by the specter of extinction, the Martians would doubtless invent all manner of marvelous contrivances for the accomplishment of their gigantic tasks. Necessity has probably goaded them to a frenzy of invention and research.”

“Don’t you think it more likely that the ‘canals’ are really giant underground ducts! I should think long, open waterways would allow most of the water to evaporate before it had reached the ends,” said Taggert.

“I do. However, the popular conception of the word ‘canal,’ is that of an open waterway, whereas any duct, passage or groove, may properly be termed a canal.”

“But such canals or tubes would not be visible from the Earth, surely,” reasoned Taggert.

“Certainly not; but the vegetation which they fed *would* be. That is evidently why, with the melting of the polar caps, the lines have been observed to deepen in color with the season. I believe this indicates the quickening of the growths in the strips of irrigated ground along the ‘canals,’ the rate of progression observed being about fifty-one miles a day.”

“How do you account for that? Gravity could scarcely be responsible, for there would be no more reason to expect all water to run from the poles to the equator on Mars than to expect the same thing to happen on Earth.”

“Exactly. The Martians must have some unusual method of pumping the water through these canals.”

Robert, who had often discussed these details with Professor Palmer during the past months, was more interested right then in the probable characteristics of animal life on the planet, and particularly the people. Were they brown-skinned or white, hairy or smooth, with features like our own, or different? He had not forgotten his curious vision of the Martian desert,

nor the spell cast over him by the maiden in his dream. *She* had been neither brown nor ugly. He wished earnestly that the Martians would be like her. These were the thoughts that ran through his mind as he watched the vast fairyland developing beneath him.

By this time the planet's disk had spread until it formed a low horizon on all sides. Although they were less than a thousand miles distant, the planet's surface still appeared to be quite level. No indication of mountain ranges could they detect from this height. At an elevation of about three hundred miles Robert checked the velocity of the *Sphere* further for fear of excessive atmospheric friction. For they should soon be entering the enter edge of the planet's envelope of gases.

The poles were no longer visible, being lost beyond the rising horizon, as the *Sphere* was now almost directly over the equator. The planet continued to revolve before their eyes. So long as the *Sphere* remained beyond the envelope of atmosphere, they were independent of Mars' rotation.

"Better follow the spin of the planet now, Robert," said Professor Palmer.

A deflection of the disk soon gave them the desired easterly drift. Gradually, the vast panorama beneath them came to rest.

As they continued to descend, a slight glow began to replace the blackness of space. This indicated the presence of some atmosphere round them. At this point they observed the first sign of vapor upon Mars. A solitary patch of opalescence partly covered one edge of a large, dull-green blotch a little to the northeast of them.

Here and there a certain roughness about the surface of the planet seemed to represent slightly hilly regions, but such places were scarce. The planet's predominating characteristic appeared to be a monotonous flatness.

The "canals" had now become broad, bluish-green bands, terminating in large circular areas of a similar shade. Robert could almost imagine he saw tree-tops.

Little was said as the *Sphere* approached the planet's surface. Each was thrilled with his own imagination and excitement over the immediate prospect of viewing, at close range, the mysterious planet which had so long baffled the experts of the world. Robert retained perfect, skilful control of the *Sphere*, aided by advice from Professor Palmer as the latter studied the distant Martian landscape intently. Taggart busied himself making notes.

The sky had now taken on a normal glow like that of the Earth's, and for the first time they felt that they were finally and definitely within the boundaries of the Martian world. The character in detail of the country below was now faintly visible. The earlier suggestion of some moderately hilly regions was emphasised by the setting sun, a low, but ragged, ridge appearing off toward the northwest. It was plain that if they wished to land by daylight there was little time to lose.

There was no longer any doubt as to the artificial character of the planet. One of the great "oases" extended toward the northeast to the now restricted horizon. From this, and almost directly beneath them, ran a broad belt of mottled green, continuing toward the southwest till lost to view.

The altimeter now registered 60,000 feet. However, they were, no doubt, much closer to the surface than this, as the atmospheric pressure on Mars was certain to be considerably less than that on Earth; in fact, their height did not appear half that great.

"Not much choice about a landing point," commented Taggert. "This country looks more like the Sahara every minute."

Indeed, there was an uninviting monotony about the landscape. With exception of the dark belt, and the oasis on the horizon, the entire country seemed one vast desert. The *Sphere* drifted slowly, as Robert had now checked its descent almost entirely.

"I think we had best land about five miles beyond the canal belt," said Professor Palmer. "We ought to be safe there from any sudden attack, should the inhabitants prove hostile. More likely we will there be visited by a cautious few where both parties will have an opportunity to look each other over carefully before making any overtures."

This seeming a sound piece of advice, the *Sphere* was steered several miles to the north of the belt. Here Robert allowed it to settle slowly.

The sun was sinking into the horizon even as the *Sphere* came to rest in the loose, yellow sand of the desert, about two hundred miles south of the equator. The trip to Mars was a reality!

“Hurrah we’re here!” shouted Taggert, executing a hand-spring, and narrowly missing the incandescents in the ceiling because of his unaccustomed light weight.

Robert and Professor Palmer accepted their triumph more quietly.

To Robert the remarkable trip already seemed as a nightmare. As he looked out on the quiet, desolate scene in the deepening twilight, he could scarcely realize that they were not still on the Earth. For despite the desolation of the vast Martian desert stretching before his eyes, and the fact that he had never before even seen a desert except in pictures, his imagination balked when he tried to believe himself on a strange planet, millions of miles from the Earth. The idea was preposterous, absurd! Robert’s more deliberate self persisted in half suspecting that they had simply miscalculated, and had actually returned to the Earth at some remote spot.

“Boys, I’m going to try a whiff of our new atmosphere,” said the professor, unlatching one of the small portholes.

Before either of the others could interfere, he had swung the heavy glass slightly inward, and sniffed the Martian atmosphere speculatively.

Whatever fears they held were quickly dispelled by the look of relief which came over Professor Palmer’s countenance as he swung the port wide open and eagerly inhaled the outside atmosphere.

“It’s all right,” he cried. “The regular stuff! Come on and enjoy some fresh, Martian desert air.”

Robert and Taggert did not need to be urged. The air within the *Sphere* seemed suddenly to have grown unbearably stale. With one accord they opened the other ports and filled their lungs with the sweet, cold air outside.

“We’ll have to close up again to keep warm tonight,” shivered Taggert.

“Thirty-four degrees above zero,” read Robert from the Fahrenheit thermometer outside.

“The air on these deserts cools very rapidly after twilight,” said the professor. “Even in the Sahara, on our own globe, the temperature

frequently drops below freezing at night. However, the temperature in the canal belts should be more uniform.”

“Let’s get outside and look around a bit,” suggested Robert.

“Good idea,” echoed the reporter. “I’d like to see what some of these oafs look like.”

“Not likely to be any of them strolling around in the desert at night,” said Robert.

“Can’t tell; I’ll bet they didn’t fail to see the *Sphere* when we slid over here. We’re likely to have an army down on us tonight.”

“I wouldn’t worry about that,” said Professor Palmer. “The Martians are obviously a people of much intelligence. I expect they will act just about as our own people would, should some curious machine land upon the Earth. They will probably wait till daylight, then come out and satisfy their curiosity.”

“And ours,” added Robert, remembering his dream.

As they stepped out upon the sand, buttoned into warm coats, a brilliant spectacle was presented to their gaze.

Low in the southeast Phobus hung like a glowing orange. Its now apparently smooth, bright disk was a decided contrast to the dark, threatening, cavernous face which had frightened them but a few hours earlier. All round them the indigo sky was studded with stars of the great brilliance that is reserved for travelers of the deserts. Behind them the comfortable flood of illumination from the *Sphere* spread its friendly radiance over the sand.

“Look over there!” cried Taggert suddenly, pointing toward the east.

Far away on the horizon’s edge a diffused white glow shone steadily.

“A city,” guessed Robert quickly.

“A group of cities—a Martian oasis,” suggested the professor.

“Why, there are some lights along the canal, too,” said Robert, becoming aware of a number of lights stretching along the endless strip of fertile land to the south of them.

“Well, I’ll be damned!” cried Taggert. “Say, let’s run over to town tonight!”

But however sanguine the professor might have been regarding the existence of an intelligent race upon the planet, he balked at a precipitate

invasion of their haunts right then. So it was decided to wait till morning for developments.

The lantern which Robert carried was almost superfluous in the bright starlight. They enjoyed the novelty of trudging about through the sand, after their extended confinement within the *Sphere*. In spite of the looseness and depth of the sand, they walked over it with amazing ease because of the decreased gravity on the smaller planet. Robert, for example, who weighed 150 pounds on earth, now weighed less than sixty pounds. Yet he retained his full strength, so that the task of walking was tremendously reduced.

“Run you a race, Robert,” called Taggert, starting out abruptly at a great pace.

The temptation was too great. Robert was a good runner and reveled in the sport. He dashed after the reporter.

His feet scarcely seemed to touch the sand as he raced after the fleeting shadow ahead of him. With giant strides, twenty feet long, he steadily reduced the distance between them.

Suddenly there came a dim shout ahead, followed by a dull thud—then, silence.

Robert slowed up as quickly as possible and looked round him. The reporter had disappeared!

Far in the rear the *Sphere* shone brightly, like a beacon. Between it and himself he could see the professor’s lantern bobbing up and down as he strode along.

As he continued bewilderedly to search the sands for some sign of Taggert, his eyes became better accustomed to the semi-darkness.

Suddenly he descried a long dark shape lying in the sand several rods away.

He approached it cautiously, only to discover what seemed to be a large log. But as he looked up another dark object ahead caught his eye. Surely that looked like the figure of a man sprawled upon the sand. Even as he looked, it moved and struggled to a sitting posture.

“Hello—that you, Taggert, old fellow?” he sang out, approaching.

“It’s me all right,” came Taggert’s voice, weakly.

“What happened to you?”

“Fell over that dashed boulder back there. About knocked the wind out of me. I must have been going about forty miles an hour,” he explained, getting to his feet with Robert’s assistance.

“What are you two up to?” cried Professor Palmer, coming up with them.

“Our stowaway just tried to break his neck over a log back there.”

“A log?” incredulously from the now recovered reporter. “Say, this is a desert, not a jungle! That was a rock I fell over.”

They walked over to the object of their discussion, and examined it in the rays of the lantern.

“A petrified log,” pronounced Professor Palmer.

“Well, who brought it out here?”—belligerently.

“It grew here many centuries ago, my boy. This is a relic of a dead forest, of which we are probably on the edge. See, there are others scattered about over that way. I have seen the same thing out in Arizona.”

Their discussion ended, they decided to go back to the *Sphere* and get a good night’s rest.

“Suppose some of these oafs have taken possession during our absence,” suggested Taggert, persisting still in so calling the as yet unseen Martians.

“If it hadn’t been for you young scamps it wouldn’t have been left unguarded,” retorted Professor Palmer.

But they found the *Sphere* as they had left it, and no one in sight.

With lights out for greater safety, they spent a quiet night. All three were up again with the dawn.

The warm sunshine streamed in at the windows cheerfully. Soon the thermometer on the shady side registered forty-one degrees and was rising rapidly. It had dropped to twenty-five the night before when they retired.

An appetizing breakfast was prepared by Taggert, who had insisted upon being the official cook. The keen Martian air and a good night’s rest had brought them all ravenous appetites, and they did the simple repast full justice.

“Come to think about it,” mused Taggert, “the night passed mighty quickly. Professor, how long are the nights and days on Mars?”

“The night seemed to pass quickly because you slept soundly. It happens that a Martian night and day together consume just about twenty-four hours and forty minutes, our time. In other words, by an odd coincidence, Martian

days and nights are each approximately but twenty minutes longer than those of the Earth.”

The professor’s last words were interrupted by Robert’s abruptly rising to his feet and pointing mutely out the window!

Moving swiftly toward them, about a quarter of a mile away, was a large conveyance which appeared to be occupied by about ten beings.

“Sink me, but these birds are certainly early risers,” grumbled Taggart. “Seems to me they might let us finish breakfast before calling. They’re no gentlemen, I say.”

Professor Palmer was eagerly studying them through binoculars.

“Just as I suspected,” he murmured presently. “They have features just like our own, and seem to be of nearly the same stature as ourselves. Let’s get ready to welcome them, boys. They don’t look like pirates.”

He put down the glasses and turned just in time to see Taggart concealing an automatic on his person. Evidently the reporter didn’t entirely share the professor’s faith in the Martians.

A few minutes later the conveyance drew up without. It came to a stop noiselessly, as though it were electrically propelled. Several of the dark-garbed occupants got out and walked toward the *Sphere*, removing their odd hats, which looked not unlike broad-brimmed tropical helmets.

Professor Palmer’s prediction proved correct. They were of ordinary human formation, and, indeed, looked much like a group of intelligent foreigners, with their olive skin and dark hair, though they were somewhat shorter in stature, averaging about five feet four inches in height.

Their clothing was simple, and evidently designed for comfort. All wore roomy, dark trousers, bound at the ankles, and small coats to match, not unlike vests with sleeves. Under these a lighter-hued blouse was worn. Neat but styleless shoes, with uppers that appeared to be finely woven grasses, clad their small feet. There was a total absence of bright color about their apparel, neutral shades of blue, gray and maroon predominating.

Professor Palmer opened one of the ports and leaned out. His appearance caused a brief flurry among their visitor, proving a signal for some hurried conversation, accompanied by excited gestures.

Then one of them walked forward and addressed him in a pleasant, soft, rolling tongue not unlike Spanish, but entirely unintelligible to the

professor.

Professor Palmer answered by pointing obliquely upward in the general direction of the Earth at that time. Then he tapped the *Sphere* significantly, and indicated an imaginary course from the Earth above back down to Mars with a slow sweep of his hand.

His audience seemed to grasp the fact that the *Sphere* had come from a distant planet. In fact, it appeared that he had but confirmed some previous guess on their part. They nodded knowingly to each other as their spokesman resumed his own gesticulations in an effort to communicate with the professor, smiling in a friendly fashion and rubbing his Roman nose with a trace of self-consciousness as he proceeded to invite them, with elaborate gestures, to visit their country beyond.

His companions also pointed repeatedly toward the foliage in the distance, and to vacant seats in their vehicle.

“What do you think, boys?” asked the professor, turning to Robert and Taggert. “How about boosting the *Sphere* over there? That will be better than leaving it out here where we can’t keep an eye on it.”

“Suits me,” Robert replied.

“That’s the stuff,” said Taggert. “Don’t let these dagos separate you from your return trip if you can help it.”

So with more gestures Professor Palmer explained their intentions to the Martians, who finally understood apparently and seemed satisfied.

By careful manipulation of the speed of the gyrostats and the disk shutters, Robert raised the *Sphere* slowly to a height of about fifty feet. The Martians looked on in wonder from their conveyance, which, getting under way, preceded them across the floor of the desert.

The broad, flat wheels of this conveyance, notched to give greater traction, carried it over the sand at a good clip. Steering seemed to be controlled by an automatic dial-and-lever device, operated easily by the driver with one hand. The usual staggering of the front wheels through loose soil seemed entirely eliminated.

Contrary to their original impression, the floor of the desert in this direction was a stony, windswept waste with much bare rock visible. The faces of these rocks were polished by grains of sand blown across them by the winds of centuries. Here and there was one with fractured and crumbling surface, probably cracked by the rapid, alternate heating and

chilling of the blazing rays of the sun and the cold nights, and not yet healed.

As the *Sphere* drew near the fertile land, they observed that it was densely wooded with trees of varied height and foliage. From their close proximity to the ground it looked like a vast, boundless forest which might extend many miles beyond.

Professor Palmer had estimated the usual width of these irrigated strips at from one to several miles, though he had mentioned observing one of nearly twenty miles in width.

At the forest's edge Robert brought the *Sphere* to rest

Here they were in a quandary as to what to do about the *Sphere*. It was obvious that they could not study the life of the planet without leaving the *Sphere*. Yet they were naturally reluctant to trust it unguarded into the hands of these strange inhabitants.

"But we have already risked far more in our journey through space," reasoned Robert. "We put ourselves in these people's hands by coming here; now I suppose we may as well trust them. We could not expect a Martian, coming to the Earth as we have come to Mars, to drag his '*Sphere*' after him everywhere he went."

"Look here," Taggart interjected, "I horned my way into this expedition uninvited. Now let me do something useful. I'll stay here with the *Sphere* until you can make some satisfactory arrangement for its safe keeping."

"That may take several days, or more," objected Robert.

"Well, what of it? Plenty to eat here. Just run along and leave it to little Hughie. I'll stay on deck until you return or send me a written order; and I'll feel a lot better about having done my bit."

So it was finally agreed that Taggart remain on guard while Robert and the Professor went on to make suitable arrangements, if possible, with the authorities. They shook hands with the reporter and left him calmly smoking an atrocious pipe which he had not ventured to put in action during their long trip with sealed windows. He seemed genuinely comfortable and well pleased with his lot. Robert and the professor took seats in the waiting conveyance, which carried them over a winding road through the forest.

Professor Palmer recognized and pointed out to Robert certain varieties of trees and shrubbery resembling the tamarisk, acacia and eucalyptus, prickly pear and aspen poplar. The latter variety, which was singularly like

the earthly specimen, predominated, and seemed to flourish luxuriantly in the loose, sandy soil.

“Not so bad,” said Robert, sniffing the sweet, clean air.

“A very healthful climate, indeed,” commented the professor.

Their evident satisfaction was observed with smiles and nods by their hosts, who were watching them closely.

They had probably passed through a mile of the great forest when they emerged into a large, rectangular clearing.

“Why, there’s a railroad!” cried Robert excitedly, pointing to what looked very like a double line of tracks running through the center of the long clearing.

A moment later Robert’s statement was proved to be true, for their conveyance was brought to a stop beside the rails, where a small but well built wooden structure apparently served as a crude station.

At the invitation of the Martians they got out, declining, however, the suggestion of entering the waiting room of the building. They preferred to examine first the wonders of their strange outdoor surroundings. The Martians gathered about and proceeded to study them with poorly concealed curiosity. Yet at all times their attitude toward the adventurers was solicitous and courteous. To Robert they seemed more and more like a delegation of learned experts sent to observe their every move and thought as closely as possible.

About the clearing stood a number of plain buildings of goodly size, with numerous windows. Several Martians in rough garb, including the loose trousers and blouse, but without the odd coats or hats worn by the first Martians, busied themselves about these buildings. In the distance there were sounds of chopping, and an occasional resounding thud, as from a tree felled. They seemed to be in a lumber camp, and this conjecture was later confirmed. They were standing then in the heart of one of the planet’s greatest forest regions.

Presently Robert became aware of a humming sound. Looking about quickly, he failed at first to see from whence it came. Then he discovered for the first time the great distance that it was possible to see in either direction along the railroad, because of its striking straightness. Mile after mile it ran straight as an arrow through a veritable tunnel of trees.

In the distance he descried a swiftly moving speck on the madder-colored ribbon of roadbed. It grew rapidly larger, evidently nearing them at a tremendous rate. A minute later a Martian train, drawn by a squat engine, ground to a stop before the station. Like the conveyance which had met them, it seemed electrically driven. Only one man was visible in the engine cab. Robert counted eight coaches, each about sixty feet long.

Their guides conducted them to the back coach, which they entered at one end. This coach differed considerably from the rest, for while the others were built with compartments similar to English coaches, this one was not unlike one of our own observation cars. It was unoccupied. They found out afterward that their guides composed a special committee which had arranged for this car in their guests' honor, the *Sphere's* approach having been discovered and observed closely by Martian astronomers as it neared the planet. Every effort and provision had been made to find and meet its expected inmates promptly, and with every consideration for their comfort.

They were scarcely aboard before the train was moving. Without a jar the luxurious coach slipped away from the little station, gathering momentum rapidly. A minute later the station dwindled and was lost from view behind them down the shrinking avenue of trees which whirled past them dizzily.

"Just like greased lightning," said Robert.

It was evident that the owl-eyed committee of Martians were eager to establish some code of communication by means of signs with their guests, but observing their desire to study the changing landscape they politely refrained. One of the Martians, however, evinced considerable curiosity over Robert's watch chain, whereupon Robert displayed his watch. Not to be outdone, the Martian pulled out a small, flat mechanism about an inch and a half square. A glance at this object convinced Robert that he was looking at a Martian watch.

The indicating scheme was essentially the same as our own method. There was a small dial in the center, with sixteen curious numerals round its edge. The hands, of the same length, but of contrasting colors, apparently indicated the time, while in the upper corners were changeable numerals, probably showing what corresponded to the Martian month and day of the month. The lower corners were not utilized, but were simply decorated with some artistic scrolls. A third, but shorter hand, also connected to the central dial, revolving rapidly with a familiar ticking sound, probably corresponded

to the second-hand on our own watches, but it was more like the long hand on a split-second stop-watch.

The case of Robert's watch seemed to interest the Martians no less than its mechanism. The watch was an old one of the bulky type which had belonged to Robert's father. The heavy case was of solid silver. From their exclamations, as they examined the silver case, Robert judged that they prized silver highly as a precious metal.

By this time the character of the country had changed. No longer were they surrounded by the fragrant forest. Vast, level, green fields stretched on either side, while in the distance the equally flat desert was visible at times.

The speed of the train began diminishing until it came to a stop beside another small station. Here they got their first view of one of the great canals or ducts upon which the plant life and the lives of every being on Mars depended.

About a hundred yards beyond the station a gang of two hundred or more men were at work in and around a deep excavation, aided by several huge digging machines. All were clad in rough garments of a dull maroon color, which Robert soon learned was as common a color on Mars as our own khaki is to us.

But it was not the Martians which attracted Robert's and Professor Palmer's chief attention. A giant cylinder lay partly exposed within the excavation, its ends disappearing into the soil at either end. In diameter it was at least seventy-five feet. It appeared to be of cementlike construction.

At intervals of perhaps fifty yards along its length, smaller tubes branched off and were lost in the sides of the ditch. Where one of these branches joined the main line, a swarm of workmen struggled valiantly to mend a break from which the water gushed as if under heavy pressure.

The scurrying Martians had checked the leak perceptibly by the time the train started again after a minute's wait.

A few minutes later they flashed by an immense structure situated near the tracks, looking like a great power plant of some kind. However, it reared no stacks to the sky. For miles at either end of it stretched a vast, flat expanse of some curious construction, which in the distance looked like a great, elongated checker-board.

"A Martian pumping station," hazarded Robert.

“I believe you’ve struck it,” said Professor Palmer. “And I believe the big ‘checker-board’ is nothing more nor less than a device for absorbing power from the sun’s rays. That alone would explain the Martians’ remarkable achievements in the face of the unquestionable dwindling, perhaps complete exhaustion, of the planet’s fuel supply.”

“How old do you suppose Mars is!” asked Robert.

“That’s a hard one; but the various planets are classified according to stages. There is the sun stage, in which the planet is hot enough to emit light. This is followed by the molten stage—hot, but lightless. Then comes the solidifying stage, with the formation of solid surfaces and ocean basins. The next stage is what we call the terraqueous stage, the age of sedimentary rocks. Our Earth is in this stage. Following this is the terrestrial stage, in which the oceans have disappeared, and, after that the dead stage, when air has departed. Mars is in the terrestrial stage, the stage following that of our planet, and preceding the dead, or final stage.”

“Then the Martians are engaged in a constant struggle against extinction!”

“They are, though, with their marvelous ingenuity, they may last a few thousand years longer. But we are witnesses of the waning of a world.”

They were now passing through a fertile farming region. Small buildings dotted the landscape, while here and there Martian farmers were diligently at work in their fields. There was a complete absence, however, of any beast of burden or toil. Everywhere power seemed to be furnished electrically. Farmers could be seen plowing with large, powerful tractors.

Frequently they caught glimpses of the checker-board devices adjoining the buildings, but on a much smaller scale than the one they had first seen. Quite likely, they decided, these were individual sun-power plants.

With the increasing frequency of houses, Robert and the professor became convinced that they were nearing a center of population. Their interest was keyed to the highest pitch as the agricultural district gave way to the outskirts of a Martian city.

The more Robert saw, the greater was his surprise at the striking similarity of things to those on Earth. Yet, on second thought, this did not seem so strange. After all, it was to be expected that the chief inventions of two such advanced worlds should, in the main, be similar. He might easily have imagined himself approaching some foreign metropolis.

One thing that they noticed particularly was the absence of the dirt and squalor which one so frequently sees from the train when approaching our large industrial centers. Buildings all seemed of substantial construction and everywhere were in excellent condition.

Now, beyond the buildings in the foreground they could see the tops of giant structures in the distance, their great white walls showing resplendent in the clean atmosphere and bright sunlight. It was certain that they were entering a very large city. They afterward learned that this city, called Parang, was next to the largest one on Mars, having a population more than four million souls.

Little conversation passed between Robert and the professor as they both eagerly studied the mysteries of Martian life passing before their eyes. Neither could yet quite bring himself to a full realization of the fact that they were actually viewing life upon the planet which had caused so much conjecture upon Earth.

“I wonder whether we shall pull in on the ‘elevated’ or the ‘subway,’” grinned Robert.

His question was answered a few minutes later as they plunged into the darkness of a tunnel. For ten minutes or more they roared swiftly through the darkness, dimly illuminated by the incandescents in the ceiling of their car. Then they emerged suddenly into a vast, brilliantly lighted underground chamber, filled with the din of noisy bustle, and came to a stop before a large barred area.

Here their hosts politely signified that they should get out.

News of their arrival had probably preceded them. A great, curious crowd thronged the area. Half of their body of guides led the way through one of the gates, forcing a passage for them through the dense, curious crowd. It was here that Robert caught his first sight of the Martian women. Unconsciously he was expecting to find the golden-haired type of his dream. But in this he was disappointed, for, without exception, they were dark-haired, and with complexions of a pale, olive tint, as were the men.

The women’s garments did not differ greatly from those of their men folk, in that they wore a variety of loose trousers like those worn by our Turkish women. Perhaps their garb is best described by stating that it was strikingly similar to that, of the women in the Mohammedan harems, without the face veil.

Passing through a massive archway, they walked up a few steps leading into a large waiting room not unlike one of ours. Through the throng they continued and entered a roomy elevator which whisked them upward. Emerging from this they found themselves once more in the daylight, and in a great vestibule leading to the street.

Outside, several closed conveyances waited, evidently engaged by prearrangement. Robert and Professor Palmer entered one of these with two of their beaming hosts, sinking down into the luxurious upholstering gratefully.

As they rolled along, Robert and the professor studied the scenes around them in suppressed excitement.

The first feature which struck Robert was the immense height of the buildings, practically all of which were of ten stories or more. Buildings of thirty stories were common, while several they passed were more than sixty stories. The rather narrow streets seemed like miniature canyons between the tall structures. Without exception the buildings appeared to be constructed of a fine cement, similar to the metallic product of which the railroad rails were made.

The street paving and walks were also made of the same material, and were in excellent condition. No poles or wires were visible; nor were any street-cars or tracks seen. Evidently with such excellently maintained pavings made possible by the weak gravity, the Martians preferred ordinary vehicles for city transportation.

Stores of all sorts seemed plentiful, though Robert could not see what kind they were, because they did not display their wares as we do. The Martians apparently had outlived the practise of wasting window space in this manner, preferring merely to advertise their wares by signs. These signs, of course, were wholly unintelligible to Robert and the professor. The lettering was made up of a number of geometrical figures, among which the familiar triangle, square and "T," with variations, predominated.

Once, when their conveyance was temporarily stalled in a traffic congestion, Robert got out a pencil and attempted to reproduce the characters which appeared in relief over the doorway of an imposing-looking building near by:

△□△▽ △△△△

No suggestion of a curve softened any of the characters, each one being made up of a series of straight lines and angles. In fact, Robert already had observed that this severe precision was a marked characteristic of all things Martian. It was particularly noticeable in the architecture. The impression borne upon him was that this people had reduced everything to a science of fine mathematics.

A ride of some twenty minutes brought them into a quieter section of the city. Here the streets were somewhat wider. The tall buildings gave way to more modest structures, which appeared to be dwellings not unlike our apartment buildings.

No sign of any growing thing was visible anywhere—not even a blade of grass. Evidently the painful scarcity of water upon the planet did not permit of floral culture for merely ornamental purposes. Theirs was a serious business of scientific economy.

In due time they turned into a broad driveway leading to an immense, official building of four stories. Here Robert saw the first bit of ground uncovered by cement or stone since they had arrived in the city. The grounds surrounding the building, and its drives, were covered with verdure similar to a fine clover, closely cut. Some trees grew about the place, but there were no flowers.

Their conductors were visibly excited as they drew up before a broad flight of stairs leading up to the main entrance of the massive building. A pompous person opened the door of their conveyance, and escorted them ceremoniously up the steps beyond which a small group of dignified-appearing Martians awaited their arrival.

As they drew nigh the group, a commanding figure of a man detached himself from it and advanced toward them. At this their escort fell back respectfully, leaving them to greet this evidently important personage.

In stature this man was several inches taller than the average Martian, being about the size of a well-built man of our own race. Imperious, resplendent in rich garments, he easily dominated the smart assemblage.

A smile played over his virile features as he stopped before them. Strangely enough at such a moment, his odd, square, and particularly luxuriant beard reminded Robert of a nonsensical little rime which he recalled, something like this:

Aha, it is as I feared,
Two cocks and a hen,
One owl and a wren,
Have all made their home in my beard.

With an air of kingly elegance, this leader addressed them for several minutes in his strange tongue as in a welcome. Though it was mutually apparent that neither understood the other's words, Robert responded briefly, thanking him for his evident welcome, Professor Palmer nodding his concurrence.

Indicating that they were to accompany him, the leader escorted them through the assembled, obsequious gathering of men and women, through the broad entrance into the building. Passing along a beautiful corridor of carved onyx, they presently entered a large, pleasant room with windows looking out upon the grounds.

Two attendants came forward at their entrance. These the leader addressed in a commanding tone, indicating Robert and the professor in a manner which convinced them that these men were to serve them during their stay at the establishment; and this they found to be the case. With a few more friendly words their host politely withdrew.

Robert and Professor Palmer found that they had been presented with an entire suite of rooms, the one into which they had been escorted serving as a sort of drawing room. The two attendants looked after their every want.

About noon a simple lunch was served them. It consisted of a dish resembling baked yams, some artichokes (at any rate they looked and tasted like artichokes), a bit of greens, and some very good wine.

That afternoon they were visited by an intelligent-looking Martian who quickly succeeded in making them understand that he was sent to act as instructor in the Martian tongue and also as an interpreter while they were learning. He signified that the emperor (the striking personage to whom they owed all this hospitality) had sent him.

The same evening they accepted the emperor's invitation to dinner.

To their relief, Robert and the professor found that the dinner was to be attended by few besides themselves. The emperor sat between them at the head of the table. The interpreter sat next to Robert. But three other persons partook of the meal with them. These they understood to be certain learned men, probably astronomers.

The food seemed to be made up chiefly of well-cooked vegetables, several of which were quite similar to our own. A savory bit of roasted meat was also served them. This dish puzzled them. It did not seem to be of a fowl, though it certainly had the flavor of one, tasting much like duck. Not until some time later did they find out that this meat was that of an animal identical with our dog! It was, however, considered a delicacy by the Martians, who raised these animals with great care, they being quite scarce.

Though conversation was as yet very difficult, they managed to exchange a surprising amount of information with the Martians, the interpreter proving quite resourceful. The emperor, especially, was intensely interested in them.

So keen was the interest of the Martians, that, after the repast was cleared away, paper of excellent texture was produced, and Robert and the professor were urged to do some sketching. Their hosts clearly were eager

to overcome the barrier of languages and to partly satisfy their curiosity at once regarding our planet. A sort of stylograph was handed to Professor Palmer, and with it he proceeded first to make a simple diagram of the universe, showing the Earth and Mars thereon in their orbits round the sun, and indicating their journey from the Earth to Mars. This was followed by prompt nods of understanding and ejaculations on the part of the three Martian astronomers, as if their expectations had been confirmed. The professor then drew maps of the Earth's continents, rivers and seas, and made sectional drawings of mountain ridges, volcanoes and ocean beds. The Martians' intelligent minds seemed to grasp everything with remarkable perception.

The three technologists apparently would have been content to keep Robert and Professor Palmer up all night in their zeal for information, but the emperor had more consideration for his guests, insisting finally upon seeing them back to their comfortable quarters, where he took leave of them for the evening.

That night they rested in the welcome softness of luxurious beds. The day's travel and excitement had fatigued them, and they slept soundly in spite of the strangeness of their surroundings.

Robert dreamed again of his maid of the desert. As before, she faded suddenly away, even as he reached out eagerly to assure himself that she was real. One vivid impression that he received and retained upon waking was that she was in peril. The look of entreaty in her eyes, the repeated startled glances that she cast over her shoulder, convinced him that she needed him.

But, why should he worry over a mere dream! His practical mind reasserted itself. Of course the girl did not exist in reality. Still, there was no denying that the girl of his first vision and of the last one were one and the same. He could never forget a single detail of her exotic beauty and charm. For the first time in his life Robert felt the awakening of real love.

He laughed. In love with a dreamgirl! Nonsense. Nevertheless, his thoughts returned to her continually throughout the day. Unconsciously he found himself hoping to see her, somewhere, somehow. And more and more, in spite of his commonplace reasoning, he came to feel that there was such a girl.

That day they saw nothing of the emperor. The interpreter labored faithfully with them part of the morning and again during the afternoon.

Already they felt that they were making some genuine progress toward an understanding of the Martian tongue.

Their every comfort continued to be administered to. As they sat before the windows looking out upon the lawn bathed in the late afternoon sunlight, it required an effort indeed for them to fully realize that they were gazing upon a strange planet millions of miles from the Earth.

“What do you think we had best do about the *Sphere*, Robert?” asked Professor Palmer.

“The very thing that has been puzzling me. These people seem to be such an intellectual race that I can hardly believe them dangerous, though there is a certain elusive suggestion about the emperor’s face that I don’t fancy. However, I think the *Sphere* would be just as safe or safer here, and with Taggart staying with us.”

“My idea exactly. Now that you mention it I think there is something in what you say about the emperor, but we’ll have to make the best of things. Let’s try to find out through our interpreter where it will be agreeable to keep the *Sphere* hereabouts and then send a message to Taggart to bring it.”

So it was agreed. Explanation to the interpreter proved a comparatively simple matter. A few sketches and gestures and he signified his understanding. A short while later he returned to inform them that the emperor had assigned them a structure near the palace in which they could keep the *Sphere*. Further, that the emperor would be delighted, not only to view the *Sphere*, but to welcome their companion.

A note to Taggart was dispatched at once by a courier supplied by the emperor. The courier was instructed to accompany Taggart back in the *Sphere*, guiding him to the palace. As Taggart had thoroughly mastered the operation of the *Sphere* under Robert’s tutelage, during the watches of their long journey through space, they felt no uneasiness about his ability to fly it to where they were.

This matter settled, Robert and the professor sat down before a sumptuous dinner served in their suite. They were becoming accustomed to the well-cooked Martian food, and they relished it.

The confinement in the building since the middle of the previous day, however, was becoming irksome. The interpreter had not encouraged any suggestion about going out, and they had politely refrained from pressing

the matter. Left to themselves after dinner, they decided to take a stroll outdoors.

Robert had observed a secluded nook behind the palace. The windows of their bedchamber opened on this enclosure, which was entirely surrounded by a wall about twelve feet high. Apparently the wall was without any gate. A number of small trees marked the smooth lawn within, casting long shadows in the gathering twilight.

It was in this nook that Robert proposed they take a stroll.

As they reached the exit leading out upon the enclosure, one of their attendants appeared rather abruptly at a door to their left. He paused there a moment as if about to speak, then disappeared as suddenly as he had appeared.

“So—we are under surveillance, eh?” remarked Professor Palmer.

“I wonder. He did look at us rather suspiciously.”

“Well, let’s go ahead and see whether they stop us.”

But they were not molested as they stepped out upon the lawn.

The cool dry air refreshed and invigorated them. It was free from the dampness of the dew which we are accustomed to feeling on our clear terrestrial nights, while the rarity of the Martian atmosphere was probably made up by a greater percentage of oxygen, as evidenced by the ease and enjoyment with which they breathed it. This they later affirmed. Tubercular diseases upon Mars were virtually unknown. In fact, the Martians were afflicted with little sickness of any kind.

For perhaps fifteen minutes they walked about, smoking cigars from their slender store which they had carried from the *Sphere* in their pockets. The fleeting Martian twilight was replaced by darkness—that is, if one can call night filled with the soft radiance of millions of stars “darkness.” Over the edge of the wall at the western end of the garden hung Phobus, one-half its disk lit in a dull orange glow. It appeared about one-quarter the size of our moon.

A strange feeling of oppression, which he could not understand, possessed Robert. Try as he would he could not shake it off. Then he realized for the first time the intense silence which pervaded the night. There was a total absence of the countless sounds of nature which we are so accustomed to associating with summer nights. Apparently there were no insects upon Mars, or, if there were, they were voiceless. Their own voices

startled them when they broke the stillness, and unconsciously they took to speaking in hushed whispers.

A few feet to the right of their doorway the palace wall ran out into the garden at a right angle for about fifty feet, thence it again turned off at a right angle to the right. The garden ran around this extension and back into a recess on the other side like the lower part of a large letter L.

Suddenly a woman's stifled scream tore the silence apart. It ceased abruptly, suggestive of the clapping of a hand over the mouth of the one who had screamed. The sound seemed to come from the recess at the far side of the garden.

In a dozen lithe bounds Robert had rounded the wing of the palace, and was in sight of the far end of the garden. He fancied that he caught a fleeting glimpse of some light, loose garment in the dark shadows of the recess. A faint sound as of the stealthy turning of a lock followed and all was again quiet.

A moment later Professor Palmer joined him.

"What's up?" he asked huskily, puffing from his exertion of attempting to keep pace with Robert.

"Don't know. Imagined I saw a woman's garment fluttering a moment ago. Suppose we have a look over there in that black corner."

A solitary dark window looked out sinisterly from the recess. Robert had an uncomfortable feeling that they were being watched from it as they approached.

As they drew close to the wall where it joined the building Robert caught sight of something they had failed to notice before in the darkness. This was a massive, closed door in the corner of the wall.

"The door I heard locked a minute ago," whispered Robert.

He put his shoulder to it, but failed to move it.

The night again was as silent as a grave. In vain they listened for some sound beyond the locked door till finally they gave it up and returned to the comfortable warmth of their suite.

That night Robert tossed about nervously until he despaired of sleeping. He envied the professor, whose measured breathing he could hear from the adjoining room. Later he dropped off into troubled slumber and dreamed once more of the maid of the desert.

This time the scene was a different one, the palace garden the setting instead of the desert. Step by step the incident of that evening was enacted again with this difference: in some strange manner he could see all that occurred in a garden which was on the other side of the wall with the locked door.

A maid—his maid—emerged from a door in the palace and hurried across the garden. Reaching the wall she fumbled among a mass of vines which clung to its side. She pulled something from behind them which he could not at first make out in the shadow. As she propped it against the wall he recognized it as a small ladder.

This she mounted quickly, looking back several times as if fearing pursuit. Just as she reached the top of the wall a man ran out of the doorway from which she had first appeared, and looked around swiftly. He was immediately joined by another. Both caught sight of the girl as she paused to drop on the other side of the wall.

Held by some invisible force Robert found it impossible to go to her assistance. He was obliged to remain merely a spectator.

The girl's pursuers dashed across the garden and scrambled over the wall after her. She tripped over her long gown and fell. Before she could recover, one of them was upon her.

Together they lifted her struggling form and carried it back toward the massive door in the wall. There they halted while one of them fitted a key to the lock. A moment later it swung open. Just as they were taking her through the door, she screamed, and one of her captors clapped his hand over her mouth roughly. Then the door shut softly, but the sound of a heavy lock shot home reached Robert.

With the shutting of the door Robert suddenly was released from his trance. With a mighty bound he made for the wall—only to find himself standing awake in the middle of the floor of his bedchamber!

The vividness of his dream had left him trembling with excitement. He felt convinced that he had just visioned a review of the actual events of earlier in the evening. Prompted by the impulse of the moment, and realizing the impossibility of further sleep that night, Robert donned his clothes and quietly passed out into the garden.

He shivered as the chilled atmosphere struck him, and turned up his coat collar. The glory of Phobus no longer lit the crystal-clear sky, but in the soft

light from the brilliant stars he could make out the wall running into the recess.

He found the stout door as securely locked as before.

Remembering his increased agility on Mars, Robert decided not to be restricted by a mere door while his reckless mood lasted. A jump and a clutching of the cornice quickly put him astride the wall with no discomfort save a bumped knee. A drop on the other side and he found himself in the garden of his dream.

“Now that I’ve arrived, what next!” he mused.

Looking round the enclosure he observed that it corresponded exactly in size and shape to the one on their side of the wall. All windows were dark. There was nothing to suggest the disturbance of the early evening. The ladder—if indeed, there had been a ladder there—was gone. But the mass of vines on the wall corresponded exactly to that in his dream!

“I suppose if I were a real hero I would dash in and rescue the distressed maiden in some way or other,” Robert muttered, scratching his head in perplexity.

As if in answer to his quandary a window above scraped lightly. A folded piece of paper fell at his feet. He looked up just in time to see a graceful, ivory-white hand being withdrawn. Was it the draperies or her garments that he saw behind the pane as the window was lowered gently!

The paper was crammed into his pocket and, after a swift glance around, he hurriedly scaled the wall, realizing the uselessness and folly of attempting there to read by the light of a match a note written in a still unfamiliar language.

“Boys, howdy,” Taggert greeted them.

The *Sphere* had been safely brought and placed in the building provided. Official greetings and curiosity had been taken care of and the three adventurers were again united and alone.

“Pretty nearly scared that little Martian stiff,” Taggert went on, “not to mention myself. We hopped off a little bit too strong and before you could say ‘uncle’ we were almost lost in the old ozone. Guess he thought I’d decided to kidnap him and take him home. He jabbered something scandalous. But I soon got things straightened out and we beat it here P. D. Q.”

The days following were spent chiefly in the learning of the Martian language by all three. Robert was particularly anxious to master the Martian code of writing, sufficiently to enable him to decipher the note which had been thrown to him in the garden. Not trusting the interpreter, he could not request him to read it to him.

He selected characters and words from the note and tactfully brought them out during the lessons until he finally had obtained a fairly accurate idea of the note’s contents. That the girl and her father were political prisoners by the emperor’s command he made out. Just why, or what her immediate danger was, Robert was still unable to decipher accurately, but he got the impression that she was threatened with a morganatic marriage to the emperor.

Until he could learn the exact contents of the epistle and grasp a sufficient knowledge of the Martian tongue to discover something about their political intrigues, Robert decided that any move he could make would result in more harm than good for her. He longed for some means of communication with her in order that he might let her know that he was but waiting the right time to help her. If he could but speak or write the same language! Then he might at least manage somehow to tell her that he had not forgotten her. Though he had not yet seen her he steadfastly believed her to be the girl of his dreams.

At the end of a week Robert, in his zeal, had so far out-distanced his companions in the mastery of the Martian tongue that they were both continually asking him about this or that word. He could already make most of their wants known through speech to the attendants.

It was about this time that one of their two attendants, a young, pleasant-faced chap, called Modah, startled Robert by stealthily handing him a small, sealed envelope when they were alone for a moment. Making a sign of secrecy the Martian hurried off. Robert pocketed the envelope as the other attendant entered just then.

The reading of this note, though longer, proved easier than the first. Robert recognized the signature immediately. However, it was not until two days later that he succeeded in completely deciphering both notes, and that only by tactful questioning of the interpreter.

The first read thus:

My friend:

I call you "friend" because they tell me you have come from a far planet and I know that one so daring can not be cruel like my captors.

My father and I are held as political prisoners here by Emperor Kharnov who is trying to force me to marry him.

If you can read this letter, will you not try to help my father and me to escape?

Zola.

The second note plainly had been written less hurriedly than the other, in which several characters were imperfectly formed. This one covered two sheets:

My friend:

Modah, the faithful, will give you this note. He has told me much about you and I feel that I may, indeed, count on you as a friend.

This is to tell you that he can be depended upon to death. He was a servant of my father's palace in the country of the snow where my father ruled. Modah secured a place in the emperor's palace to help us. My maid, who came with me, also is loyal.

The emperor rules supreme over the entire planet, but he usurped the crown during my father's youth, giving my father the governorship of one of the polar regions instead.

He feared my father, and, on the pretext of a conference, invited us here only to make us prisoners. He threatens to take my father's life if he does not publicly renounce his title, and will try to force me to become a morganatic wife to himself.

Help our just cause, O my friend, and our gratitude will not be lacking.

Trust Modah.

Zola.

The notes made a profound impression upon Robert. He pictured the girl of his vision again for the hundredth time. Could it really have been she whom he had already dreamed of thrice? But no matter who she was, he was firmly determined to find some means of helping her.

By this time he had a fair command of the Martian writing— enough, at least, to write an intelligible, if elementary, message. That night he succeeded in passing a brief note to Modah, unobserved by Numid, the other attendant. In this note Robert asked Modah to see him in the seclusion of the East Room as soon as he could elude the sharp-eyed Numid.

He had selected the East Room, a sort of library in their apartment, for the rendezvous because it had but one entrance. Here Robert waited anxiously for what seemed hours. Finally there came a light, furtive tapping on the door. Then Modah slipped into the room quickly.

It had been the professor's suggestion that Robert talk to Modah alone, since it was evident that it was Robert in whom his interest and that of his mistress was centered.

"Numid wasted much time, *Elah*, Talk, talk, talk. I waited till he slept— then came. He must not know. Would get suspicious."

"I want to help your mistress," Robert told him. "How will it be possible for me to see her?"

"I have arranged to take you to her this night, if you wish."

"Good work, Modah. Shall we start now?"

"At once, *Elah*. Follow me."

So, with quickened pulse, Robert accompanied him, having first informed Professor Palmer and Taggart of his mission. Through the apartment they passed silently and entered a small antechamber at its rear. From this chamber several doorways gave egress, one of which led into a long, dark and narrow passage where Robert had to feel his way cautiously. The uncomfortable suspicion occurred to him that this might all be but a scheme to do away with him quietly. But he reconsidered, realizing how completely they were all in the Martians' power; it did not seem reasonable to suppose that they would take such elaborate pains to do away with him.

Presently they emerged into another anteroom similar to the one adjoining their own apartment. A soft glow from a shaded lamp illuminated the chamber. Through a broad archway on the right floated the soft strains of

some stringed instrument. Robert halted unconsciously. Never had he heard sweeter music.

Modah's insistent gestures finally roused him from his reverie. Having succeeded in attracting Robert's attention, he passed through the archway, motioning him to follow.

The heavy portieres parted before them. A large room of luxurious tapestries and upholsterings dimly illuminated by rose-shaded lights met his eyes. As they entered, the music ceased.

Then Robert saw a vision rise from beside a large harplike instrument at the other side of the room and approach them. As she passed a lamp its rays bathed her head in its glow. His dreams were come true. She was the girl of his vision! Not for nothing had he thrice dreamed of her. He would have known her among a thousand.

The majesty of empresses and the grace of a nymph were hers as she moved toward them. Serenely, and without hesitation, she came directly to Robert and placed her hands in his. Her lovely eyes looked into his trustfully.

"I knew you would come," she said simply.

Her low, clear voice was a joy. The loveliness of her held him speechless.

"And I knew that I should find you," he answered, finding his tongue at last.

She led him to a couch and commanded him to sit beside her. Modah had vanished.

"We may be surprized at any moment by a visit from the emperor or one of his spies," she said. "We must make our plans quickly."

"I am at your command, princess," said Robert.

"There remain but two days in which to escape. The time set by the usurper for my father's final decision expires then. My father will never bow to him. I shudder to think of what may follow."

Robert pondered a minute. Obviously the *Sphere* offered the most promising avenue of escape; but how to gain access to it at night without arousing suspicion and probably fatal resistance was the question. A plan came to him. He could visit the *Sphere* the following afternoon under some plausible excuse without exciting suspicion, and could remain tinkering in

it, ostensibly to make some repairs. Unless disturbed, he could wait till dark, then cautiously bring it over to the garden.

The scheme was simple enough, but the possibilities for failure were numerous. Should the emperor become suspicious and have the *Sphere* watched, his plans would be of no avail. A dozen other possible obstacles occurred to him but no better plan.

Swiftly he outlined it to Zola, obtaining her promise that she and her father would be in readiness to leave the following night. Her absolute confidence in his ability at once disturbed and inspired Robert. He felt that he must succeed at all costs. Here was probably his only chance of saving the maid of his dreams. He must not be found wanting.

“Quick! The emperor!” said Modah, tensely, appearing suddenly, as from nowhere.

Pressing the princess’ hand in hasty farewell, Robert followed Modah quickly. Down the long hallway they hurried. At a dark and forbidding doorway Modah halted and signaled Robert to follow. The next moment they had plunged into the darkness of a narrow passage leading off the main one. Groping blindly along the wall in the wake of Modah’s hurrying footsteps. Robert narrowly missed flattening his nose against the opposite wall as they turned a sharp bend. A moment later, however, a faint light glimmered ahead, revealing Modah scurrying along, a few yards in advance.

A moment later Modah stopped. Together they listened for some sound of pursuit. But the silence of the massive, walled passage was unbroken save for their own bated breath. Once more they hurried ahead, and soon Robert recognized with relief the passage leading into their apartment.

Both Taggert and the professor were eagerly awaiting him. Briefly Robert explained the result of his excursion, and suggested his plans for their escape with the princess and her father. It developed that Modah had taken a different and roundabout route back for greater safety, though the divers dim passages all had looked much alike to Robert. He was only certain that it had taken them longer to return. He resolved to substantially reward the vigilant Modah at the first opportunity. Had the emperor discovered him with the princess, all would have been lost.

Early in the morning Taggert left to examine the *Sphere* and survey the ground for an escape. Ostensibly he went to obtain some needed articles of

wearing apparel. Robert and the professor took the precaution to instruct him as to what articles they wanted at breakfast, while Modah and Numid were both present.

Immediately after breakfast, and during Taggert's absence, they were visited by the emperor. Fearing to hear of his own clandestine call upon the princess, Robert prepared himself for the worst.

But the emperor apparently was ignorant of the matter, for he merely inquired regarding their welfare and begged them to accept an invitation to attend a meeting of his learned men to discuss life on the two planets; he seemed much pleased over their progress in the Martian tongue.

While he was there Taggert returned with the articles mentioned that morning. Those he made no effort to conceal. After a few words with him the emperor bid them adieu.

"When is this meeting he's spouting about?" Taggert inquired anxiously.

"Tomorrow evening," said Robert. "Too bad we must miss it. Professor Palmer could gather a great deal of interesting data regarding Mars from those men."

"He would if those pirates would give him a chance to ask a question," said Taggert. "But the chances are that they'd keep him dizzy satisfying their own curiosity."

"Never mind, boys. I'll no doubt have other favorable opportunities to gather all the data I want from our polar friends when we reach their country. What is more important now is the condition of the *Sphere*. You haven't told us what shape you found things in, Taggert."

"Oh, everything looks all right, thank goodness," answered Taggert. "I took a peep into the petrol tanks, too, and turned the gyrostats over a few revolutions. Saw a couple of these runts hanging around, but I guess we'll be able to pull off our party O. K. I tried to act as if there was something out of order for their benefit in case they were spies."

"Good idea. That will offer a plausible reason for my going over this afternoon," said Robert.

Having laid all their plans carefully, Robert and Taggert visited the *Sphere* late that afternoon, and together they tinkered and tested for the benefit of the two or three Martians who were ever about, as well as for their own benefit to see that everything was in working condition. Fortunately everything was in the same shape as when they had left it.

About sunset Taggert left, taking word that Robert was engrossed in making some delicate repairs and would not return for dinner.

As the twilight rapidly faded Robert observed with uneasiness that three Martians remained within the building where the *Sphere* was placed. His suspicion that they were spies was strengthened when one of them engaged him in conversation and inquired whether he might accompany him within the *Sphere* to learn something of its intricacies. The note of insistence in the Martian's voice convinced Robert that a refusal would be useless. He granted the request gracefully, biding his time until he felt sure the princess would be ready. Then it would be up to him to get rid of the spies in some manner. Just how he was to accomplish this he did not yet know.

The time agreed upon was immediately after the first sign of the rising moon, Phobus, above the horizon. Explaining the various features of the *Sphere* as vaguely as possible, Robert watched closely for the first glow of the planet's little moon. The several windows in the western end of the building offered a fair view of the horizon in that direction.

As the fateful hour approached, Robert determined to get the *Sphere* outside the building in some manner, in order that he would be best prepared to make a dash to the palace at the proper moment.

Accordingly, he informed the Martian with him that he intended to take the *Sphere* outside for a trial flight before leaving it, asking him whether he desired to accompany him or not. He was hoping, of course, that the Martian would say "no." He did not wish to seem desirous of getting rid of him for fear that his suspicion would be aroused.

As Robert had half expected, however, the Martian informed him politely that it was the emperor's wish that the *Sphere* not be taken out of the building for the present. It was plain that Robert would have to resort to strategy to carry out his plans.

Accepting the Martian's statement with apparent readiness and good nature, he remarked that he could just as well test the *Sphere* some other time, and prepared as if to leave it. He succeeded in maneuvering so as to follow the other out, though the Martian preceded him with evident reluctance.

Upon reaching the exit in the outer shell of the *Sphere* the spy turned as if minded to request Robert to precede him there. Realizing that it was now or never, Robert leaped upon him suddenly, delivering a sharp blow to the spy's jaw that sent him reeling down the few remaining steps and through the open trap on to the floor of the building. Simultaneously with the Martian's cry of baffled rage, Robert slammed the trap-door shut and raced up the steps. A moment later he had the gyrostats in motion.

His heart sank, though, as he saw all three of the Martians tugging at the massive door that stood between the *Sphere* and freedom. Already it was sliding to with their efforts. His escape was cut off!

For a moment Robert was tempted to press into service the automatic that reposed in a locker but a few feet away. But the thought of firing upon them in cold blood was repugnant to him. Apparently the Martians were unarmed. During their sojourn on Mars, Robert and his companions had not found out what kind of weapons the Martians used. Two of the Martians remained inside the great door, shouting to him to come out of the *Sphere*. The third had slipped out, no doubt to inform the emperor of his attempted escape.

The Martians' probable ignorance of the power of his automatic made it impracticable for Robert to attempt forcing them to open the door again. Then, suddenly, he remembered the tremendous lifting power of the *Sphere*, and a possible solution came to him. Carefully he maneuvered the *Sphere*, raising it slowly to the roof. As he felt the jar of the shell against the rafters he increased the lift. He could hear the building groan with the immense pressure. Then he opened the disk's surface to its maximum and switched on the current!

For a moment the *Sphere* swayed uncertainly as the stout rafters writhed against the tremendous power of the disk. Robert caught a glimpse of the Martians flattened against the door in consternation. The next instant the *Sphere* shot upward with a crash as the roof of the building collapsed behind it!

Instantly Robert shut off its lifting power, and began maneuvering the *Sphere* back downward and toward the palace.

Fortunately the *Sphere* seemed undamaged by its unusual exit from the building. Within a minute after his escape Robert had it immediately over the palace gardens where the princess and his companions were to join him.

The gardens were dark and forbidding-looking beneath him, though the ruddy orange rim of Phobus already was visible over the horizon's edge. As yet there was no sign of pursuit.

Softly the *Sphere* settled like a round, dark cloud. As it reached a level of perhaps thirty feet Robert saw a darker blotch on the expanse of lawn in back of the princess' quarters. Being sufficiently familiar with the garden to know that this was not a part of the shrubbery, he suspected and hoped that it was his friends awaiting him. He dared not show a light yet, but continued to settle as rapidly as was safe.

Then, to Robert's relief, a small, carefully shaded flashlight winked twice at him from below. This was the signal agreed upon.

A moment later the *Sphere* came to rest on the turf. A glance through the windows was sufficient to recognize his friends. Taggart came close and rapped on the glass. His low "Hurry" reached Robert in spite of the double panes.

Hurrying down the narrow steps he opened the manhole. The next instant he was helping the princess into the dark passage which he dared not light yet. The flashlight which he also carried for this very emergency, however, enabled all to see their way quite well.

But their good fortune was not to continue indefinitely. Hardly had the princess and her father stepped into the *Sphere* when there came a sound of excited pursuit on the other side of the garden wall which divided the princess' section from theirs, and the garden gate burst open!

Urging the professor and Taggart into the *Sphere* hurriedly, Robert slipped the automatic out of his pocket, thanking his lucky stars that he had brought it with him. Without hesitation he blazed away over the heads of their pursuers as they rushed forward, hoping to halt them for an instant. But they seemed undaunted. On they came in great bounds, shouting excitedly.

Having done what he could to keep them away from the *Sphere*, Robert hurried into the passage just as the leader came up. As the latter darted after him Robert fired. The Martian went down, but the others pressed forward, brandishing saberlike weapons. Unable to lift the trap-door into place because of the body of the wounded Martian upon it, Robert was about to shout to Taggart to raise the *Sphere*, when one of the Martians leaped over his prostrate fellowman and struck at Robert with his weapon.

By an unlucky bit of chance the blade struck the automatic from Robert's hand even as he fired again. He could hear the gyrostats humming and had just presence of mind enough to yell to Taggert to raise the *Sphere* when there came a roar and a flash by his ear and the vicious Martian's weapon clattered on to the steps as he staggered back, blocking for a moment the others behind him. The rays from the small globe in the ceiling now lighted the passage.

"Close shave, Robert," came Professor Palmer's voice, and once more the roar of his firearm deafened them in the narrow passage.

The Martians' rush was checked for a moment, and in that instant they had lost. The *Sphere* trembled. As a great shout of baffled rage welled up from without, Robert knew that they finally were off. Stung by their failure, two Martians rushed at the manhole. One of them managed to grip the rungs outside. The next instant he was precipitated forcibly back to the ground as the *Sphere* shot upward, wrenching the rungs from his grasp. But Robert's earlier assailant still sprawled, wounded and partly dazed, upon the lower steps. Then, abruptly, before they could prevent it, he rolled off into space with a wild cry!

Slamming the manhole cover shut, Robert and the professor made their way up into the main chamber. The princess was waiting for Robert with great anxiety. There, for the first, time, Robert met Hakon, her father, a dignified and robust man of middle age, with an imperious though pleasant manner. The beard, as affected by the emperor, was absent, but he wore a short, gray mustache that matched his heavy, graying hair.

"I can not express my gratitude for the service you and your companions are doing for my daughter and me, my friend," he said in a pleasing, resonant voice. "I can only say that you shall all be fittingly rewarded upon our arrival in my country."

"The privilege of rescuing your Highness and the princess is sufficient reward," replied Robert.

"Can't we go back and lick their army for you or something?" asked Taggert, looking at the princess in vast admiration.

"Our zealous companion possesses more enthusiasm than discretion," said Professor Palmer, at which they all laughed.

Reaching a height of about a thousand feet, Robert started the *Sphere* toward the region of the planet's north pole, having taken his direction from

the slowly rising Phobus in the west after consultation with Professor Palmer and Zola's father.

Taggart volunteering his culinary services, they all indulged in a light repast of canned salmon, fresh baked biscuits, in which Taggart excelled, jelly and tea.

The distance from their previous position just south of the equator being roughly 3,500 miles from the northern pole, Robert estimated that they could easily reach Zola's country within six hours. However, as it would be more convenient as well as interesting to arrive after dawn he did not push the *Sphere*. Their royal passengers were greatly interested in its mysteries, and far into the night Robert explained the intricacies of the wonderful machine to Zola and her father. Finally, however, they accepted Robert's invitation to rest a few hours in one of the small rooms adjoining the main chamber, upon his promise to awaken them at dawn.

Shortly after they had retired, the gyrostats developed an overheated bearing, with the result that Robert was obliged to descend to a convenient expanse of desert to safely correct the trouble.

The remedy proved more difficult than expected and dawn was almost upon them by the time the *Sphere* was again ready for flight. Thereupon Robert decided to call their guests before taking flight.

The princess, in some manner unknown to most women, appeared looking fresh and lovely as she stepped from the *Sphere* out into the rosy flush of the new dawn.

"Oh, it's so good to get out into the open once more, and to know that we are free," she cried. "It seems ages since we were made prisoners by that frightful usurper. How can we ever thank you enough?"

"Ah, princess, your happiness more than repays me," replied Robert, who found himself strolling beside her.

Together they watched the eastern sky grow rapidly brighter while the others stood apart as if understanding their unvoiced desire to be alone.

"A strange dream came to me recently," she murmured. "This brings it all back. I dreamed that I was alone in a vast jungle of horrible, twisted growths of huge thorns from which I could find no escape."

"Then you came upon a small clearing," Robert interrupted before he thought.

“Why—how did you know?” she breathed, her eyes alight with astonishment.

“I—I was there. That is, I, too, dreamed that I was there and that you came.”

The sun’s dazzling edge peered over the rim of the horizon, its warm rays nestling in her pale gold hair.

“And did—did you—oh, I know it sounds absurd—but did you not see me one night off in space as you journeyed to me in your wonderful *Sphere*?”

“That is how you knew I would come to you? How well do I remember! But then, as before, you slipped away from me. I have a terrible fear you will fade away, leaving me but a dream to remember you by.”

“No, no, no!” she cried earnestly; and then, as if suddenly embarrassed by the subtle admission just made, she flushed and turned away. As they walked quietly back toward the *Sphere* where the others waited there was a song in Robert’s heart.

Once more they embarked on their journey toward the polar region. The gyrostats now operated faultlessly. Robert rose slowly to an altitude of about 2,000 feet.

Since their arrival on Mars Robert and Professor Palmer had established the existence of the canals, or ducts, thereby settling for all time—provided they succeeded in returning safely to Earth—the controversy over the most noted characteristic of Mars as seen from the Earth, the curious network of lines. Incidentally, they had seen the Martian deserts as well as both rural and city life on the planet. They were informed as to the correctness of their original conjecture as to the method of pumping the water from the poles to all sections of the globe. This was accomplished by means of electric pumping stations at intervals along each duct, the rays of the sun being absorbed and converted into electric energy for this purpose by the big checker board structures.

But the feature which, so far, had defied plausible explanation by all experts of our world, still remained a riddle to the adventurers. This was the “carets,” or triangular-shaped points which existed wherever the canals were seen to connect with one of the large, shaded regions.

These shaded regions also were a riddle. A theory that the latter were seas was exploded by the fact that in some instances the canal lines were

visible running across them. For another thing it was agreed by practically all experts that the seas on Mars had long since dried up.

It was, therefore, with considerable excitement that Professor Palmer observed that they were nearing one of these regions. From a height of several thousand feet one of the “carets” could be seen plainly as they approached it.

As Robert lowered the *Sphere* toward it, Hakon showed signs of uneasiness. Still he held his peace until it became quite evident to him that his hosts intended approaching close to this section. Zola, too, was evidently perturbed.

“There is danger there, gentlemen,” said Hakon, finally. “I beg of you, do not go too close.”

“What is it?” questioned Professor Palmer, now intensely interested. Both he and Robert received the impression that some terrible physical force connected with the phenomenon threatened their safety. But in this they were quickly disillusioned by Hakon.

“We are now in the region of the usurper’s strongest support,” he told them. “It was the hotbed of traitors before he stole the throne, executing many of my loyal leaders. These people have, no doubt, been notified to be on the lookout for us and will try to destroy us with their *hilwai* if we approach within range.”

Hakon explained further that *hilwai* were a kind of large gun, electrically operated, in lieu of the obsolete kind in which explosives had been used in previous centuries, before the planet’s mineral supply had been virtually exhausted. These guns had a range of about twenty miles horizontally. Though this was not a very great distance, considering the weak gravity on Mars, it was sufficient to make it uncomfortably dangerous for them if they undertook to view the “caret” at close range.

Hakon was surprised to learn of their curiosity regarding the dark blotches and their caret-shaped junctures wherever they were joined to a canal, and readily explained this phenomenon.

“The dark regions are vast, low sections which were covered by seas ages ago,” he informed them. “The carets are formed by two branches from the main canal where it enters the low region. From these fanlike branches, and the main canal which crosses the bottom, the low country extending beyond is fed. Thus the lowlands, which formerly formed the sea bottom,

are cultivated, resulting in the largest, richest and most productive regions on the planet. Even small, precious lakes are formed in the lowest spots, and occasionally there are light falls of rain, which are unknown in any other parts of the planet.”

Circling this region at a comfortable height, the adventurers examined it in turn through a telescope. What they saw tallied with Hakon’s explanation. The denser vegetation, fed directly from the main canal running across the bottom, appeared as a slightly darker strip. Two small lakes were visible. Both of these appeared to be only a mile or two in circumference. The western edge of the low region was hidden in a light cloud bank.

Leaving this region behind, Robert pushed on toward the polar region ahead at full speed. After an hour had passed thus, Hakon suggested descending closer to the surface, that he might watch for landmarks to direct them to their objective. Though the country here looked much like that near the equator, there seemed to be fewer canals; at any rate, there were fewer latitudinal canals, for they had crossed very few during the last half hour. However, they had the opportunity of observing one of the “double” canals, which was really two canals running parallel, some seventy-five miles apart. Hakon explained that this was in order to take advantage of exceptionally productive districts which justified this deviation from the regular scheme.

In this manner they neared what was perhaps the most northerly intersection of canals on the planet. Hakon directed Robert to land within the metropolis, which he called Svergad, situated in the center of the great oasis at this intersection.

Any doubt which Robert and his companions might have had regarding their reception in this new region was dispelled by the happy expression of anticipation on the princess' countenance as they approached the large oasis which extended over the entire section between the two parallel canals where a third canal intersected them. It was plain that she was overjoyed at the prospect of returning to her beloved country.

They dropped swiftly. Guided by Hakon, Robert finally brought the *Sphere* to rest in a large, open area paved with huge, grayish stones. These formed a vast courtyard behind a huge building of similar material. All around them reared impressive buildings, while the roofs of still others extended as far as they could see in all directions. They appeared to be in the heart of a metropolis that compared favorably with the one in which they had been captives. The large building adjoining the courtyard Robert correctly judged to be Hakon's palace.

There was almost a total absence of foliage. This, Robert subsequently found out, was due to the odd indifference of the people of this country to the beauties of foliage, though water in this region was more plentiful than in the equatorial regions.

A number of men appeared from within the palace and approached them as Robert assisted the princess to alight from the *Sphere*, followed by Hakon and the others.

As those approaching recognized them they hurried forward with evident delight, greeting them with demonstrations of deep regard. Hakon presented Robert and his companions to those assembled, telling of their aid and praising Robert especially in highest terms. That they were from another planet seemed no news. It was evident that the information as to the *Sphere's* arrival on Mars had already reached its farthest corners through the

excellent system of wireless communication of which the Martians were masters.

They were promptly established in the palace as guests of highest honor. While separate apartments were offered each of them, at their mutual request Robert, Professor Palmer and Taggert were once more installed in one apartment where they could be constantly in one another's company. Every convenience and luxury were theirs. At Hakon's suggestion, an instructor called upon them every day to help them improve their knowledge of the Martian tongue, which service they were glad to avail themselves of, particularly Robert, who wished to be able to talk to the princess with less awkwardness.

The princess and her father visited them once or twice every day, while Professor Palmer became a great favorite with the learned men of the city, to whom he delighted in talking of the two planets, comparing astronomical data. In addition, the princess insisted upon Robert's being with her every evening, and their acquaintance ripened into a great friendship. Robert found himself deeply in love with her, as indeed did everyone who knew her. But with Robert it was different. He felt as if he had known her always. He was ready to make any sacrifice for her, and began to wonder where all this was leading.

Meanwhile, Hakon had dispatched an ultimatum to the emperor, and was awaiting his response. Controlling, as he did, the water supply to virtually the entire northern hemisphere and part of the southern one, Hakon possessed a powerful weapon, though the usurper had by far the larger army.

It was during this period of tense waiting that Robert asked Hakon whether he had any objection to their visiting the reservoirs from which the canals were supplied, before the winter season was upon them. With customary acquiescence the latter readily offered to furnish them with a guide and any other assistance which they required for the trip.

Robert took leave of the princess with considerable reluctance. Never had he been able to entirely overcome a strange foreboding that sometime they would be separated and never see each other again.

Something of the same premonition must have been present in the princess' mind that morning, for she seemed reluctant to let him go.

"I am sure that I'm selfish in saying it, but I do so wish you were not going on this journey. I won't rest until I see you back."

Her solicitude touched Robert deeply. As yet he had not told her of his love. Who was he to declare love to a princess? But he was sorely tempted to take her in his arms then and there—to tell her what was in his heart.

"It is but little more than a day's journey there and back," he replied, his voice a-quaver with the lure of her. "This is one of the chief features of your planet which the professor determined to defy the dangers of space to see. We must not disappoint him."

"But it has proved fatal to many, and even your wonderful *Sphere* may not return. Will you not let them go alone—Robert?"

She pronounced his name quaintly, like "Roe-bert." It was the first time she had attempted to call him by his name, which she had heard his companions use. His pulse beat madly.

"Tell me," he breathed, "do you care—Zola?"

She blushed adorably.

"Yes—Robert."

His arms were suddenly about her. She buried her head on his shoulder with a little sigh, and clung to him.

"Now, you will not go?" she asked presently.

"I must, beloved," he replied, tenderly. "They still depend upon me to operate the *Sphere* safely in an emergency. I can not refuse the professor this service. Why, if it hadn't been for him, I could not have come to Mars, and you would still have been the emperor's prisoner."

"You are right. I'm afraid I am just a selfish girl with foolish fears. You must hurry now and come back to me quickly, dear."

So they parted, Robert promising to come back to her as soon as possible, each striving to conquer a nameless fear that they would never see each other again.

A few hours later found the *Sphere* approaching the pole at a tremendous speed. Accompanying Robert and his companions were three Martian experts. Two of these were astronomers; the other, a professor of physics, acted as their guide.

Far away to the north a comparatively small expanse of white indicated what remained of the great polar cap of ice and snow that, during the winter, extended over the entire region.

“How is the water from the melting snows accumulated for distribution?” Taggert asked.

“As the spring advances, and the edges of the vast polar cap of snow begin to melt, the water is collected in the sea basins,” explained the professor of physics.

“But the task of running the great ducts beneath the shores of these seas in order to connect with and drain their lowest points seems almost an impossible one.”

“Quite true. That plan was finally rejected for that very reason. A system of siphons was built instead. Even this must have been a tremendous task—our history records that it required more than a century to complete it; yet so remarkable was this construction that the original work, now many centuries old, still remains in perfect condition.”

Robert dropped the *Sphere* to within a half mile of the surface. They were passing over what appeared to be a great basin of some sea. Continuing northward they passed beyond the first great basin and over several smaller ones. The white polar cap was less than thirty miles away. As they neared it a small expanse of water in the lowest depressions of the basin became visible.

“Ah, that must be one of the siphons!” exclaimed Professor Palmer, pointing.

A small, sinuous ridge ran from the south along the bottom of the basin, disappearing in the little patch of water.

“Yes, that is one of them,” confirmed their guide. “No doubt it is perplexing to you as to how each canal is supplied with water without

interruption as the water in the seas and the polar cap is consumed and recedes.”

“That is true,” admitted Professor Palmer.

“The people of our planet have long been masters of electricity. In fact, it has been the only available source of power on Mars for centuries. This power is utilized to manipulate the valves in the various siphons which feed each canal, insuring its usefulness until the last portion of the polar cap is virtually exhausted. Accurate maps of all polar sea basins exist, of course. The pumping station nearest the pole, on each canal, has one of these maps indicating every siphon and valve feeding that canal. As the water from the nearest basin is exhausted, the valve of that feed is closed by manipulation of a corresponding switch in the station, and one of the other feeds is opened. A chart of the retrogression of the seas with the declining season enables the attendant to drain each sea basin completely in its proper turn as the melting snow recedes.”

“Wonderful!” exclaimed Robert and Professor Palmer in unison.

Taggart was busily scribbling in his notebook.

In a short while the *Sphere* had reached the edge of the snow-cap. Immediately below, and stretching away some distance to the south and west, was a small sea, still well filled but covered with a smooth sheet of ice which reflected the rays of the low sun in a blinding, yellow blaze.

At the suggestion of their guide, Robert raised the *Sphere* till it was perhaps a mile high. From this point of vantage they could see the full extent of the polar cap. It seemed to be about twenty-five miles in diameter. They fancied they could see some of the canals beyond its farthest edge, though they could not be certain, because of a slight murkiness in the atmosphere in that direction.

Robert now lowered the *Sphere* almost to the surface of the snow and for a time they hovered over various parts of the cap. Quite different was this polar region from that of the Earth, for it was almost entirely one continuous, level sheet of snow and ice, without the great, towering icebergs and the mountain peaks which we associate with the arctic regions of our own world.

So far, their polar excursion had proved quite tame. Remembering the princess’ anxiety, Robert wondered whether it had been due entirely to imagination. He questioned the Martians as to the possible dangers of the

region. They replied that, during the winter season, the region was noted for its sudden and terrific blizzards, which lasted for weeks at a time. According to them, the approaching winter was not expected to break for some weeks yet.

By common consent, the *Sphere* was landed on the shore of the sea which seemed to surround the shrunken cap. A layer of crystallike ice stretched off in three directions, unbroken except along the shore, where it was evident that the water beneath the ice was still being withdrawn, for the ice cakes were deposited in huge blocks on the sloping shore for several hundred yards beyond its present edge.

As they stepped from the protection of the *Sphere*'s comfortable warmth the cold surprised them in its intensity, in spite of the heavy coats which they had donned for the occasion. A sharp breeze blew from across the polar cap. The horizon in that direction was partly hidden by the increasing murkiness which they had observed before from above.

A short walk along the shore before resuming their way back to the capital was suggested. Accordingly the little company picked their way over the rugged bottom above the broken incline of ice.

Queer shells and curious bits of rock were strewn about underfoot. Robert picked up a quantity of the prettiest specimens and put them in his pocket. Some were of the most beautiful tint, ranging from a delicate pink to an intense cerise, while others were of equally delightful shades of almost every color in the rainbow.

While their progress was slow over the rough ground, they found that they had covered nearly a mile at the end of some twenty minutes. The seabottom offering little further interest, it was decided to return to the *Sphere*, especially as the sky was assuming a peculiar aspect. It seemed as if they were surrounded by a shrinking circle of darkness. The evident uneasiness of the Martians was anything but reassuring to Robert and his companions. All were plainly anxious to get back safely into the protecting walls of the *Sphere*.

"I hope one of these Martian blizzards doesn't surprise us," remarked Taggart, eying the horizon speculatively, as they hurried along.

The blackening horizon seemed to be racing toward them.

"Frankly, I don't like the looks of it," said one of the Martians. "I believe we are in for a winter storm, which is likely to be none the less severe for

its earliness.”

“I have observed the progress of one of these polar storms from the Earth,” contributed Professor Palmer. “Within twenty-four hours it had transformed this very pole from a cap of insignificant size to an expanse of snow extending over nearly a quarter of the northern hemisphere.”

A sudden flurry of large but scattered flakes of snow seemed to burst out of nowhere as he spoke.

“Well, it looks as if you are going to have a taste of one anyway,” Taggert cried.

The *Sphere* was still quite plainly visible in the distance. But, probably because of the falling snow, the *Sphere* seemed rather to have got farther away during the last few minutes instead of closer. Robert was conscious of a strange foreboding as he quickened his pace over the rough ground with the others.

The flurry quickly became a determined snowfall. The beautiful big flakes swirled round them dizzily, all but obscuring the *Sphere* entirely from view. However, the ice would serve to guide them in case the *Sphere* became invisible.

Resort to this method of guidance soon was a necessity. It proved to be not so simple as expected; for, as the flakes fell faster and faster, the great cakes of ice were not visible until they were almost upon them, and then they had an awkward habit of appearing only after the adventurers had fallen over them.

Stumbling along, with Professor Palmer keeping pace with difficulty beside him, Robert felt that surely they must have passed the *Sphere* already. It seemed to him that they had walked two miles or more since they had turned back, and still the *Sphere* could not be seen. He turned an instant and looked back half expecting to catch a glimpse of it behind. Taggert trudged along a few feet back; the others were strung out two or three yards in the rear.

A minute later Robert heard a muffled cry behind him. He turned just in time to see Taggert fall and slip from view in the blinding snow. Robert took a quick step to the reporter's aid. The next instant his feet shot from under him and he tobogganed down the slippery incline of ice toward the sea.

One thought reassured him as he felt the smooth surface racing past him: the level, unbroken expanse of ice over the sea would surely stop him when he reached it. But, hard on the heels of this thought, came the realization that, with all sense of direction lost out on the expanse of ice in a blinding snow, he might blunder farther and farther away from the *Sphere*. He earnestly hoped that the end of his slide would at least bring him close to Taggert.

Presently he reached the bottom of the incline and shot out over the smooth, frozen surface of the sea. Here his momentum was quickly checked. As soon as he could regain his feet he stood up and peered round

him. But only the driving snow, which all but obscured his hand before him, rewarded his gaze. He shouted several times, listening intently after each call. The snow seemed to muffle his cries in the making. Nothing was visible round him but a mil of snowflakes. His ears tingled with the bitter cold even under the fur cap he wore. He shouted again, removing his cap to listen afterward.

A faint answering cry floated back to him; but whether this was simply the rebound of his own cry from the dense wall of snow he did not know. As he continued to listen the same faint cry again came to him, this time a bit stronger, seemingly from away to his left. He clapped his cap on and strode off in that direction.

Several times he called out again, stopping to listen each time. Again that faint echo mocked him, but this time it seemed to come from behind. The well-known difficulty of determining the direction from which a sound comes in a fog came to Robert's mind, and he despaired as he realized the same difficulty in attempting to find a distant call in the falling snow.

Just then the cry sounded again, and this time it was undeniably plainer. Robert shouted in reply and was overjoyed to hear it once more increased in volume. He hurried toward it, shouting and listening alternately.

A moment later Taggert strode from out of the white veil.

"The wandering echo found at last," Robert greeted him, while they pounded each other on the back in joyous relief.

"Where in Sam Hill are we?" Taggert wanted to know.

"Question is, in which direction is the *Sphere*?" cried Robert.

"Well put. What's the answer?"

"Have you any idea from what direction you slid out here?"

"Nope. Have you?"

They stared at each other blankly. Then the ridiculousness of their cross-questioning struck them and they laughed together. For the moment the seriousness of their plight was forgotten. The white flakes swirled about them, settling upon their heads and shoulders till they looked like snow-men.

An idea suddenly occurred to Robert.

"This storm arrived from ahead of us when we were on our way back to the *Sphere*, didn't it?" he asked.

"Believe it did," agreed Taggert.

“Well, then, unless the direction of the wind has changed, we have only to push on at right angles to it, with it blowing on our left, to eventually come upon the shore which we left.”

“Right you are!” exclaimed Taggert, after a moment’s reflection.

So they pushed ahead in the direction indicated for several minutes, making fair progress in spite of the rapidly increasing wind which swept the smooth ice clean, leaving a difficult footing. Each minute they expected to come upon the slope up the shore, where they hoped to find a crevice in the raised cakes which would offer sufficient footing to scramble up the slick incline to the ground beyond. Still the blank wall of driven white revealed nothing but the level floor of ice, ever stretching a few feet ahead of them as they shuffled along.

“Seems as if we should have raised something by this time,” said Taggert presently.

“The wind must have shifted,” said Robert. “If it hasn’t changed much, though, we should reach the shore anyway soon.”

So they continued onward, half blinded by the snow, the bitter wind whipping round their bodies. With each step the hope of reaching the *Sphere* became weaker. The princess’ anxiety recurred to Robert as he plodded doggedly on ahead of Taggert. He wondered dully whether he should ever see her again. Well, at any rate, she was once more safely within her own country. He was thankful for that much. His reflections were cut short suddenly as he sprawled forward upon the ice, tripping up Taggert, who was following him closely. Caught unawares, Robert fell on all fours, knocking the breath out of him.

“’T’ell!” spluttered Taggert, scrambling up.

As Robert also struggled to his knees he saw that he had tripped over a crevice in the ice. It was the edge of a slightly higher block which sloped upward. The realization came to him suddenly that they had finally reached their first goal!

After some difficulty they managed to clamber up one of the perpendicular cracks between the slippery blocks.

Upon reaching the ground, where the snow had now collected several inches deep, they looked about eagerly but in vain for some sign of the *Sphere*, or of their erstwhile companions.

“There is nothing for us to do but to plod on against the wind until we reach the point where we left the *Sphere*,” said Robert. “We should be near where we slipped down before.”

“Lead on,” said Taggert.

Once more they plodded ahead, keeping close to the ice blocks on their left. The now fine snow drove directly into their faces with stinging force, making it almost impossible to keep their eyes open enough to see.

For half an hour they pursued their way painfully. Only the exertion kept them from freezing in the increasing cold. The fierce wind whined about them hungrily, pitilessly, as if eager to make an end of them, while the eddies and drifts of snow round each depression or rock grew steadily deeper and more nearly impossible for them to plow through as their strength waned.

Robert realized that ere this they should certainly have reached the vicinity of the *Sphere*, but the snow restricted their view to a radius of less than ten feet. Unless they were fortunate enough to run right up on the *Sphere*, their chances of sighting it were remote. Even now they had lost touch with the border of ice, and, but for the uniform slope of the ground, and the general direction of the wind, would not have known whether they were still following the shore in the original direction or not. They shouted together many times but got no response. The snowfall and howling wind so muffled their cries that they despaired of being heard. Having continued on for some minutes longer they decided finally to go no farther, as they were convinced that they had already passed the point where they had left, the *Sphere*. A convenient rock, some ten feet high and of about the same breadth, offered temporary shelter.

“If Professor Palmer fired a shot we should hear it here,” Taggert suggested.

“I doubt it,” replied Robert. “This hubbub and the muffling effect of the snowfall might drown a report within a hundred yards.”

Here a startling thought occurred to Robert. What if the Martians took advantage of Professor Palmer’s isolation and made off with the *Sphere*?

“Why the silence and corrugated brow?” queried Taggert “An idea?”

“An idea, but no good,” Robert responded, forcing a grin.

“Well, you needn’t grin about it. I don’t see anything comical about the prospect of being buried under several tons of snow,” chided Taggert.

The wind had fashioned a sheltered hollow in the lee of the rock where they had taken refuge. The intense cold which prevailed in spite of the heavy snowfall, however, made it imperative that they keep in motion to avoid being frozen. Already Robert recognized a warning feeling of drowsiness. He shook himself alert with an effort.

“Can’t sit here,” he said, suiting the action to the word by rising and stamping his feet. Stabbings as of a thousand needles seemed to run through them at first. If only there were some fuel! Matches they had in plenty.

Taggert struck a listening attitude. A familiar humming was faintly audible above the whine of the blizzard!

Together they listened with bated breath as the humming grew plainer. Alas, a few moments later it passed away, and with it went their hopes.

“Missed us,” ejaculated Taggert, with an involuntary oath.

The realization that the *Sphere* had just passed them by in a vain search for them brought their already drooping spirits to zero for a while. Here had been safety and comfort within perhaps a few rods, and they had been unable to make their presence known. Robert pictured Professor Palmer’s anxious gaze as he peered downward into the veil of flying snow.

“Cheer up, Tag,” Robert admonished, with an attempt at enthusiasm which his feelings belied. “The professor will be doubling back trying to find us. He’ll run up on us yet.”

“Maybe he will—if he doesn’t bounce the old ball into the lake,” replied Taggert, doubtfully. “Say, I wish one of us had brought a ‘gat’ along so that we could signal him if he gets near us again.”

For several minutes they stamped about their cramped shelter, beating their arms round their bodies in an effort to keep up their circulation. As the time slipped by without further sound of the *Sphere* their hopes dropped still lower. The situation was becoming desperate.

Their dismal reflections were abruptly interrupted by a resumption of the humming sound, heralding the approach of the *Sphere* again. Both men stiffened, listening intently, the spark of revived hope burning again within their breasts as the fleeting moments passed. Would the *Sphere* come close enough in this game of blind man’s buff to discover them? Or would it pass them by again, leaving them finally to their doom?

The prospect of freezing to death in the arctic region of a strange planet seemed a dreadful thing. In the heat of battle a man may find death in the midst of wild enthusiasm and patriotism. But here, hemmed in by a wall of beautiful, but deadly, flying flakes, there was no excitement to mask the death awaiting them—only a fearful realization of their fate, millions of miles away from their countrymen—alone! Yet in those fateful moments Robert's thoughts were mainly of Zola. Would he ever see her again?

With leaping heart he realized that the *Sphere* was coming closer—closer than before. He strained his eyes as he endeavored to pierce the intangible walls of their vast prison. Ah, what was that dark blur hurtling through the white froth? It was passing them by again. He joined with Taggert in his shouts. Fools, to think their muffled cries could rise above the tumult of the gale and the whirl of the *Sphere's* machinery, piercing its thick, metal walls!

The fast-fading blur seemed to pause in its flight a moment. But even as they dared to hope, it passed out of sight again. Then quite abruptly it appeared again, this time moving toward them slowly, less than a dozen feet above the ground. The *Sphere's* bulging walls plowed through the snow as it swooped down, sending up a great flurry of the fine flakes. Robert caught himself idly likening it to a cannon ball fired into a great bin of flour.

His next recollection was of stumbling up the short flight of metal stairs into the comfortable warmth of the *Sphere's* main chamber, aided by Professor Palmer.

The Martians seemed genuinely concerned over their plight, and offered a confusion of advice. Fortunately, neither Robert nor Taggert had suffered any serious damage from their severe buffeting in the elements and were soon quite comfortable.

Robert, unable to shake off a strange feeling that the princess was in danger, and anxious to return to her in all haste, insisted on operating the *Sphere* again. The big metal ball fairly quivered as he utilized its maximum power to reduce the distance between them and the capital as rapidly as possible.

Exerting the full power of the disk, Robert shot the *Sphere* upward. Suddenly they emerged from the gray twilight of flying flakes into the sunshine. Beneath them the turbulent mists boiled and tumbled as if with anger over the escape of their prey.

A few minutes later Robert checked the *Sphere's* upward flight, darting southward toward the metropolis and his princess at a terrific speed.

Darkness came on quickly after leaving the pole as they passed into the night which had long since enveloped that part of the planet toward which they were headed. Like a cannon ball the *Sphere* roared southward, the dim topography of the planet swaying dizzily below.

Prevailed upon by Robert, the others had decided to snatch some rest after a light repast. But he felt no fatigue in his anxiety to reach the princess.

The sun's edge was peeping over the horizon once more as they neared Svergad. The others woke up and watched their approach interestedly. Soon the spires and domes of the northern metropolis appeared in the distance. A few minutes later they were drifting preparatory to landing.

Suddenly one of the Martians gasped.

"The enemy!" he cried, pointing.

All eyes followed his outstretched arm. Thousands of tents dotted the level plain beyond the city's distant boundary, extending in all directions and partly encircling the city!

Upon their arrival in the palace, Robert and his companions found the royal household in an uproar. The emperor's army, vastly superior in numbers and weapons, had already fought its way round two sides of the city and was fast surrounding it. Striking with unexpected promptness, and transported swiftly the greater part of the distance by rail, the advance forces had appeared before the city shortly after the *Sphere's* departure for the pole. Thus did the emperor elect to reply to Hakon's ultimatum. In another day the Svergadians expected to be in a state of absolute siege.

It must not be supposed that the Martians had done away with arms and soldiers, even though they enjoyed one universal government. Rebellions were not unknown, and the central government had its hands full keeping all factions quiet. Emperor Kharnov, himself a usurper, maintained a strong army, ostensibly for policing and preserving order, but in reality to protect himself against the aspirations of his numerous enemies.

With but some seventy hours of food supply within the city boundaries, the situation seemed desperate. So far, Kharnov had made no specific demands, but it was expected that the conditions which he would dictate after encircling the city would hinge upon the possession of Zola.

It was in this hour of impending disaster that Robert had greatness thrust upon him. With implicit confidence in Robert's ability, Hakon begged him to take charge of his forces to resist the invader. Zola's trustful belief in him and her urging decided Robert. He became chief commander of the besieged forces.

Immediately upon his appointment Robert in turn appointed Taggert as his aid and placed Professor Palmer in charge of the palace defenses. His next step was to organize an immediate surprise counter-attack upon the invaders. Though he had not slept in many hours, during which time he had endured much, Robert felt no desire for sleep. The excitement and emergency confronting him buoyed him up for the time. He refused to rest until forced to.

Another hasty ascent in the *Sphere* revealed to them the exact situation. The emperor's main army had pushed northward along the fertile region of one of the two great parallel canals. It had stopped south of the metropolis, sending out smaller contingents in either direction to begin the encircling movement. This movement had already made considerable progress, encountering little resistance in the nearly deserted, small outlying towns whose loyal defenders had all rallied to Hakon's main forces in Svergad. The right spur of the invaders had made the most progress, having succeeded in pushing more than half-way round the metropolis.

It was at this right spur that Robert decided to strike first, in an attempt to take it by surprise, cutting it off from the main army, with the hope of defeating it before the main army could come to its aid. He also planned to launch an attack simultaneously upon the main army in an effort to divert its attention from the spur.

A brief council of war with Hakon and his leaders followed, in which the exact point of attacks were determined upon. Also, orders were wired to all loyal pumping stations situated near the southern edge of the regions loyal to Hakon to shut off the water supply. This would deprive Kharnov's supporters in the region of the equator of all water supplied by ducts leading from the northern polar cap; thus the emperor's armies and his present supporters would soon be confronted with the difficult problem of existing without water in many regions. An ultimatum, demanding the immediate withdrawal of his forces and informing him of the discontinuance of the water supply until his retreat was well under way, was dispatched to Kharnov at once.

A system of signals from the *Sphere* was decided upon. It was agreed that, as a weapon of destruction and direction, the *Sphere* was their most powerful aid. The greatest concern was over the dwindling supply of petrol in its tanks, and the extreme difficulty which Hakon's best chemists were having in obtaining enough of the planet's meager supply of mineral oils to produce any quantity of this fuel. Robert was facing the alternative of either remaining to help defend Zola and her people with the possibility of never being able to return to the Earth, or of returning at once with the now barely sufficient supply of petrol to make the trip. He felt that he owed his companions the chance to decide, though he himself was determined to stand by the princess in any event.

“Old man, I’m with you to the finish. Aren’t we all partners in this adventure anyway?” was Taggert’s reply when Robert told him of their quandary.

And Professor Palmer echoed him.

Without further delay the attacks were launched. Lacking any means of observation from above, the invaders were at a disadvantage in spite of their superior numbers. The separation of the spur from the main body was accomplished as planned, and a second body of Hakon’s men striking at the head of the column of invaders completed their rout, killing many and taking the rest prisoners.

The arms captured were a welcome acquisition; but the problem of feeding several thousand prisoners was a difficult one. Fortunately, a small food supply train was captured with them. They were immediately put to work throwing up earthworks and conveying heavy equipment about.

One general and several other high officers were captured. A special guard was set over these valuable hostages, for their possession would be of advantage to Hakon when terms came to be considered.

During the attack Robert had directed his forces by means of a simple smoke tube apparatus with which the *Sphere* was equipped, and by means of which long or short puffs of dense black smoke were ejected by manipulation of a plunger. The smoke was produced in a small oven chamber from ignited briquettes.

As the attack progressed perfectly, Robert did not utilize the terrible destructive force of the Norrensen Tube. Owing to the small reserve of petrol and consequently of electric energy depending upon it, he decided to withhold all knowledge of this weapon from their enemies until a particularly opportune time for its use presented itself. The enemy was not so considerate of him, however. Several times projectiles fired by their artillery screamed uncomfortably close past the *Sphere*. Fortunately these had no means of bursting, there being no explosives available on the planet.

The immediate effect of their successful attack upon the invaders was the latters’ withdrawal of their left encircling spur, and an evident preparation for a concentrated frontal attack.

Having looked to all important details of preparation to resist such an attack, Robert snatched a few hours of sleep. He lay down with his clothes on, exacting the promise that he would be awakened upon the first signs of

an actual attack. Pickets were posted so that the enemy could not approach closer without their knowledge. The *Sphere* had been brought down again and Robert and his companions rested in their quarters within the palace.

It was not till close upon dawn that Robert was aroused by Taggert's shaking him vigorously.

"Hate to do it, Bob," he was saying, as Robert sat up suddenly, "but those Wops are getting ready to raise hell again. Old Kharnov has sent a note of defiance, stating that as he will shortly be in possession of Svergad the temporary shutting off of the water supply doesn't matter."

Robert was on his feet in a moment and was striding through the doorway when he became aware of a hesitant restraining hand upon his arm. He turned to see Zola at his side. She crept into his arms.

"Darling!"

"Ah, my Robert, they would not let me come to you because Zola's love does not matter when everyone is so busy fighting. But I couldn't let you go away again without seeing you."

"Little ghost! You come and go like a dream. How did you get here at this time?"

"Taggert, he helped me. No one else knows but my maid."

"Good old Tag! But now I must hurry, dear, to help fight for you."

"Take me with you, Ro-bert."

"Not now, sweetheart. Stay here in safety till I return, and then you shall go back home with me if you wish,"

"And be with you always, my Robert?"

"Always!"

Here Taggert appeared suddenly, as from nowhere.

"Sorry to interrupt your tête-à-tête, people, but we must be getting on."

A hurried leave-taking followed. Zola placed in Robert's hands a small package, which he did not have time to examine then.

A thick haze hung over the dark landscape, which was just visible in the dim starlight. A tense quiet prevailed.

The *Sphere's* stall was readied quickly. All was well, and a few minutes later Robert and Taggert were rising rapidly in it above the sleeping city. The package given Robert by Zola was discovered to contain a small bottle of excellent wine and some *lana*, a nutritious cake, both of which were quite welcome, as he had not eaten anything for several hours.

Rising to a height of about four thousand feet, Robert allowed the *Sphere* to drift slowly over the enemy's lines. With the ports open, the sound of activity below came to them with startling clearness. It was plain that preparations were being made for an attack on the city.

Robert had reduced the speed of the gyrostats to a minimum in order that the invaders might not be aware of its proximity. Every light in the *Sphere* was extinguished. It hung above the busy Martians—a great, dark globe of unsuspected menace.

Suddenly a sharp command rang out below. Immediately there followed a series of flashes accompanied by a corresponding series of dull thuds—sounds of the curious Martian electric artillery in action. Robert's heart sank as he thought of Zola's danger.

With the first thuds of the artillery below and the answering fire from the metropolis, the entire plain seemed fairly to seethe with activity. Bugles were blown, and commands rang out above the other sounds.

Robert switched on the *Sphere's* giant searchlight. A blinding shaft of light shot down upon the broiling mass of humanity beneath. A great cry of consternation was wafted up to them.

“Not exactly elated at our presence,” remarked Taggert, dryly.

“We'll be about as welcome as a plague a few minutes from now,” remarked Robert.

Swinging the reflector about, he followed every movement of the army of invaders, pausing at intervals to flash instructions to the defenders by a prearranged code of signals corresponding to the smoke signals of the day. Already the enemy was swarming upon the small force of defenders in the outskirts of the metropolis.

Several projectiles sang past the *Sphere*, indicating that the invaders were not going to submit to its presence idly. A number of smaller searchlights were directed upon it from beneath, almost blinding Robert and Taggert as their eyes sought to pierce the darkness below. The invaders were not without defense.

Adopting a swift, staggered course, Robert managed partly to avoid the blinding beams of light. Instructing Taggert to continue these tactics, Robert prepared to use the *Sphere's* powerful weapon, the Norrensen Tube, against their enemies. He prayed only that it would not fail him now, and that their power would hold out long enough.

Calling out to Taggert to steady the *Sphere's* course, Robert carefully aimed the tube into the thickest of the invaders where Taggert played the searchlight's shaft back and forth. Robert pulled the lever releasing the lightning bolt!

A blinding flash, and a rending crash followed! The *Sphere* vibrated like a violin string. Only the goggles which they wore made it possible for Robert and Taggert to look upon the jagged pillar of incandescent flame which spattered upon the plain below with terrific force. So swift and brilliant was its course, that almost ere it had registered upon the retina of their eyes, it was gone, leaving them nearly powerless to see.

When they were again able to see and hear with fair distinctness, they became aware of a great hubbub below. The firing of the artillery had ceased and the attack of the invaders had turned into a rout. Hakon's men were driving them back like sheep!

"God!" ejaculated Taggert, for once jarred out of his habitual sangfroid.

Robert was silent.

A vast pit was visible where the bolt had struck, and the bodies of hundreds of men were strewn round it. The demoralization of that division for the present was complete. The destructive force of the bolt was appalling, but the spectacle it presented had been even more so. The invaders were terrified beyond control by panic fear of this leaping bolt of fire from above.

Robert had no desire to take lives unnecessarily: If the first bolt was sufficient to cause the invaders to retreat or to cease fighting, he determined not to release a second one. He studied closely the movements of the armies. As if afraid to antagonize the *Sphere* further, the invaders had ceased to play the beams of their searchlights upon it.

In the east the first faint flush of dawn was visible. Already the tall spires of the metropolis were touched with coral.

Robert became aware that the *Sphere* had settled too close to the ground for safety in case of further firing from the hostile artillery.

"Better raise her another thousand, Taggert," he called, turning to the latter, who was operating the machinery.

"Just what I was trying to do, old man, but she doesn't respond."

A swift examination proved Robert's fears—that their reserve of power had been virtually exhausted by the tube! The speed of the gyrostats was

perceptibly slackening. The *Sphere* was sinking!

Feverishly they turned to the engine which drove the dynamo. With this running, enough current soon would be generated to lift the *Sphere* out of danger. It was then that the extent of their calamity was discovered. The engine stubbornly refused to start, for one of those mysterious reasons to which engines are addicted. It persistently defied their combined efforts to diagnose the trouble.

In order to avoid crashing upon the plain, now less than a thousand feet below, Robert swung the disk upright and opened the shutters from its entire face. Even with its weakened power, unaided by the dying current, it would be almost sufficient to balance the *Sphere's* weight, as long as the gyrostats were turning slowly. Thus the crash might be averted, but at the same time it was impossible to utilize any side pull of the disk to carry them beyond the enemy's lines. They were facing capture—perhaps death!

Finding that each was in the other's way, Taggert, who had a knowledge of engines, continued to tinker with it while Robert looked on from a window, where he kept an eye also on the plain below.

The *Sphere* continued to settle slowly, though they knew not what moment the gyrostats might stop, allowing them to crash down upon the plain. With each hundred feet of descent their hope grew stronger; but that they would fall into the enemy's hands was now inevitable, for they were sinking into the very center of the camp.

Off to the south the sounds of fighting had ceased. Stretcher bearers were bringing in the wounded from that direction. Robert wondered dully how long it would be before he and Taggert would fall into the hands of the invaders.

Curiously enough, the Martians below were equally perturbed at the approach of the *Sphere*. Having witnessed its terrible destructive power, they felt absolutely at its mercy, and feared even to train a gun on it for fear that they would be wiped out immediately in retaliation. So far they did not suspect that the *Sphere* was not descending at the will of its occupants.

Realization of this occurred to Robert.

"Maybe we can bluff it through, Tag," he said, hopefully. "It's our only chance."

By this time the *Sphere* had settled to within a bare fifty yards of the ground. Its buoyancy was now virtually the same as the slightly denser

atmosphere at this height. It drifted slowly for several minutes, the gyrostats continuing to revolve softly.

The Martians had crowded round in a generous circle below. Mingled fear and curiosity were written upon their upturned faces, which could be seen quite clearly in the rapidly brightening daylight. A pompous officer pushed forward into the open space directly beneath the *Sphere*. Robert leaned out and addressed this individual.

“As emissaries of his excellency, the Governor of Svergad, I demand the immediate withdrawal of these forces,” Robert began, affecting a confidence which he was far from feeling.

“Fine. Go to it, boy!” encouraged Taggert in a hoarse whisper.

The officer did not seem surprized at Robert’s demand. But, unfortunately, in spite of his pompous manner, it appeared that he was not of sufficiently high rank to treat with so important a matter. Every moment was precious, for at any instant the *Sphere* might betray their weakness by settling to the ground.

The officer politely stated that their demand would be referred at once to the generalissimo. Realizing the importance of satisfactorily completing negotiations before the *Sphere* descended, Robert decided to take a bold stand.

“The generalissimo will not do. I must speak with the emperor quickly—here.”

As expected, the officer was visibly taken aback. Several valuable, fleeting seconds passed before he recovered from his surprize and indecision.

“Your request will be communicated to the emperor at once through the generalissimo,” he answered presently.

The minutes seemed like hours as Robert and Taggert anxiously awaited the reappearance of the official. Meanwhile the *Sphere* very slowly settled lower and lower. Taggert’s efforts to start the engine had availed nothing. They could but wait idly with what patience they could summon. To have attempted utilizing part of the disk’s power to carry them beyond the enemy’s lines would but have resulted in disaster, for their present buoyancy was maintained only with the aid of the disk’s power (directed absolutely perpendicularly).

A small group moving toward them heralding the probable approach of the emperor or generalissimo revived hope in their breasts. They might yet maintain their strategic position long enough to drive home a bargain that the Martians would honor when they were obliged to descend and attempt to bluff their way through a difficult situation somehow.

“The old boy himself,” ejaculated Taggert as the group drew near.

Robert recognized the emperor in a car pulling up below. Seated opposite him was the officer to whom they had first talked, while beside the emperor was a much decorated and bedecked individual who he hoped was the generalissimo himself.

The latter stepped out of the car after the officer, and assisted the emperor to alight. The three of them slowly paced beneath the *Sphere* as it drifted lightly with the soft morning breeze. The emperor was plainly angered and making a poor attempt to conceal the fact.

“What means this rude summons which his Highness, the emperor, has chosen to humor?” said the generalissimo; for that this was the generalissimo they quickly discovered. His voice carried up to them clearly. Barely thirty feet separated them now.

“I will speak directly with the emperor himself,” Robert began. “As emissary of his Highness, the Governor of Svergad, I demand the immediate withdrawal of all forces from this region. Unless such a movement is begun without delay, we are prepared to wipe out your entire army by means of the terrible weapon which you have seen.”

The emperor raised a countenance dark with passion. The generalissimo checked him with a nervous grip on his arm as he was about to speak, addressing him earnestly in an undertone.

The emperor calmed himself with evident effort.

“You have the advantage of us,” he said. “Therefore, my armies will retire at once; but I warn you that you shall be conquered in the end and be made to suffer for this.”

“Very well, your Highness. I suggest that you give orders immediately for their retirement as, unless it is commenced within a few minutes. I shall resume their destruction.”

Again the generalissimo was obliged to restrain the emperor’s burst of passion. A moment later, at a curt nod of assent, from the emperor, the

generalissimo dispatched the officer with a command to begin an immediate general retirement of all forces.

The first contingent was soon moving northward. The emperor and generalissimo had departed ominously for their headquarters. The *Sphere* continued to drift idly, settling ever closer to the ground. The enemy forces still stretched more than a mile between the *Sphere* and the city. How Robert and Taggert yearned for a stiffening of the breeze that they might be swept safely beyond the enemy's lines before the descending *Sphere* placed them in their power! Even now soldiers below paused in their preparations for retreat to gaze up at them curiously, albeit anxiously, evidently wondering why the *Sphere* hovered so low. Should the true reason be guessed, their temporary victory would be lost.

The next ten minutes seemed like an eternity to Robert. It had brought them only a little nearer the city and their friends, while scarcely thirty feet separated them from the enemy. Some of the troops were already marching off toward the south beneath them, but it was a physical impossibility for the thousands still between the *Sphere* and its goal to move out of the way in less than half an hour. And half an hour spelled almost certain disaster for the *Sphere*.

"Looks as if we'll have to indulge in a little more diplomacy," Robert remarked.

"Drop in upon his Royal Highness to bid him goo'bye, so to speak," said Taggert.

"Something like that."

At this moment the suspense was broken by a sudden lurch of the *Sphere*. The gyrostats had stopped!

With but the disk's diminished power to break its fall, the *Sphere* dropped into the sand with a jarring thud.

The Martians had carefully avoided the section beneath the *Sphere*. Consequently it fell clear. For several minutes they kept their distance. It was evident that they all feared this mysterious machine greatly.

Presently a pompous officer strode into the space about the *Sphere*. Clearly he saw in the situation great possibilities for his own future and he intended making the most of it. Yet, like the rest, he was not certain that the *Sphere* was as helpless as it seemed. Before he had given an order. Robert stepped out of the *Sphere*.

Robert interrupted the officer as he was about to speak.

“I come to give warning that unless the retirement of all the emperor’s forces is accomplished more rapidly, I will resume their destruction,” he said, imperiously.

The Martian was visibly undecided. Perhaps he and his regiment would be wiped out if he attempted to hold the *Sphere*. He observed Taggart glaring at him balefully from a window. Nevertheless he was a brave man, especially when he thought of the brilliant future he might enjoy if he got away with this. A disagreeable smile disfigured his sharp-featured countenance.

“I think the emperor will soon have changed his mind about the retirement—just as soon as he knows that you have honored him by landing within his lines.”

“The emperor is too wise a man to delay the withdrawal of these troops,” answered Robert, “for even at this moment, should I choose, every man in sight would be blasted into atoms at a signal from me!”

The cocky officer’s assurance again was visibly shaken. Neither knowing that the death-dealing Norrensen shaft was rendered useless on account of its outlet being buried in the sand, nor being certain that the *Sphere*’s power was exhausted, he must have feared instant annihilation if he persisted. A cunning light came into his eyes suddenly, and was as quickly veiled. Affecting a friendly attitude he strode over to within a couple feet of Robert. To have backed away from him would but have advertised their weakness, and any retreat from the circle about them was impossible. So he stood his ground and waited the officer’s next move.

As if about to take Robert into his confidence, he placed his hand upon Robert’s shoulder. Then like a flash he rapped out a command, at the same instant covering Robert with his *lahan*, the Martian electric automatic.

Immediately the ring of Martians closed in upon the *Sphere*, while the officer’s two aids hurried to his help. Counting on his immunity from a possible blast from the *Sphere* while in close proximity to Robert, the crafty officer had dared to defy the *Sphere*. But he had not counted upon his prisoner’s hot-headedness.

Even as he whipped out the *lahan*, Robert started a dashing uppercut to his head. Before the surprised officer could fire or dodge, the blow caught him on the point of his jaw with a beautiful precision. He staggered and fell

just before his anxious aids could reach him. Notwithstanding the gravity of the situation, Robert could not repress a laugh over the comical appearance of the little, pointed red beard on his late opponent's chin, which was tilted serenely skyward, as he sprawled on his pompous back.

A few minutes later Robert was securely bound and led roughly away. His last glimpse of the *Sphere* showed him a ring of shouting Martians round it, but no sign of Taggert.

Robert was taken to a large tent near by, where, without ceremony, he was securely fastened to the large center pole by means of chain and anklet like a criminal. Two armed guards were set to watch him, after having searched him and taken away the automatic with which he had been armed when he left the *Sphere*.

At noon a plate of coarse food was brought to him, and a cup of water. He found the food palatable and, being hungry despite the gloomy outlook, ate all of it. His guards stared at him intently all the while, evidently regarding the eating of an Earthbeing as a great curiosity.

The chain, a stout-linked affair which would have weighed heavily had he been on the Earth, allowed Robert little freedom of movement. The wooden stool which he had been furnished affording no rest for his back, soon grew dreadfully tiresome. He tried every position he could conceive for relief, each of which his guards regarded with fresh suspicion. Once he tried to engage them in conversation, but he succeeded only in drawing grunts and uneasy stares from them.

Sounds of activities without drifted to him. There seemed to be considerable movement of troops past the tent, all in one direction. Although he was not certain in which direction Svergad lay, Robert knew that they were in all probability moving back to the attack, now that the menace of the *Sphere* was removed. He wondered dully what had happened to Taggert.

He was just shifting his position for about the fifty-first time when the flap of the tent was raised and the pompous officer of the morning's encounter stepped in, followed by two fellow officers. The guards came to attention stiffly.

The pompous officer scowled ominously. Plainly he intended dealing with Robert as severely as possible. He strode over to him.

"Well, how does it feel to be in chains, my vicious friend?" he rasped.

"Quite all right, you old toad," Robert answered pleasantly in English.

"What?"

“I said, ‘Quite comfortable, old chap,’” said Robert, this time in the Martian tongue, interestedly surveying a discolored lump on his captor’s right jaw where he had landed his fist that morning.

The officer’s face went dark with passion. He raised his cane as if to strike Robert. Before he could do so, however, Robert jerked up his stool to protect himself. Simultaneously the two guards sprang forward to protect their superior. But the latter’s courage failed him at the menace of the stool and he let his hand with the cane fall to his side. He contented himself for the time with a murderous look at Robert, who put the chair down, but remained standing for greater freedom of action. The guards fell back, visibly relieved.

“You shall be handled later, sir,” barked the officer. “Meanwhile, do you know the fate in store for your friends?”

“The privilege of beating hell out of your rotten army,” Robert shot back at him. He could have sworn that he saw a brief flicker of amusement in one of the other officer’s eyes as he said this.

Once more the angry officer seemed on the verge of striking Robert, but his eyes fastened upon the stool and he changed his mind. He controlled himself with an effort.

“Another day and you will change your song,” he managed to sputter. “Until then *you* will have the ‘privilege’ of reflecting here in comfort, while a suitable punishment is devised to repay you for your destruction of our men!”

With that he turned and stamped out of the tent, followed by the other officers.

The two guards viewed Robert with evident awe after this interview. Clearly they were unused to hearing their high officers abused so roundly and carelessly. Perhaps they were saying to themselves something to the effect that this Earth-being must be *some* guy.

Their one-cylinder brains were further startled here by Robert’s tumbling the dreaded stool over, preparatory to reclining upon the sandy floor. They exchanged glances uneasily. One of them started as if to object, then seemed to think better of it, relapsing again into a stupid dumbness.

The afternoon sun was well down in the sky when the curious sound of distant firing of the Martians’ *hilwai* came to Robert’s ears—the first he had heard since his capture. His heart sank as he considered the probable result

of the bitter struggle which it heralded. If only that confounded engine had not failed them at the critical moment!

The sounds of battle continued until after sunset, when his guards were relieved by two others whose intelligence and curiosity appeared to be about on a par with those of the first two. One of them possessed a rather remarkable, rudderlike nose which fascinated and diverted Robert for a time, to its owner's evident discomfiture.

Somehow Robert managed to pass the night in troubled dozing on his hard bed of sand, tormented by anxious thoughts of Zola. He welcomed the first flush of dawn with thanksgiving, despite his weariness. The sound of firing was resumed about this time.

Shortly after his breakfast Robert was visited by Kharnov.

The ruler's crafty features were disfigured with a scowl. He entered the tent alone, commanding his escort to remain outside.

"You have returned hostility for hospitality, and the penalty shall be death," he said.

His tones fell with deadly precision, like the pronouncement of a sentence by a criminal judge. An involuntary chill swept over Robert in spite of his effort to appear unmoved.

"I was but protecting a helpless girl from your unwelcome attentions," he replied, ashamed that he could control the nervousness in his voice only by a great effort. "That you have chosen to pursue her with an army to be resisted is your own folly."

"I am not here to discuss fine points with you, sir," returned the emperor sharply, "but to offer you a means of saving your life. You shifted your loyalty once; you can do it again. Aid my army with the *Sphere* and you may return to your planet in peace. Refuse and suffer the consequences!"

"My loyalty was and is only to the right," retorted Robert. "Your proposition doesn't interest me."

He almost forgot his personal fear as he realized with a thrill that Kharnov's latest attack must have been successfully repulsed or he would not have made this offer.

"Beware! I am offering you your life."

"I don't care to buy it at that price."

The emperor was patently more perplexed than angered by Robert's rejection of his terms. His code did not permit of an understanding of anyone's refusal of life for principle. He was concerned merely in deciding just what concession it was necessary to hold out to Robert in order to secure his acquiescence. A crafty thought occurred to him.

"You will also be given as many of the diamonds, so highly valued on your planet, as you can take with you. Does that interest you? Of course, I should require that two or three of my men accompany you while you were operating against the government's enemies."

An inspiration came to Robert. Once he got the *Sphere* in midair again—three or a dozen Martians—he might frighten them into submission and land within the city. It was worth trying anyway, and, if it failed, they could but take his life, which it seemed certain he would lose anyway if he flatly refused the emperor's demand.

"You have the advantage of me," he said finally. "I accept your terms. But my friend must accompany me to assist in the operation of the *Sphere*."

"One of my men will assist you after you have instructed him. Your friend must remain a hostage till you have carried out your part of the contract. No harm shall come to him unless you fail."

This put the matter in a different light. Here was an obstacle that would require real diplomacy to hurdle.

"I fear your man can not learn quickly enough, but I'll try to teach him," said Robert, hoping to find some solution later. "However, the *Sphere* can not be operated until I have discovered and corrected some defect in its machinery."

"Ah, so that explains your strange landing."

"Exactly."

Kharnov pondered a moment.

"Very well," he resumed. "I shall select the men to accompany you and return later. Meanwhile I will see that you are made more comfortable here."

A sinister smile played over his face as he turned to go, giving Robert a feeling of unknown, impending evil.

True to Kharnov's promise, however, the tent promptly was equipped with various comforts, including a couch, table and chair, and other welcome additions. Good food and excellent wines also were brought him.

The soft couch and soothing wine, after his night of sleeplessness, produced a pleasant drowsiness which Robert found difficult to resist. He slept, and it was dusk when he awoke.

The distant sounds of battle had ceased again. Dinner was brought to him and two hours passed without further word from Kharnov. Once more drowsiness overtook Robert and he dozed. His two guards, who had so far watched him with eyes of hawks, relaxed their vigilance somewhat. They envied him his privilege of sleeping.

“Brother,” said Rudderbeak, finally, “it is not necessary that two of us watch over this sleeper. Let us decide by chance which of us may rest also.”

“Words of wisdom, indeed. Let us do so by all means.”

Fate decided against Rudderbeak. The pebble which he cast at the line drawn on the sand in the tent rolled a trifle too far. So, grumblingly, he took up the long night watch while his companion stretched out upon the floor with a sigh of contentment. He was to be warned in ample time at the approach of anyone.

Here fate again took a hand. Rudderbeak found standing too great an effort with two peacefully slumbering beings before him. He felt that he was at least entitled to the luxury of sitting down. So he availed himself of the comfortable chair so lately brought in for their captive.

The chair creaked as he settled his weight in it. Had it not been for this, another sound just outside the tent might have attracted his attention. Even as he stretched his long legs in extravagant comfort a shadow without flattened itself upon the ground.

A little while after that Robert woke abruptly to see a face within a few inches of his. The lantern suspended in the tent top cast weird shadows around the enclosure, giving the features of the countenance above his a horrible aspect. An involuntary cry arose in his throat, but was suddenly checked. For there was a strange familiarity about the glint of red in the intruder’s hair! The shadow-blurred features quickly lost their ferocity. It was Taggert.

Their hands met in a clasp of glad reunion. No word passed between them as Taggert proceeded to examine the length of chain with which Robert was fastened. He shook his head as he saw the heavy links.

Robert saw his mouth straighten into a firm, hard line as he stared at the sleeping guards. He removed the gun from the side of the sitting guard.

Then, drawing an object from his pocket, he walked over to where Rudderbeak lay. He hesitated.

“Got to be done,” he muttered. Raising his arm he brought the object down upon the slumbering guard’s head forcibly. The Martian slumped deeper into his chair. His gun slipped to the sand. Stooping quickly, Taggert picked up the fallen gun, retaining it and handing Robert his automatic. The other guard, stretched upon the ground, had not stirred yet.

Deftly Taggert searched the stunned guard. But the key he was looking for was not in the Martian’s possession. He turned his attention to the other guard, who still slumbered. As Taggert meditated, the sleeping guard stirred uneasily. In spite of the reporter’s efforts to avoid noise the disturbance had evidently been sufficient to break into the sleeping guard’s subconscious mind. He opened his eyes suddenly, looking directly at Taggert.

Before the Martian could, gather his senses, however, Taggert pressed the *lahan* against his chest.

“Not a sound!” he commanded in the Martian tongue. “Now give me the key to this anklet—quick.”

The Martian was wide awake enough to realize his helplessness. He promptly pulled the desired key out of his pocket and handed it to Taggert.

Robert covered the Martian with the automatic while Taggert fitted the key to the anklet. A moment later he was free.

They were now faced by the problem of binding and gagging the conscious Martian. No rope was available within the tent.

“His shirt,” suggested Robert, following his inspiration by pulling the garment off the now completely cowed Martian.

Swiftly they bound, gagged and blindfolded him, the twisted arms of the shirt making fair substitutes for cord, though there was not enough to bind him as thoroughly as they wished. They locked the anklet upon him after shortening the chain by twisting and knotting it round the big center-pole. The unconscious Martian they left as he was.

Warning the bound Martian with dire threats, they slipped out of the tent into the night.

Had Phobus been shining then they would surely have been seen at once; but the little Martian moon had set an hour past. Creeping painfully past the clustered tents they came at last to the edge of the camp, which, fortunately, was not a great distance away.

Here it was necessary for them to pass the pickets. The brilliant starlight and level, open desert made their escape extremely difficult. How far apart the pickets were stationed they did not know, but one paced slowly across a stretch just ahead of where they lay partly concealed in a slight depression or wave in the desert's floor.

Cautiously they wormed forward to another shallow depression while the near-by sentry's back was turned. Here they waited anxiously as he paced back and again turned away. One thing in their favor for the present was that the sentry directed his attention chiefly in the opposite direction, toward the city. That would become their disadvantage, however, when once they succeeded in getting past the sentries and between them and the city.

No sooner had the sentry turned his back again than they were scrambling feverishly toward a distant, faint strip of shadow, which indicated their next scanty haven of temporary safety. Their arms and knees were weary to the point of exhaustion; but they pressed on desperately. Still the little line of shadow ahead seemed far away. Would they make it before the sentry turned and discovered them? Surely he would notice the track where they had pawed their way through the loose sand.

How they finally reached their goal neither could remember. It seemed that they had crawled and crawled for eternity—a sort of dreadful nightmare in which their limbs moved unwillingly while they remained in the same spot. Both were exhausted when they slid stiffly into the scanty haven of the little dip in the sand. For the moment they cared not whether they were captured or not. They longed only to lie panting till their parched throats had cooled.

When they dared peep at the picket he had started serenely back on his walk away from them again. He had noticed neither them nor their tracks! With revived courage and strength they resumed the grilling struggle toward safety. Once more they reached a welcoming shadow without discovery.

“Boy, howdy!” gasped Taggert. “When do we—quit this caterpillar glide?”

“If our luck—holds, we’ll—soon be beyond—the danger zone,” puffed Robert, resting on the flat of his back.

Taggert's escape and the fate of the *Sphere* were still puzzling Robert, as they had not yet had an opportunity to mention these things. At Robert's query now Taggert enlightened him briefly.

"When I saw them overpower you and lead you away," he recounted, "I realized that I could help most by holding the *Sphere*, and coming to your assistance later if I succeeded. So I drew the manhole trap to and waited for developments. Peeping cautiously, I was fortunate in being able to spot the tent they took you to.

"Well, they howled round the outside for a while, but made no attempt to break in. I suppose they feared to tamper with the *Sphere* after its exhibition of its destructive powers. Finally they withdrew at a command from an officer. Still I was careful not to show myself. I tinkered with the engine some more, quietly, but without results.

"Finally darkness came on. The Martians were camped about the *Sphere* but all seemed anxious to give it a wide berth, for the nearest were a hundred paces or more away. They were scattered pretty well, so I took a long chance and slipped out into the darkness, snapping the trap-door shut. Luck was with me. You know the rest."

"But man, you were banking on a chance in a thousand of getting through!"

"Oh, chances. Like kisses, they are to be taken."

They both laughed a little at this, and it lightened their spirits. The situation did not seem so hopeless after all.

"Then the *Sphere* is still unharmed," mused Robert.

"Not only unharmed but not far from us right now. Look over there."

Robert looked off to their right as indicated by Taggert. A familiar dark shape reared itself above the level of the desert, probably a quarter of a mile distant. He breathed a sigh of relief. Though useless to them just then it relieved him to know that the Martians had not wrecked it.

The still glare of the stars piercing the narrow Martian belt of clear atmosphere shone coldly upon them.

Robert wondered which of them was the good old Earth. Ah, to be safely back there again!—the professor, Taggert—and Zola. Would they all see each other again? Could he expect Zola to give up her father and other kin, her friends—everything she had learned to love in her own world? She had said she would go with him. Would her decision withstand the final parting

from pleading friends and kinsmen, even if her father would let her go? Would she be happy on his Earth, if he succeeded in rescuing her from her present peril? Could he honestly urge her to leave with him?

These questions raced through his mind as his strength surged back to him. There was but one answer: if they could get away he would take her with him if he could possibly get her to go willingly. That he could make her happy he felt certain, for he would devote his entire life to doing so.

He became aware of Taggert's prodding him vigorously in the ribs.

"'Can' the star-gazing, old dear. It's time we were wriggling merrily on our way," he whispered.

They bestirred themselves reluctantly from the latent warmth of the sand. Waiting until the picket was once more on his way from them, they staggered to their feet and made a dash of it.

When they had traversed perhaps three quarters of a mile they slackened their pace to a walk. About two miles ahead of them lay the city's outskirts.

Approaching the lines of the city's defense they faced a danger similar to that from which they had just escaped. They might be mistaken for foes and fired upon when discovered.

Discovery came sooner than they expected. A Svergadian picket popped up suddenly from a pit dug in the sand and challenged them sharply, training his gun the while upon Robert, who was in the lead.

Robert explained who they were and asked permission to go to Hakon. The sentry seemed dubious but finally summoned an officer, who heard their story and promptly promised to have them escorted into Hakon's presence.

Their escort was dismissed upon their arrival at Hakon's quarters, by his aid, who recognized Robert and Taggert at once.

"The governor will see you at once, I'm sure," he told them. They waited in a small, sumptuously furnished anteroom, while he went to summon Hakon.

A few minutes later Hakon stepped into the room. He wore a long, loose garment which he had thrown over his sleeping clothes. His face was pitifully haggard. He seemed to have aged terribly since they had seen him last. A faint, sad smile softened his features as his eyes fell upon them.

"Ah, my young friends, I am glad to see you safely back and out of the enemy's clutches. First, I want to thank you with my whole heart for the timely aid of the *Sphere*, which really turned the tide of the battle in our favor at a most critical time. Now I am grieved to hear that, through no fault of yours, it has fallen into the hands of the enemy. But, I, too, have a sorrow. My beloved daughter has been tricked into captivity."

"A captive!" gasped Robert, clutching at his heart. His Zola a prisoner of that ogre, Kharnov!

"Here is an ultimatum just received from the beast," resumed Hakon, dully, handing Robert a folded document.

Robert unfolded the paper.

"Your daughter will come to no harm provided you capitulate by noon tomorrow," it read in effect.

"The dog!" cried Robert. "How did he contrive to get her into his possession?"

"This was found in her room," said Hakon, handing him a small crumpled piece of paper.

To his surprise it had his name at the bottom, although it was written in Martian. Zola did not understand his own tongue. On the paper was written: "I am a prisoner. Unless you come to intercede for me with the emperor, I shall be put to death at dawn. Robert."

Surely love is blind! Else she would have known that he could not have written such a wantonly selfish plea.

“This is a false message, of course,” said Robert, wondering whether Hakon believed him the author of such a note.

“I know it, my boy.”

“We must strike quickly,” said Robert. “Let me lead an attack at once. Nothing short of a complete routing of their entire army will satisfy me.”

Hakon smiled at his fierce enthusiasm.

“Even now an attack on a tremendous scale is organized,” he said. “It will take place just before dawn at a signal to be given all divisions simultaneously. You may direct the center.”

“Say, where do I come in?” broke in Taggert.

“Come along with me, old man. We’ll scrap together.”

“Fair enough. We’ll knock his Nib’s royal block off.”

Sleep that night was out of the question. As dawn approached, Robert absorbed the details of the gigantic offensive at a conference with several of the leaders. Everyone was on the alert. The governor’s daughter was a great favorite and greatly admired for her character and beauty. Every man knew of her danger and, with such an incentive to success they would be all but invincible.

When finally the signal for the general advance came over the wires, the big army started forward in three giant columns, cautiously at first until their attack should have been observed. The absence of aircraft—because of the thin Martian atmosphere which made them impracticable—made it possible for them to make considerable progress before the enemy was aware of their supposed victims’ offensive.

It was agreed that upon the discovery of the advance of any column, that column would at once fire a rocket as the signal for a general rush upon the enemy from all points.

Robert’s column had advanced unobserved well over half-way to the enemy’s lines. Suddenly a rocket flashed heavenward from the right column. His men needed not Robert’s shouted command as they dashed madly after him toward the startled enemy less than a mile ahead.

With the rapid pace made possible on Mars by the weak gravity, they covered the intervening distance in about two minutes.

The dull drumming of the enemy's awakened fire was punctuated by occasional thuds near by as their bullets found marks in the onrushing column.

The Svergadians met a brief check as they encountered the outpost. Then onward they swept like a great wave upon the dismayed besiegers.

The growing flush in the east bathed the battlefield in pale rose, touched here and there with purple dusk. Overhead the fading stars twinkled faintly as if shrinking from this scene of wicked strife.

In spite of their surprise the enemy rallied to a stiff defense. They had the advantage of greatly superior numbers, and knew it. And Robert's column had charged directly into the main body of their forces.

The morale of few armies, however, could have withstood the fanaticism with which the Svergadians charged that morning. Their long, slender bayonets flashed viciously as they plunged forward fearlessly again and again. Every man was fighting to avenge his princess, whom the leader of these men had abducted. The enemy was dismayed. They were given no time to rally. Surely these were fiends who attacked them with no care for their own lives! Dismay became consternation—rout!

In vain did the enemy's officers struggle to stem the mad retreat. The front ranks turned in panic from the vicious line of steel, and stampeded over the troops supporting them.

Side by side fought Robert and Taggert. Long since, they had emptied their automatics. Armed with bayonet guns picked up from fallen Martians, they charged blindly with the rest. That they both continued unwounded in the foremost ranks was a miracle.

But the goddess of chance is a capricious deity. She selected a moment of comparative safety to strike. It came as Robert and Taggert were vigorously following up the rout.

"Keep them on the run, men," shouted Robert, turning and setting the example.

As he sprang forward, a thud at his right caused him to turn sharply. He was just in time to catch Taggert as he swayed and pitched forward!

"Where is it, Tag, old man?" he sobbed.

A crooked smile struggled to the reporter's livid lips. He fumbled at the right side of his breast. A fleck of bright-colored foam showed on his lips as they moved feebly. Robert stooped close to listen.

“Running like—hell, ain’t they—Bob?”

“Like hell,” Robert assured him, choking.

Feverishly he ripped open Taggert’s shirt to reach the wound, but the latter restrained him with his last remnant of strength.

“No use, Bob. I’m—done. Please listen—closer. That’s better. Take papers—inside coat pocket—send them—*The Chronicle*—if you get back. Let money go—to Mother. Picture here. Tell her—and Sarah—good-bye.”

His body went limp. The last word was barely audible. His gallant spirit had flown.

Robert let Taggert’s body down reverently. Poor, happy-go-lucky fellow! Three weeks ago he had been a stranger, a stowaway, an outsider prying into their affairs. Now he seemed like a lifelong acquaintance—a brother!

The swift tide of battle had swept on ahead. Near by a large, officers’ tent reared high its peak. Strangely it had survived the fierce struggle, which, but a few minutes before, had raged round it. To this tent Robert carried Taggert’s body, and placed it softly upon a cot inside. Choking back a lump in his throat, he drew a cover up over the cot and turned away. A bright blue sash caught his eye—one of the rare, brilliant-hued bits of apparel which only the most well-to-do Martians can afford because of the scarcity of minerals for dyes. This he tied conspicuously on the outside of the tent to identify it.

With these precautions for later recovery of Taggert’s body, Robert dashed on after the receding line of battle. So hot was the chase and so overwhelming the enemy’s rout, that he had difficulty in gaining the front again.

Once more in the front rank he fought furiously, for to his original grievance was now added that of the death of a pal. The resistance of the enemy’s center was completely broken. Its officers no longer had any control over it. Whole companies surrendered rather than be slaughtered.

Suddenly, however, the headlong retreat of the enemy was checked. Those in the rear still scattered in consternation, or abjectly surrendered, but ahead there was a confusion and congestion—some obstacle against which the retreat floundered, swirled—and finally rallied.

Once more Robert found himself in the thick of the fray. Somehow, unaccountably, the enemy’s retreat had been halted. Those in contact with

what had been the rear of the retreat, were now actually on the aggressive, fighting like rats, with their backs to a wall.

Goaded by the thought of Zola's danger, Robert fought furiously. His gun he had discarded in lieu of a saber, which he now wielded with terrible destruction. His strength, superior to that of the slightly smaller-statured Martians, was augmented by his passion to destroy, to kill, until he should reach the very heart of this resistance which was keeping him from *her*. His very fierceness was a protection, his whirlwind attack striking terror into the hearts of the opposing Martians near him. His followers, too, inspired by his example, fought with great vigor. Like the head of a wedge they hewed their way steadily into the enemy's ranks.

Once more their opponents were routed. Like chaff they were swept back, leaving but the core of their temporary resistance—a small knot of picked men round whom they rallied briefly though bravely.

Against this group Robert charged with his followers.

A terrific struggle ensued. By their uniforms Robert knew the stubborn group to be the emperor's picked guardsmen. His heart leaped with fierce exultation as he realized that he was probably about to face the crafty, deceitful ruler.

It was at this juncture that Kharnov himself appeared suddenly from out a sumptuous tent!

In a belated effort he attempted to rally his remaining guards in a futile counter attack. By an almost superhuman effort Robert fought his way through the ring of defenders to the false emperor. A blaze of intense hatred leaped into the latter's eyes as he recognized his former captive. Eagerly he sprang forward to meet him.

Because of his unusual stature among his own people, Kharnov was an equal match for Robert. In skill, with the saber, he was Robert's superior. Both men were fired by an intense passion, but Robert's wrath had the advantage of a righteous cause.

Warily they circled each other, the tempered blades clashing as they parried each other's vicious thrusts. In their furious aggressiveness, both men were quickly wounded, though lightly, several times. Their shuffling feet thrashed the loose sand into spume as they circled each other swiftly. Their breath whistled hoarsely in their throats as they lurched back and forth, each eager to cut the other down.

As their first fierce vigor became dulled somewhat by their terrific exertions, both men settled down to deliberate, crafty fighting, each keenly watching for an opening which might prove fatal to the other.

Robert was handicapped by his unfamiliarity with the saber. Fortunately he was a skilled fencer. Only his skill with the foils enabled him to parry the slashing attacks of his opponent.

His eluding of Kharnov's furious attacks only enraged the latter the more. He was overtaxing his endurance. It began to dawn upon Robert that, if he could successfully withstand Kharnov's attacks a little longer, he would soon have a big advantage over the berserk ruler.

His opportunity came unexpectedly. Missing his footing, he all but lost his balance. Quick to take advantage of his misfortune, Kharnov launched a terrific blow at him. Had it struck home it must surely have rent Robert in twain! Fortunately, however, it missed him—but narrowly. The tip of the razorlike blade whined through the air but a fraction of an inch from Robert's throat.

The delivery of this terrific swing momentarily unbalanced Kharnov. Recovering at this moment, Robert lunged swiftly. Precipitated upon the blade partly by the unchecked force of his wild swing, Kharnov was mortally wounded, the broad blade piercing his breast deeply. With a dreadful oath he dropped to the sand.

Robert quickly knelt at the emperor's side to render assistance. But it was clear to him at the first glance that the unfortunate man was done.

With a supreme effort Kharnov raised himself upon his elbow. His lips twitched as he tried to speak. Then suddenly he slumped back lifeless upon the sand.

Having convinced himself that the hapless man was beyond need of aid, Robert looked to his own wounds. The battle line had swept far ahead during their private struggle. He found himself unmolested and unaided.

His cheek was bleeding profusely, but the wound proved to be superficial. The cloth of his coat, too, was wet from a wound in his arm which pained him cruelly. An attempt to remove this garment resulted in a twinge of pain almost unbearable. He swayed faintly, gritting his teeth in an effort to retain his senses.

Once more he tried to take off his coat. His head swam. Then everything went black before his eyes.

How long he lay unconscious Robert could not know. When he opened his eyes again, the din of battle had ceased.

He found himself extremely weak, and it was with difficulty that he got to his feet. For several minutes he swayed uncertainly, his knees all but refusing to hold him up. The rare Martian atmosphere seemed like a vacuum. His senses reeled drunkenly.

An ornamented flask, hung at his late adversary's belt, caught his eye. Uncertainly he stooped and unfastened it. His parched throat seemed afire as he twisted at the flask's cap with clumsy fingers. As the cap dropped to the sand, he raised the neck to his hot lips and drank.

The fluid in the flask all but choked him as he gulped it down. He recognized it as *gao*, a vicious wine distilled from the peculiar sea-weed growth of the marshy regions. Yet the craving to drink was so strong that he absorbed a generous portion of it before putting down the flask.

So potent was the wine that he instantly felt invigorated. His nausea was gone. He seemed to have taken a new lease upon life.

Discovering that his arm was bleeding profusely, he once more attempted the removal of his coat, this time with success. He ripped his left shirt sleeve to the shoulder. There was a deep gash above his elbow. Deftly he twisted a torn strip of the sleeve below the wound, thereby checking the flow, and bandaged the cut as well as he could. This done, he looked about him curiously.

He was apparently alone upon the battlefield, over which a deathlike quiet, now prevailed. Here and there a tent stood, while the rest were collapsed upon the ground where they had been knocked down during the fierce hand-to-hand struggle that had surged round them. So precipitate had been the retreat of the late emperor's army that it had not had time to strike its tents in the rear before being swept far back of its original position.

Out on the tent-dotted plain ahead of him Robert suddenly observed another living being. This person, who was perhaps a quarter of a mile distant, was progressing in his direction by a series of curious dashes from

one standing tent to another. The sun shone in Robert's eyes, making it difficult for him to see plainly.

As he squinted painfully, he made out a second figure, beyond the first, seemingly in pursuit and gaining. Determined upon solving this latest riddle, Robert tottered off to intercept the fugitive. His strength seemed to return to him with the use of his legs, for his knees soon steadied.

As they neared each other the fugitive flitted suddenly behind a tent and remained there. Evidently Robert's presence had been discovered and had startled him. The pursuer came on swiftly, running directly toward the point where his quarry had hidden. He brandished some object resembling a long whip. From time to time he shouted something unintelligible, which Robert could hear but faintly.

His curiosity now thoroughly aroused, Robert himself moved behind a near-by tent in order that he might observe without being observed.

The pursuer came on quickly, but finally stopped as if confused by his quarry's disappearance. It was plain that he did not know which tent the other had taken refuge behind.

At this moment Robert saw the fugitive dart out into the open again and make directly toward the tent behind which he himself stood. Simultaneously he heard a shout from the other, who in another moment appeared in pursuit. It was at this juncture that Robert suddenly perceived that the fugitive was a woman!

Before he could think further the fugitive dashed round another tent and past his. She halted uncertainly, then, unaware of his presence, darted quickly to his side. Her hand, as she flattened herself quivering against the wall of the tent, touched his. Quick as a flash she wheeled upon him.

Even as her involuntary cry of fear rang out, she recognized him. It was Zola!

The next instant her pursuer panted round the tent's edge, his evil face distorted with passion.

"You devil," he hissed, raising a murderous-looking whip to strike her. His hand dropped abruptly as he espied Robert.

Before the Martian could recover from his astonishment, Robert swung at his jaw with every ounce of his remaining strength. His fist landed with crushing force, driving the Martian's suddenly inert body before it like that

of a stuffed dummy. Several yards away it plowed through the sand and lay still. Weeping softly, Zola crept into his arms.

“Oh, my Ro-bert,” she faltered, “I thought—I should—never—see you again.”

As for Robert, he could but hold her hungrily to him.

“You will never—never—let them keep me—from you again?”

“Never, dearest!”

She sighed contentedly. Her arms crept round his neck. She raised her face to his. Her soft, tremulous lips met his tenderly. Then she caught sight of the reddening bandage upon his arm. Instantly all her mothering instinct was alive. She wriggled free from his embrace like a sinuous kitten.

“You have been wounded!”

Robert chuckled.

“But not nearly so badly as our friend, the late emperor,” he said.

“He is——?”

“Dead. He had the misfortune to stop the sharp end of my blade during the skirmish between us.”

He said nothing of poor Taggert’s death. She would be sadly grieved over the jolly little reporter’s brave finish. Time enough when she questioned him, or noticed Taggert’s continued absence.

“You fought him for me, my brave! Ah, I should have died had you fallen!”

He took her in his arms again, while only the sun looked on.

“It was well that he died!” she broke out fiercely, without warning. “He lured me from out of our lines with a forged message which I thought you had written, Ro-bert. I was bound, taken to his tent, and a guard set over me. He made love to me, and laughed when I slapped his face. Then he left, but said he would soon tame me. When the battle turned against him, and his army was finally put to rout, his two guards set upon me. One, I stabbed to death, but *this*” (pointing to the Martian in the sand) “wrested the blade from me. I fled and he pursued me here.”

She shuddered, but went on with her self-appointed task of improving upon Robert’s indifferent bandaging.

“Little dove,” said Robert, irrelevantly—and was fittingly rewarded.

Hand in hand they turned toward the city. Before them lay several miles of sand, much of which was strewn with the bodies of those unfortunates

who had fallen in battle. Wearily they picked their way, sadness in their hearts over the thought of the many brave hearts stilled, but, withal, a feeling of wonderful peace in having found each other safe.

An atmosphere of utter desolation hung all about them like an envelope of dread. Here and there a poor, wounded warrior raised himself to call for water or plead for aid. They gathered up some of the full canteens from those fallen, and from them gave relief to a number of thirst-tortured ones on their way. Where she could be of assistance, Zola stopped to apply a bandage, to bathe a fevered brow, to speak a word of comfort, and Robert assisted her.

Thus they had traversed nearly a mile when a commotion off to the southeast attracted their attention. A column of soldiers was marching toward them. The flag of the metropolis waving at its head dispelled their first fears. Robert guessed that it was a part of the right column returning with prisoners from its victorious attack. This conjecture subsequently proved correct.

The column soon overtook them. Room in an officers' conveyance was quickly and eagerly made for them by its solicitous occupants, and they rolled back through the city gates in short order.

The approach of the victors had already been observed, and the great city fairly bristled with gay flags. Welcoming, hysterical throngs greeted them at the gates with deafening cheers and shouting.

Among the first to greet them were Hakon and Professor Palmer. They arrived astride a pair of fine *lunas*, as the victorious troops were entering the gates. These animals, which resemble our horses, were of a small, precious herd owned by Hakon, they being quite rare.

With a glad cry Zola leaped out of the conveyance and ran toward her father. Likewise he jumped to the ground and ran with pathetic eagerness to meet her. They embraced each other while the soldiers and public looked on in sympathetic understanding. Scarcely one of them but had heard of their princess' latest abduction with sorrow almost as keen as her father's, for she was dearly beloved by all.

A lump rose in Robert's throat as he noted their touching reunion. Could he—should he—pluck this Martian flower from her kin and country, to take her back with him to a strange world? Once more he wondered whether he could hope to make her happy—to make her forget. He feared not, and his

heart was heavy with the realization that he must give her up—that an attempt to transplant her would prove a sad failure. Rather than see her pine away in a strange world he would go back alone, even though it broke his own heart. He turned to greet Professor Palmer as that big-hearted and lovable man swung down off his steed and rushed upon him with open arms. Truly, here was such a friend and companion as few men ever were blessed with.

“Lad, lad, it is good to see you again!” And he hugged Robert with such fervor that it was with difficulty he managed to return his greetings and assure him that he was, indeed, quite safe, and delighted to be back.

Hakon was even more effusive in his greeting. Leading his daughter by the hand, he approached Robert almost humbly, albeit with an unconscious dignity—the inevitable bearing of one born to rule.

“You have brought back my greatest treasure to me, sir; and victory, too. Nothing you may ask of me is too great a reward.”

“I but shared in the glorious triumph of your brave men, sire,” Robert answered.

The ruler made an imperious gesture born of habit.

“Enough, sir! We owe everything to you. Ask anything you will, and it is yours.”

Robert was seized with considerable embarrassment. The cynosure of countless worshiping eyes, including a pair of very blue and very trustful ones beside the governor, he wished devoutly that he could escape. His was not a bombastic nature. Naturally of a somewhat retiring disposition, this sudden lionizing temporarily robbed him of coherent speech.

He thought of poor Taggert, who had given his life. There was but one thing that he wanted—and she was denied him. He had definitely determined not to sacrifice her future happiness for his own. Her eyes tempted him sorely. They seemed to reproach him. He realized that she really loved him and hoped that he would ask for her hand. He also felt assured that Hakon would readily consent to his marrying her, if he were to remain upon Mars. But his first duty was to the professor and Taggert. He had been entrusted with a mission by the loyal-hearted reporter as the latter was dying. That mission he would fulfil to the limit of his power.

“If you will persist in a reward, sire, then let it be in *tyrnir*, the yellow metal which we call gold, and which is so plentiful here but so rare on our

planet—or in *rahmobis*, gems of great value among our people, who know them as diamonds.”

“It shall be as you desire,” answered Hakon. “You shall have as much of both as can be carried in the *Sphere*.”

Even as Robert spoke he saw the happiness fade from Zola’s countenance. A look of gentle, pathetic reproach came into her eyes. She looked away as if to hide it from him.

It flashed to Robert’s mind that perhaps, after all, they might not be able to return to the Earth. Would they find the *Sphere* intact? He thrilled guiltily, realizing that the answer to this question might yet make the princess his.

Slowly the procession threaded its way back through the cheering populace toward the palace. Robert, astride Hakon’s mount, rode beside Professor Palmer, while Zola and her father followed in one of the luxurious motor carriages.

Her strange quiet disturbed her father.

“My daughter is not contented?” he ventured anxiously.

“I am but fatigued,” she replied, forcing a smile.

“Ah, of course you are, my dear. You must place yourself in the care of my physicians immediately upon our arrival at the palace.”

Nevertheless, he hazarded a shrewd guess as to the real cause of her lassitude.

As for Zola, her heart was heavy. Did Robert care more for precious metals and gems than for her? She would gladly have gone to the utmost ends of the universe with him unhesitatingly, with implicit trusting, yet he seemed already to have forgotten his recent avowal of love. He had even avoided her eye guiltily.

Ahead, Professor Palmer was listening sadly to Robert's tale of Taggert's death.

"Poor lad. We must carry his story back to his paper and the world. He forfeited his life for it. It is little enough."

"Little enough," Robert agreed. He thought of the sad message he would bear to the girl who would be waiting for Taggert.

They rode on in silence.

At Robert's suggestion upon their arrival at the palace, Hakon promptly issued orders for the *Sphere* to be sought out and brought back. A huge conveyance and two-score men were dispatched for this purpose.

Preparations were already under way for a great feasting to commence that night. An abundance of every kind of food produced by the little planet was prepared by skilled cooks. Every delicacy known to the Martians was procured for the occasion—even meat of the rare *mihida*. The *mihida* was the only animal still raised for food on Mars, on the pitifully small acreages of available pasturage irrigated from the ducts of precious water. None but the richest could enjoy this one available meat; and even those but occasionally. Many casks of *irel*, an excellent, mildly intoxicating vintage made from a fruit similar to our grape, were iced and tapped in readiness for the approaching festivities.

The great jubilee lasted not one night merely, but all of the next day and night, though Hakon and his immediate party, including Robert and the professor, withdrew with the first dawn.

Robert and Professor Palmer were each awarded a medal cross, highly prized by the Martians as an emblem of supreme valor. Their presentation was attended with much ceremony and a tremendous ovation. Seated between the princess and her father, Robert and the professor were the cynosure of all eyes and the envy of all the noblemen gathered.

The princess and Robert sat side by side, and their eyes were all for each other. Frequently their hands stole into each other's. Several times Robert caught Hakon watching them covertly, a quizzical smile on his face. What

was behind that smiling mask Robert knew not, but he thought he detected a trace of sadness in it.

During the height of the celebration news arrived of the formal recognition of Hakon by the leaders of the dead emperor's government, as their new emperor. Convinced by the overwhelming, disastrous defeat of Kharnov's forces, and by the popular demand of the people at large, these leaders were glad to hail Hakon as their new chief and ruler.

It was on the day following their participation in the festivities that Robert was informed of the *Sphere's* discovery and safe return. Examination of it, showed it to be unharmed.

Once more Robert wrestled with the stubborn engine which, in spite of the apparent absence of any mechanical defect, persistently refused to start. It finally developed that the petrol line from the tank to the carburetor was clogged with sediment. With this removed, the engine immediately ran as well as ever.

Hakon's chemists had finally succeeded in refining a considerable quantity of petrol—almost enough to fill the *Sphere's* reservoirs completely. The *Sphere* was in readiness at last for the return to the Barth.

The days following the public acclamation brought many proposals to the new emperor from the first nobles of the land for Zola's hand in marriage. This news Zola told Robert, and it was evident that she was wondering why he did not speak to her father for himself. The emperor, however, made no secret of the proposals. He even discussed them with Robert. Contrary to the general rule, his gratitude survived his successful acquisition of the throne, and his head remained unturned by the sudden fawning and praise from men who formerly had been his bitterest enemies. In spite of the high rank and the large fortunes of his rivals, it was becoming apparent to Robert that the emperor was inclined to favor a match between him and Zola. But he felt certain that there would be a stipulation in that event that he must not return to the Earth.

As the time decided upon between Robert and the professor for their departure drew closer, Robert decided to have a heart-to-heart talk with Hakon. Accordingly he sought an interview with him at the first opportunity.

He found him in excellent spirits. In fact, so carefree did the new monarch appear, that Robert hesitated to broach the subject; but concluding

that it was a case of now or never he put his temerity aside.

Hakon heard him out calmly. It was apparent that he had been expecting this.

“My son,” he said, finally, “this is no surprise to me. The days of my youth are not so distant that I do not recognize the symptoms of love.” He sighed. “I can’t blame you for loving her. She is her mother over again.”

His fine eyes softened as he spoke of his deceased wife. Robert did not presume to interrupt his thoughts. He waited patiently while the emperor sat in silent reminiscence.

Presently Hakon resumed, putting memories from him with a visible effort.

“You are brave, my boy, and deserving of her great love—you see, she has already told me. Duty calls you back to your world, many, many leagues distant. But it is a younger, more luxuriant world. I will not selfishly deny her happiness, though she is my greatest treasure. I would that you could remain with us, but, if you must go, she may go with you if she wishes. Let her decide. I make but one condition; if she can not be happy in your world, bring her back to me if you can.”

“I promise, sire,” said Robert, touched too deeply at the emperor’s sacrifice to say more for the moment.

A soft step caused them both to look up abruptly. Zola stood before them. She had stolen in while they were talking. Her eyes were brimming with misty happiness.

“I heard what you were saying, you dears,” she murmured.

“And your decision, Zola?” Robert faltered.

She pressed a white hand to her breast, swaying like a frail blade of grass.

“I must think—I must think,” she said, faintly.

And she fled from the room.

That night brought no sleep to Robert. Torn between compassion for Zola's father, and fear that he himself would lose her, he tossed about incessantly. When finally dawn came he fell into a sleep of utter mental exhaustion.

When he opened his eyes it was with no recognition in them of anyone or anything. The delirium of fever had laid hold upon him. The severe strain and exertions of the past several days had reduced his vitality, and the mental anguish of the night following his interview with Hakon regarding Zola had proved the last straw.

For three days he remained delirious. During this time Zola nursed him almost constantly. It was with greatest difficulty that she was induced to snatch rest occasionally. And only to Professor Palmer would she relinquish her post.

Hakon came to see Robert twice daily. His own physicians were in continual attendance upon Robert. No effort was spared to bring about his recovery if possible. On the fourth day, with the crisis safely passed, Robert recovered his senses.

His first recognition was of Zola, to her unbounded delight. She was seated at his bedside. During his delirium he had spoken her name many times. At first he feared she might be another vision. He reached out to touch her and reassure himself of her reality, only to sink back weakly. She caught his hand.

"Do you know me now, Robert, darling?" she whispered, with eager tenderness.

Robert pressed her hand happily, nodded, and promptly fell off into peaceful slumber—his first normal rest in many hours.

When he again opened his eyes he was stronger and able to take some nourishment, which Zola fed him. She had not left his bedside since his first return to consciousness early that same morning. By the doctors' orders she would not permit him to talk. But for lovers there are other means of communication than mere words. Both were infinitely happy.

The effects of Zola's continued vigil of the past three days and nights were visible in her face. Only at Robert's insistence, and for fear that he would excite himself into a relapse, did she finally consent to take to her bed for sleep. She slept the entire afternoon and night without waking, and rose feeling greatly refreshed but with bitter reproachment on her lips for those who had permitted her to sleep so long.

By this time Robert, much improved, was allowed to talk. Zola perched herself on the edge of his bed.

"We are to be married as soon as you are up," she announced, bending and kissing him as he started to splutter some inane reply. The emperor, coming in at the moment, laughed outright and made his exit quickly.

"And I shall see and know that wonderful world of yours," she continued.

Her calm assertion swept Robert's last scruples away. In his heart was a song of joy, and his boyish enthusiasm and anticipation ran riot. The thought of transplanting this desert flower from an unlovely, withered planet to his own luxuriant world was a prospect of boundless, delightful possibilities! It would seem a wonderland to her. She would be the happiest and most appreciative girl alive—and his!

"You bet you shall, sweetheart," he agreed. "You shall see our wonderful, rugged mountains, and beautiful green valleys; the winding rivers, the vast oceans, and the great lakes of water, the very drops of which are so precious here. Our clouds, the mysterious storms that will frighten you with their magnificence, and the silver rain; all these wonders and many more shall be yours."

"Do you really have big bodies and rivers of water, open and unprotected from the sun's rays? Why doesn't it evaporate, or sink into the soil and become lost?"

"You shall see, sweetheart. You shall ride upon oceans more vast than your deserts, where nothing but rolling water can be seen."

Zola shook her head in perplexity and with a certain measure of doubt. All this seemed virtually impossible to her. Only her implicit confidence in Robert enabled her to believe, and even in that belief she was unconsciously prone to reserve. Well, she would see what she would see. No doubt it was a wonderful world; but ———. However, she was a diplomat.

“Truly these are wonderful things you tell me of, my love. I am wild to see them.”

At this point they were interrupted by the doctor.

“You children must be quiet awhile now. I forbid my patient to excite himself by talking any more till this afternoon.”

And as this doctor was an autocratic soul, accustomed to having his way, they were forced to forego their conversation till later. In the heart of each, however, there was a bewildering flutter of joy and happiness.

During the next few days Robert grew rapidly stronger, and soon was permitted to be up and around.

Taggert's body had been recovered, and now rested in state within one of the royal vaults, where it had been placed with great reverence by the Martians at the command of the emperor. Elaborate and touching were the ceremonies which attended the procedure. Robert had not been able to attend the ceremonies, but Professor Palmer, accompanied by the emperor and Princess Zola, witnessed them together.

Resigned to their determination to return to their own planet, taking his beloved daughter with them, the emperor bent his efforts toward loading the *Sphere* with both *ty nir* and *rahmobis* in large quantities.

Of the *ty nir* it was simply a question of how much the *Sphere* would be able to lift safely. More than two and a half tons of the precious metal, in small ingots and in heavy sacks, were stacked on the floor of the main chamber—virgin gold, every ounce of it.

The supply of *rahmobis*, or diamonds, though not so plentiful, was a far greater treasure even than the precious yellow metal, although most of these were in the rough. They averaged in size from half a carat to several carats, with here and there a specimen running ten or fifteen carats. Of these assorted, uncut stones there were nine sacks, each about the size of a five-pound sack of sugar. In addition there were several packets of finely cut and polished gems, the product of skilled Martian cutters. These varied approximately from a quarter of a carat to two carats, but a dozen or more fine stones weighed more than ten carats each! Some excellent emeralds and rubies were included among the cut stones, but only a few, because, while the white diamonds were quite plentiful on Mars, the green and the pigeon-red varieties were very rare. Truly the *Sphere* was to carry back a ransom of kings!

But of all this treasure none was so precious to Robert as his princess.

With Robert's complete recovery, a great pageant was arranged in which the emperor, princess and all the nobles were to participate. Robert and

Professor Palmer were invited to ride with Zola and her father in the procession.

Elaborate preparations were made for this event which was to typify the recent victory and the reunion of all factions, and the gratitude of the Martians for the timely aid by their visitors from Earth. Great ornamental arches were hurriedly built, and large quantities of the various kinds of Martian flowers were accumulated in readiness for the event. The gathering of these flowers was no small task, since the restricted growing areas of the waning planet permitted of but little deviation from the grim task of producing enough food to sustain its populace.

The pageant was also to serve another purpose. At its termination the emperor was to announce the giving of his daughter's hand in marriage to Robert and her subsequent departure for Earth with him. Some resistance was anticipated from various nobles, particularly those who were eligible for Zola's hand. It was because of a possible demonstration against, the princess' departure that the emperor, with excellent foresight and admirable sacrifice, had commanded that the wedding take place quietly at the palace immediately after the pageant, and that the *Sphere* start on its long journey with his most precious possession immediately afterward.

The day of the great pageant dawned with the same wonderful brilliance that heralded 680 of the 687 days of the Martian year.

All preparations of the royal party for the pageant were completed before noon. At midday Robert and the professor partook of a simple luncheon with Zola and her father. With the specter of separation so near, conversation languished, and it was with real effort that the professor maintained at least a semblance of cheerfulness within the little group through his persistent but tactful patter of small talk.

Early in the afternoon the nobles began to arrive. Within an hour the assembly of plumed and gayly dressed riders had formed in marching order, and with a great clattering of hoofs rode through the big archway leading from the palace terrace to the main road.

A company of guards led. The emperor and his daughter, accompanied by Robert and Professor Palmer, followed them. Behind them came the chief nobles of the great empire.

From the time of the earliest formation on the palace terrace, it became apparent that some peculiar unrest pervaded the assembly. This grew more

tense as the time passed, and was only temporarily relieved when the column had ridden out from the palace. Several times as his mount shied, Robert fancied he surprized secret communications between certain of the nobles. The ostensibly unconcerned looks upon their countenances, and their abrupt cessation of whispered confidences as he caught their eye, somehow forced an unpleasant conviction upon Robert that these communications not only concerned the emperor and his party, but presaged evil for them. He wondered if, in some manner, advance news of his impending marriage to the princess and of their intended departure had got abroad. Anticipation of such information likely would produce resentment among the young-bloods who had hoped to obtain the princess' hand themselves, and they might endeavor to stir up trouble to prevent the match and the departure of the princess. He determined to keep a sharp outlook for any sign of treachery.

Into the main thoroughfare they swung. Here they halted briefly while the rest of the procession promptly formed behind. Then they moved on again toward the heart of the city.

Soon they passed beneath artificial arches over flower-strewn streets lined with dense crowds of eager-eyed, cheering Martians who were gathered to greet their new emperor and to see his mysterious aids from the planet Earth, who had put their powerful enemies to rout at the eleventh hour. Robert could not suppress a feeling of exhilaration as the deafening acclamations of the populace swelled about them. Fully half the demonstration was for the professor and himself. He glanced at the princess—*his* princess—riding close beside him, her lovely cheeks aglow with excitement. Her eyes were turned toward him in rapt admiration. Small wonder that Robert's head swam a bit with pride and keen enjoyment in this, his moment of supreme triumph and popularity. The professor, too, seemed not without his appreciation of the moment.

The procession finally reached the Galpraæ, a huge amphitheater situated in the eastern end of the city. Here, flanked by his guardsmen on one side and the nobles on the other, the emperor spoke briefly to the people. Robert, the professor, and Zola occupied positions of honor near him.

The people listened to his speech with marked respect and interest to its conclusion, when they burst into wild cheering lasting many minutes. The emperor held up his hand for quiet, till finally the demonstration ceased.

Then, calmly, distinctly, he announced his daughter's early nuptials and departure with Robert.

For some seconds after this statement a deep silence reigned. Then, suddenly, one of the nobles rose to his feet!

He pointed dramatically at Robert and Professor Palmer.

"Shall we permit these Earth-beings to carry off our own princess to another planet? Shall we permit her to wed one of these common beings while the best, the noblest, blood of all Mars is offered for her hand? *No!* A thousand times no! Our emperor's better judgment has been swayed by some strange influence of these beings. Brethren, let us not stand by idly and permit this outrage!"

As if by prearranged signal, about half the nobles sprang to their feet. Drawing their sabers, they rushed upon the little group about Robert.

At the same moment, pandemonium seemed to have broken bounds. The fickle audience in the great enclosure leapt to their feet as one and surged forward, shouting madly! The guardsmen, who fortunately were all picked men and loyal to the core, dashed forward to protect their emperor and his guests, but were prevented from joining them by the resistance of the immediate group of traitorous nobles. A few of the noblemen who were loyal joined the guardsmen in the instant melee.

Though Robert was on the alert for something of this sort, the suddenness of it left him momentarily aghast. There seemed no escape. His saber and the emperor's flashed from their scabbards together. The next instant the professor and they with two guardsmen who had somehow managed to hew their way through to them, had formed a ring of steel round Zola. Against this vicious circle the furious noblemen charged.

For minutes that seemed hours, the unequal combat raged about these five staunch men and the trembling princess. The guardsmen and loyal noblemen were more than holding their own with the larger part of the rebels. But the little group in the midst of it all was facing annihilation before aid could reach them. Already Hakon was wounded, while one of the guardsmen was down. Robert, too, was wounded, though fortunately not yet seriously.

Suddenly Robert felt the pavement give way beneath his feet. The next instant he was precipitated downward. A hard surface seemed to rush upward and strike him. He sprawled painfully. Then darkness!

For a moment he believed oddly that he had just sustained a blow which had knocked him unconscious, mistaking the sudden quiet and darkness for oblivion in his bewilderment.

Abruptly the mantle of blackness surrounding him magically dropped away. As he scrambled stiffly to his feet he perceived that he stood with others within a tunnel of masonry dimly lit by a series of incandescent lights. An exclamation of relief burst from his lips as he saw Zola sitting on the floor a few feet away. She gave a glad little cry as she recognized him. He quickly helped her to her feet. At the same moment he saw Hakon and Professor Palmer, and, with them, the surviving guard who had fought so valiantly. On the pavement lay one of their late enemies, strangely still.

"This is a secret passage leading to the palace," Hakon explained hurriedly. "Its existence and the automatic trap-door entrance above us with its rebound feature alone has preserved our lives thus far. Lead on, Dyarkon."

The guard addressed, obediently led the way down the passage, the others following. Above, faint sounds of the conflict still raging seemed far away. Zola placed her hand in Robert's trustfully. They had proceeded several rods when the emperor, who was second in lead, swayed uncertainly. He would have fallen but for Robert's timely assistance. Zola also rushed to his side with a startled cry.

"Ah, my children, I fear I am too badly wounded to go on. Leave me and escape while you may."

"We go on only with you, sire," said Robert, firmly.

Gently he and Professor Palmer lifted the protesting monarch between them. In this manner they resumed their march down the long passage, led by the faithful Dyarkon. Zola followed closely in the rear.

In silence they made their way through the long tunnel beneath the city's streets. Except for the shuffle of their feet, an oppressive, deathlike stillness reigned. At intervals Hakon begged them futilely to put him down and hurry on to safety without him.

Though the passage led in almost direct line from the amphitheater to the palace, it was a considerable distance. The emperor was no slight burden and Robert's muscles ached with the continued strain. In spite of his years, however, the professor seemed to be bearing his part of the monarch's weight without great effort.

A touch on his shoulder caused Robert to look round sharply. Zola was directly behind him, her hand upon his arm.

"Wait!" she whispered, glancing apprehensively over her shoulder.

Robert and Professor Palmer halted. Dyarkon, proceeding a few paces farther, also stopped as he perceived they were not following.

"What is it?" Robert asked. His gaze followed hers down the dim passage stretching off behind them in ghostly emptiness. He failed to discern any cause for her uneasiness.

"Listen! Did you not hear footsteps?"

They all listened tensely. Only the beating of their own hearts disturbed the deadly underground quiet. An icy touch on his neck caused Robert to start. But he discovered that it was only a drop of water, fallen from the sweating roof. Here, possibly, was the origin of the sound which had startled Zola. Every little sound within the long tunnel was magnified a hundred times by the reverberation from the dead walls. The shuffling of a foot brought muffled shufflings from the farthest recesses of the passage, dying in soft, throbbing whispers that slipped from wall to wall faintly.

"I thought I heard footsteps following us," Zola explained a trifle shamefacedly, but with a little pucker of perplexity on her forehead.

"Just the echoes, my dear," said her father.

They resumed their march toward the palace. His ears keenly alert for sounds of pursuit, Robert, too, fancied several times that he heard cautious footsteps following in the distance; but he finally concluded that what he heard was nothing more than the countless rustling echoes from their own footsteps.

At last they reached a winding stairway. Up this they followed Dyarkon till it brought them to another level stretch of paving.

At a command from the emperor the guard stopped and fumbled along the base of the right wall. A door in the masonry swung outward. Through this they all followed quickly, closing the door behind them.

They now stood within another passage exactly like the first, but running at right angles to it. Was it imagination that caused Robert to believe he heard a scurry of footsteps along the passage they had just quit?

“Did you hear?” murmured the princess, clutching Robert’s arm.

He nodded. Then he was right. They had just quit the other passage in time!

The little procession moved on again. Another short flight of stairs brought them to a stop before a blank wall at the end of the passage. Here Dyarkon repeated his former performances and the wall opened.

A brilliant stream of sunlight burst upon them. The abrupt contrast with the dim glow of the passage all but blinded them for a few seconds.

An involuntary exclamation burst from Robert’s lips. The *Sphere* rested within fifty feet of them! They were standing inside the broad wall of the palace courtyard!

Instantly his mind formed a plan of action. They would make a dash for the *Sphere*. Once safely inside they could rise quickly and observe the actions of the crowds. Then they could lay their plans at leisure.

Rapidly he outlined his plan to the others, who acquiesced at once. If their pursuers had already reached the palace they had not a moment to lose. The courtyard was yet closer.

Hakon was able to stand, though his wounds had left him pitifully weak. Dyarkon and the professor now assisted him while Robert hurried ahead to open the trap-door entrance into the *Sphere*.

As they emerged from the wall a loud outcry greeted them. Without stopping to ascertain its source they hurried toward the *Sphere* with all possible speed. Fortunately the trap operated readily, and a few seconds later they were all safely shut within.

The outcry was now explained. Into the courtyard from the palace poured a score of nobles with drawn sabers, shouting for them to stop.

Robert jerked the control over. The *Sphere* leapt from the ground with such sudden force that all except Robert and the staunch Dyarkon were thrown to the floor. A minute later they were soaring far above the heads of their late pursuers.

“Phew! Close shaves are getting to be our specialty,” exclaimed Robert, recovering his breath for the first time in many minutes. “Now for our observations and conference.”

He checked the *Sphere*’s ascent and turned to the others.

Zola was already busily binding her father’s wound. Professor Palmer had just brought her some water and a supply of bandages from the first-aid chest. Fortunately, though Hakon was weak from loss of blood, his wound was found not to be serious.

Hakon was staring intently groundward from his position by a window. Following his gaze, Robert saw a dense mob round the palace. Even at this height he could hear the Martians’ cries faintly. Evidently the rebel noblemen had succeeded well in working the masses up in revolt.

Sadly Hakon viewed the disorder below. It was now clear that it would not be safe for him to return.

“Let us all go to Earth, my dear father,” said Zola. “There we can be happy together.”

The fugitive ruler pondered for many minutes, while the others maintained a respectful silence. Finally he sighed resignedly. A faint smile played over his countenance as he turned to his daughter.

“Ah, my dear, I was a very foolish old man to think of letting you go alone. We shall, as you say, be far happier together. We shall have riches and contentment in this world of Robert’s—if, indeed, he and Professor Palmer will share a little of their fortune with us.” He smiled as he nodded toward the bullion stacked on the chamber floor.

“You are the spokesman, Robert,” chuckled the professor.

“The treasure is yours and Zola’s, sire, now that you are with us,” said Robert.

“I have given it to you and Professor Palmer, my boy, and it remains yours, except for what small portion you might wish to assign me—and Dyarkon, if he decides to go with us. As for Zola, she will share with you as your bride. What say you, Dyarkon—do you wish to go with us?”

“Oh, sire, I shall go if you desire it; but I was to have been married shortly. My heart is there.” He pointed below.

“Then you shall be permitted to return, my man. Accept this, my present to your bride; and may you have great happiness.” He handed the guard a string of beautiful emeralds which he had been wearing.

The faithful Martian was speechless with gratitude.

“I suggest, then, that the treasure be divided into four equal parts,” said the professor, presently; “one quarter for each of us. There is sufficient wealth here to make every one of us overwhelmingly rich on Earth.”

So it was agreed.

The question of provisions was the next consideration. At Hakon’s orders, large quantities of evaporated fruits and vegetables had previously been placed within the cupboards of the *Sphere*. A goodly quantity of the *Sphere*’s original supply of food tablets, etc., remained. Fortunately, too, the oxygen tanks contained enough gas to purify the air in the *Sphere* for a long while. It only was necessary to replenish their water supply, when they could also leave Dyarkon.

The latter task was not so easy as it sounds. For there are no convenient, open streams on Mars. They must either chance landing at some power station or farm, or fly to one of the poles and there obtain water from one of the giant reservoirs. The elements at the nearest pole being very treacherous at this season, it was decided to chance a visit to some farmhouse.

A hurried trip was accordingly made to a small farm, a sufficient distance from the scene of the rebellion to be reasonably safe. Here the astonished farmer, who had not yet heard of the rebellion and who did not even recognize the emperor and the princess, eagerly helped these distinguished visitors to fill the water tanks of the mysterious *Sphere*. This the farmer had heard of, and both he and his wife gazed upon it with mingled wonder and dismay. Afterward they followed it with their eyes until it had passed beyond their vision. This farmer, and his wife and Dyarkon, had the distinction of being the persons on Mars who last saw their emperor; though the two first named did not know this till Dyarkon presently told them.

After the filling of the water tanks, Robert steered the *Sphere* straight toward the distant pale star which he and Professor Palmer knew was the Earth. Despite their anticipation and resignation, Zola and her father gazed back upon their erstwhile world in silent awe, and not without some sadness, long after it had ceased to be more than a mere ochre and rose disk.

Through the eternal night sped the infinitesimal world with its population of four. And through the long hours of Robert’s watches, Zola was at his side always. Their love was as an immortal thing, born of space and

eternity. Hand in hand they fled across the universe to their future world of promise.

Profiting by their previous experience with gravitation, or rather, an absence of gravitation and stabilization, Robert and the professor properly manipulated the disk and gyrostats on this trip, avoiding the danger which had so nearly proved their undoing before. Robert prevented also the recurrence of another unpleasant experience, by cutting short pieces of stout cord, one for each of them, and particularly cautioned Zola and her father to tie them about their bodies at night and secure the other end to a rung or some other stationary object at a safe distance from the whirling gyrostats.

It was not long after that they had a taste of air-floating, and the cords proved their worth. This sensation, the continued sunshine out of a black sky and other phenomena, were all new to Zola and her father. The time passed rapidly.

A deck of playing cards was got out and Hakon and Zola were initiated into the mysteries of the Earthmen's card games, which they learned readily and seemed to enjoy keenly. They then proceeded to show Robert and Professor Palmer some of their own games. These, being played with cards not greatly different from our own, were easily adapted to the cards they were using. In fact, one of their games, called *Agahr*, was virtually identical with our own simple game of casino.

So it did not seem long ere they were within a day's journey of the Earth. Not a single mishap had delayed their progress so far. Barring the unexpected, they should be but a day longer in returning than the period covered by the trip to Mars, in spite of the considerably increased distance between the two planets by this time. Nearly three months had elapsed since the departure from the Earth.

As the Earth's disk expanded before their eyes, Robert pointed out to Zola and Hakon the outlines of the continents and oceans, the mountain ranges and rivers. Their genuine wonder and delightful anticipation were a source of keen enjoyment to both the professor and Robert.

"It surpasses my wildest imagination to vision an expanse of water so vast that one can not see its boundaries!" exclaimed Hakon, excitedly. "I can scarcely contain myself till we shall actually see these wonders with our own eyes."

"And think, Father, of the great forests of trees where one can really get lost; the mysterious clouds in the sky; the rushing rivers and waterfalls! Oh, how could I have thought of letting you stay away from all this! How happy we can be, can't we, Robert?"

"Indeed we can, sweetheart," he replied, with a feeling that his measure of delight was far more than he deserved.

Closer, closer drew the big world—his world and hers. Its great disk swelled and swelled, until it was no longer a disk but a vast expanse stretching away in all directions.

Robert had reduced the *Sphere's* speed until they approached the surface, now less than fifty miles away, at about the speed of a fast passenger train. As they drew closer he reduced their speed still further. A big cloud bank obscured their view of the Earth's surface now, but he knew that they were above the Atlantic. He had already given the *Sphere* the spin of the swiftly revolving Earth, before entering its envelope of atmosphere. They now drifted serenely, high above the clouds.

As they slowly drew near the cloud bank, Zola made a natural mistake of thinking it the ocean, till Robert told her different. Her astonishment and delight were great as they plunged through the fluffy mist and emerged above the water. A big sea was running, and Robert permitted the *Sphere* to drop within a hundred yards of the tall crests.

The continual rolling of the water mystified Hakon and Zola. This was explained to them with some difficulty. Robert opened two of the *Sphere's*

ports, for the first time since leaving Mars. They all filled their lungs gratefully with the keen, salty air as it blew in upon them. The main force of the gale was not felt, however, because the *Sphere* was being driven before it. Once an eccentric gust sucked the *Sphere* down abruptly. A mountainous wave, rearing hungrily toward the big metal ball, slapped forcibly against it, causing it to rebound high into the air with a suddenness that upset everyone. After that Robert kept a safe distance above the seething waters.

For a while they scudded swiftly along under the hypnotic spell of the restless sea. Its hissing turbulence was a source of continual awe and wonder to their guests. Finally Robert closed the ports and sped the *Sphere* toward the Jersey coast.

It was in the early afternoon when they passed over the coast line. Here their appearance was first noted and news of the *Sphere*'s safe return flashed all over the world. Later, as they sailed over New York, a droning of many whistles heralded their arrival, while a blimp, a big seaplane, and several airplanes glided and cavorted over, under and round them.

"*Sphere* ahoy!" shouted one venturesome chap, a reporter on the *Times*, as he whizzed by, a dozen feet away, in a two-passenger airplane. "What news?"

But the drone of his engine drowned a possible answer as the distance between them widened rapidly.

Leaving Manhattan, Robert steered the *Sphere* toward L— and Professor Palmer's estate. This was at the latter's request, and in response to his cordial invitation to Robert and both their guests to make their home with him for the present.

Their arrival at the Palmer estate found the place already overrun with reporters and photographers in anticipation of their return there. Even the resourceful Henry could not stem the tide. Motion pictures of them all were run off and rushed to headquarters for early projection upon the silver screen all over the world.

Hakon, and Zola, more charming than ever, both accepted the situation with jolly good nature. Praises of the beautiful maiden from Mars were many, and their sincerity was reflected in the headlines and articles which appeared as by magic in the afternoon papers throughout the country the very day of their arrival.

The party rested at the Palmer estate for several days. Many were the delightful strolls which Robert and Zola took in the lovely grounds. The soft, luxuriant grass under foot, the tall trees, the beautiful shrubbery and flowers were as a fairyland to the princess, with her fairy prince at her side. As for Robert, he was in a veritable seventh heaven.

The emperor and Professor Palmer, now great cronies, were constantly together. Halton never tired of the professor's tales of the Earth's resources, its history and people; and of our long observation of and conjectures regarding his own planet, Mars.

Negotiations were opened with a firm of expert diamond cutters in New York for the cutting and polishing of the stones brought from Mars. Their representatives, escorted by a heavy guard, arrived promptly and departed with the first valuable consignment of the rough gems.

The balance of the treasure, in bullion and stones, had been safely deposited in the vaults of three different banks for greater safety. The bullion, however, was rapidly converted into cash and deposited in equal shares to the individual credit of the four adventurers and one other person. This person was Taggert's sweetheart, a Miss Sarah Daugherty, who had waited faithfully for the valiant reporter's return. By mutual consent, a fifth and equal share of the treasure was allotted her. Taggert's mother, poor woman, had not lived to see the return of the *Sphere*. She had contracted pneumonia and passed away a month before her son's death. One of the first things Robert had done upon his return was to seek Mrs. Taggert and Miss Daugherty, after delivering Taggert's notes to the *Morning Chronicle* with an additional report on the events following the lion-hearted reporter's death. He obtained the publishers' ready consent to turn over all salaries and bonus due Taggert, to Miss Daugherty.

From the moment of their return, Robert and Professor Palmer were lionized by the world. Eminent scientists from everywhere sought interviews with them. Even the former opponent of the Palmer theories, Professor Margard, came to Professor Palmer with sincere congratulations. They were besieged by learned societies to lecture at gatherings for their enlightenment. Capitalists and promoters begged them to consider offers of enormous sums for their patents on the *Sphere*'s remarkable gravity-defying principles.

Construction of a huge device for flashing messages to Mars by means of reflection of the sun's rays was commenced in the Sahara Desert. A code

furnished by the emperor was to be used. Though wireless had been considered, the enormous distance was judged to be too great to make that method of communication practicable, even with the most powerful apparatus then conceivable.

“They’ll be betting on each other’s stock markets soon,” laughed Henry, when he heard of the project.

On the day of Robert’s and Zola’s wedding, the emperor presented his daughter with a magnificent, perfect ruby, which he had had set, and hung in a pendant, with the connivance of the professor. The gem was uniquely cut, similarly to what we know as table-cut. He also presented them with a packet of three remarkable stones, in the rough, which he had secretly brought with him. One of these was a black diamond of twenty-one carats; another was a white diamond of slightly larger size.

The third stone was also a white diamond, but of astounding size. It was several times larger than the famous Koh-i-noor; it even exceeded in size the Great Mogul in the rough, as it balanced at a trifle under 1,115 carats! Properly cut and polished, without the unfortunate bungling which both the Koh-i-noor and the Great Mogul had suffered, it should weigh considerably more than these two famous gems together, they weighing 106 and 280 carats respectively after their final cutting.

“It should be named,” said the professor, when shown this enormous stone. “What are you going to call it?”

“Let us call it the *Ragnarok*, which means ‘the twilight of the gods and the doomsday of the world’—in memory of the waning world from which it came,” suggested Robert, after some thought.

And so it was named.

The little vine-clad church in the village saw the wedding of Robert and Zola on a delightful, soft autumn afternoon a few days later. Her father gave her away, and Professor Palmer was the best man. Futile attempts at fittingly describing the glorious vision presented by the princess were attempted. But perhaps none was more apt than that ventured by the professor’s housekeeper, a kindly soul, who had helped Zola choose her dainty bridal gown and charming trousseau. “A true daughter of the gods,” was the rather surprising expression of this normally prosaic woman.

More surprising, however, may have been the choice of these two young beings of the scene of their honeymoon. Not a tour of Europe, nor of the

natural wonders of our own great country. They simply disappeared into the great Canadian wilderness. There, if one could have followed them, they might have been discovered happily paddling a well-loaded canoe up a winding stream of still, friendly, wooded shores. Above, the clear blue sky rivaled the crystal transparency of the rippling stream. A hawk drifted across the ribbon of blue and was lost again beyond the maze of tall pines. Somewhere a woodpecker drummed stoutly upon a dead limb.

Softly, easily, the slim craft rounded a bend to the even thrust of two pairs of vigorous, willing young arms. Like the hawk, it was soon lost to view—lost in a twilight wilderness of love and peace.

[THE END]

Transcriber's Note: This story appeared in four consecutive issues of *Weird Tales* magazine starting in November 1925.

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A WORLD ***

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